Global Uncertainty and Fiscal Space: Who Can Still Afford to Act?

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ABSTRACT

Global uncertainty is at a record high and public debt ratios remain above their pre-pandemic levels, so this paper asks which economies can still afford to act. In an annual panel of 139 economies over 1990–2025, a global uncertainty shock is interacted with each economy's predetermined fiscal position and traced through the debt identity, with a Bohn-type reaction function, smooth-transition local projections, a news-based fiscal-policy-uncertainty index from domestic and foreign outlets, and the pandemic as an identified shock. A global shock erodes the indebted economy's fiscal space within two years, through growth and prices before the budget, by roughly 2 percentage points of gross domestic product across the interquartile range of debt. The cyclical stance is unchanged by uncertainty. What changes is the price of the stance, which the exposed vulnerable sovereign pays. In the pandemic, space bought less debt and more output, and status explained nothing beyond position. Affording to act is a question of debt, not of income group.

KEYWORDS: uncertainty; fiscal space; public debt; fiscal reaction function; local projections; GDELT

JEL: E62; H63; H68; F34; C23

1 Introduction

Uncertainty raises the demand for fiscal support and erodes the means to provide it. A government facing an uncertain year is asked to insure households and firms against outcomes it cannot forecast, at the moment when growth, prices and the cost of its own debt are moving against it. Which governments can still afford to act is the question of this paper, and its answer turns on fiscal space rather than on the income group a government belongs to.

The question is not hypothetical. Figure 1 plots the gross domestic product (GDP)-weighted World Uncertainty Index (WUI) over 1990–2025 together with a news-based index of fiscal-policy uncertainty

(FPU) at the world level, monthly from 2016. The second index is new. I construct it for this paper from the Global Knowledge Graph (GKG) provided by the Global Database of Events, Language, and Tone (GDELT), as Appendix B describes. In 2025 the global index stood at 0.824 mentions per thousand words, 2.1 times its previous peak in 2019 and 7.8 times its mean of the 1990s, and it stayed there once the pandemic component was removed (0.823), whereas in 2020 that component had accounted for 51% of the index. The fiscal-policy index peaked with the US tariff announcements of April 2025 at 301 on a scale on which its 2016–2025 mean is 100, above its pandemic peak of 210, and the episode registered as a domestic peak in each of the six economies of Figure 2, which Kalemli-Özcan et al. (2026) date and measure. The shock arrived on debt ratios that were higher than before the pandemic: the median gross debt ratio of the economies in the panel was 48.8 percent of GDP in 2019 and 52.6 in 2025, with 39% of economies above 60 percent against 32% before.

Three questions follow. Does global uncertainty erode fiscal space, and through which channel of the debt dynamics? Does it change how governments respond to their debt and to the cycle, and does fiscal space protect an economy when the shock arrives, and against what?

Prior work bears on each question from a different direction. The measurement of uncertainty and its macroeconomic effects runs from Bloom (2009), who establishes that uncertainty shocks are contractionary, through the newspaper-based policy-uncertainty index of Baker et al. (2016), the World Uncertainty Index of Ahir et al. (2022), which counts the word in Economist Intelligence Unit country reports for 143 economies, and the geopolitical risk index of Caldara and Iacoviello (2022), to the fiscal-policy-uncertainty database of Hong et al. (2024, 2025), built from US, British and Canadian outlets for 189 countries, which they find raises sovereign spreads and lowers industrial production. Fernández-Villaverde et al. (2015) estimate the output cost of fiscal volatility for the United States, Mumtaz and Surico (2018) the role of policy uncertainty in aggregate fluctuations, and Bhattarai et al. (2020) the spillover of US uncertainty to emerging-market spreads and output. I take the global shock from the World Uncertainty Index and add a fiscal-policy index that separates what the domestic press and the foreign press say about the same economy.

A second strand concerns the fiscal reaction function and the space it implies. Bohn (1998) reads the response of the primary balance to debt as a test of solvency, Mendoza and Ostry (2008) estimate it in panels of advanced and emerging economies, and Ghosh et al. (2013) add a cubic that captures fiscal fatigue and derive model-based debt limits, of which Ostry et al. (2010) give the estimates for advanced economies that Heller (2005) had defined fiscal space against. Kose et al. (2022) assemble the cross-country fiscal-space database used here. On uncertainty and the public finances, Nguyen et al. (2023) find that the World Uncertainty Index worsens fiscal balances on average, Estefania-Flores et al. (2023) that debt forecast errors are larger under uncertainty, and Renne and Pallara (2019) that market-implied fiscal space falls with policy uncertainty in four advanced economies, while Lian et al. (2020) find that global volatility raises interest rates most where debt is high, and Mendoza and Oviedo (2004) model the emerging-market sovereign as a tormented insurer under macroeconomic uncertainty. To my knowledge, the effect of uncertainty on the reaction function itself, on the slope with respect to debt, has not been estimated, and the question of whether space or status carries the heterogeneity has not been put to a panel of this breadth.

A third strand treats fiscal space as a buffer and fiscal policy as state dependent. Romer and Romer (2018, 2019) show that governments of the Organisation for Economic Co-operation and Develop-

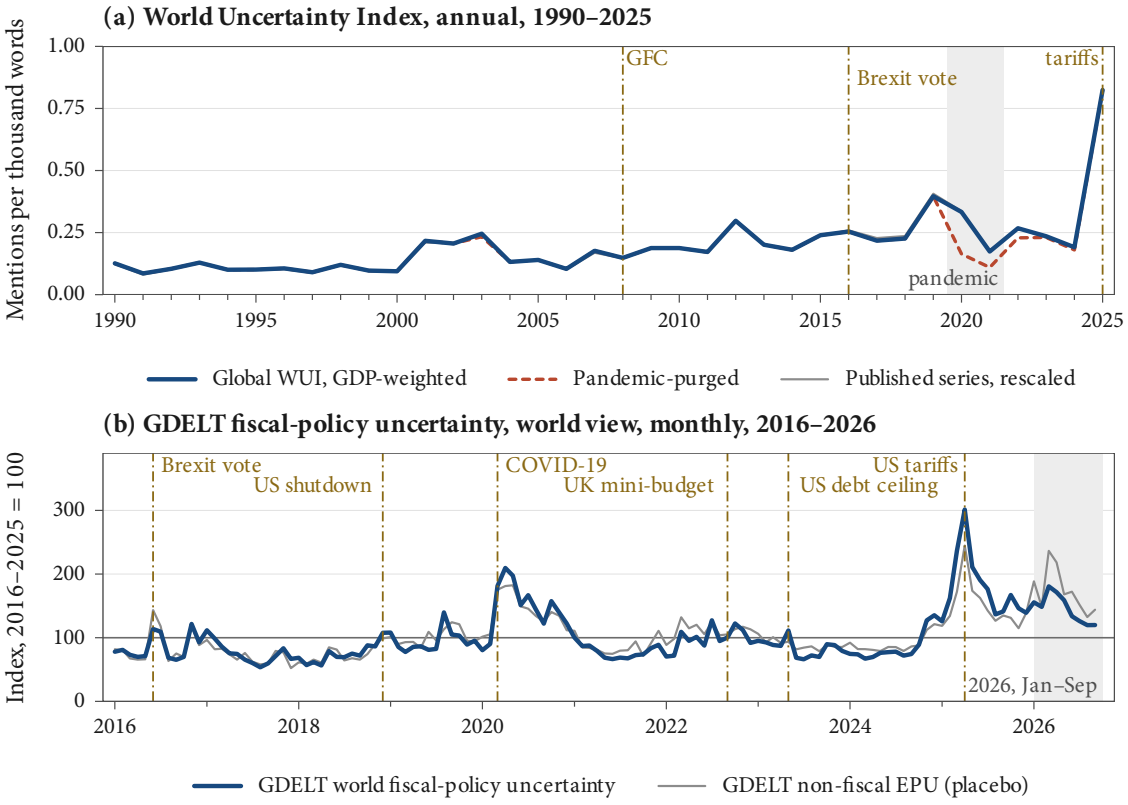


Figure 1. Global uncertainty, 1990–2026

Notes: Panel (a) plots the GDP-weighted mean of the country World Uncertainty Index (WUI; lagged nominal GDP weights; mentions of uncertainty per thousand words), its pandemic-purged version (the GDP-weighted mean of max(WUI − WPUI, 0), where WPUI is the World Pandemic Uncertainty Index) and the published GDP-weighted global WUI divided by 10^5; 2020–2021 shaded. Panel (b) plots the GDELT world fiscal-policy-uncertainty index and an index of non-fiscal economic policy uncertainty (EPU) built from the same news articles, monthly, 2016–2025 mean = 100; 2026 (January–September) shaded.

ment (OECD) with less debt supported their economies more in financial crises and suffered milder aftermaths, Huidrom et al. (2020) that fiscal multipliers depend on the fiscal position, Aizenman et al. (2013) and Augustin et al. (2022) that sovereign spreads price fiscal space, the latter in the pandemic. The smooth-transition designs of Auerbach and Gorodnichenko (2012) let the response depend on the state, and the evidence on whether fiscal policy works better in uncertain times is mixed for the United States (Alloza, 2022; Fritsche et al., 2021) and points the other way for public investment (Gbohoui, 2021). The fourth strand is the procyclicality of fiscal policy in emerging and developing economies, documented by Gavin and Perotti (1997), Kaminsky et al. (2004) and Ilzetzki and Vegh (2008), from which many economies have graduated (Frankel et al., 2013), and which Bianchi et al. (2023) derive from sovereign risk in a model and confirm in a panel of 70 countries. Whether uncertainty makes the stance more procyclical where space is short is the question this literature leaves open.

I make two departures from this work. The first is to identify a global uncertainty shock from the world index and interact it with each economy's predetermined fiscal position, under country and year effects, so that what is estimated is the difference in response between an indebted and a lightly indebted economy, and I read that difference through the debt-dynamics identity, so that the erosion

of space is attributed to growth, inflation, interest, the primary balance or valuation. The second is to bring a news-based measure of fiscal-policy uncertainty with a domestic–foreign split into the same panel, together with the pandemic as a shock that every government faced in the same year, so that the insurance that space provides can be tested on an identified crisis as well as on average turbulence.

The results suggest four conclusions. First, global uncertainty erodes fiscal space, and it does so through the denominator of the debt ratio before it reaches the budget. Two years after a one-standard-deviation shock, an economy with debt at the upper quartile of the sample carries roughly 2 points of GDP more debt than one at the lower quartile, a gap that growth and inflation open and that the primary balance sustains only later. Because the advanced economies hold the larger debt ratios, the same world shock costs them the most debt, so that affording to act is a matter of position on the debt distribution rather than of income group. Second, the stance of fiscal policy over the cycle is invariant to uncertainty. Governments lean against the cycle in turbulent years as in quiet ones, with or without space, and weak-instrument-robust sets rule out the hypothesised reversal. Country-level uncertainty bends the response of the balance to debt in the direction of fiscal fatigue, by a margin the data leave open, and the bend reverses when the uncertainty is global or voiced abroad, a split reported with its qualifications. Third, the cost of carrying debt is what uncertainty changes. An exposed vulnerable sovereign's effective interest rate rises after a world shock while an exposed safe sovereign's falls, a gap of roughly 0.6 percentage points, and the budgets of the two regimes are too imprecisely estimated to tell apart; a sharp debt threshold appears only in the level of the response to the shock, at a low debt ratio, and not in the gap between the regimes. Fourth, the pandemic shows what space is for. Economies that entered it with less debt came out with less debt and more output, economies with market access spent more on support whatever their debt, and once the fiscal position is held constant advanced-economy status explains nothing further. Space insures against the price of a routine global shock through the regime it places an economy in, and against a crisis, and not, on this evidence, against the real effects of routine turbulence.

The papers closest to this one differ on named dimensions. Hong et al. (2024) build a fiscal-policy-uncertainty index for 189 countries from the foreign view and study spreads and activity, whereas the index here adds the domestic press in each economy's own languages and is used to ask what uncertainty does to the public finances rather than to markets. Nguyen et al. (2023) estimate the average effect of world uncertainty on the fiscal balance, whereas here the balance is modelled as a reaction function with the cycle instrumented, so that uncertainty's effect on the slope with respect to debt is separated from its effect on the level. Ghosh et al. (2013) derive debt limits from the reaction function. Here the limits are not identified, because the interest–growth differential was negative for most economies over the sample period, and the fiscal-space-at-risk reading of Figure 4 takes their place. Romer and Romer (2019) document insurance against financial crises, and here the same question is asked of routine global shocks and of the pandemic and gets different answers.

The rest of the paper is organised as follows. Section 2 sets out the measures, the identification principle and the five models. Section 3 describes the data and presents the findings on erosion, on the conduct of fiscal policy, on regimes, insurance and the pandemic, and on their sensitivity. Section 4 re-argues the conclusions against prior estimates, and Section 5 concludes.

2 Empirical Framework

2.1 Uncertainty, fiscal space and the identified shocks

Five measures of uncertainty enter the models as regressors. The first is the World Uncertainty Index of Ahir et al. (2022), which counts the word "uncertainty" in the Economist Intelligence Unit's quarterly country reports, scaled by their length. I average it over each economy's fiscal year, so that the uncertainty a government faced when it set its budget is the uncertainty in its reaction function, and denote the result $U_{i,t}$. The second and the third are a news-based index of fiscal-policy uncertainty built from the GDELT Global Knowledge Graph in the lineage of Baker et al. (2016), the share of all articles that mention the economy, fiscal policy and uncertainty together. It is computed separately from the outlets of the economy itself (the domestic index $U^{\text{dom}}_{i,t}$) and from outlets abroad writing about it (the foreign index $U^{\text{for}}_{i,t}$), annual over 2016–2025 (Appendix B). The fourth is the global shock $\varepsilon^{\text{g}}_{t}$, the innovation of an autoregression of order two fitted to the GDP-weighted mean of the country WUI over the economies the index covers, with lagged nominal GDP as the weight. The fifth, a domestic shock $\varepsilon^{\text{d}}_{i,t}$, is the innovation of a country autoregression of order two of the WUI with the global index as a contemporaneous regressor. Both shocks are divided by the residual standard deviation of their autoregression and are in standard-deviation units by construction. The level measures are standardised within the main estimation sample (Appendix A). A coefficient is therefore the effect of one standard deviation.

Fiscal space is measured by the lagged gross debt ratio $d_{i,t-1}$, in percent of GDP. Where it is interacted with a shock it is standardised within the model's sample, which I write $\tilde{d}_{i,t-1}$ (a higher value is more debt), and responses are evaluated at its 25th and 75th percentiles, which I call the high-space and the low-space economy. Three further measures serve as checks and as regime variables. The composite vulnerability index $v_{i,t-1}$ is the equal-weight average of five standardised indicators from the fiscal-space database of Kose et al. (2022) (the debt ratio, the deficit, the foreign-currency share of debt, debt maturing within a year, and the sovereign rating with its sign reversed). It is oriented so that a higher value is a more vulnerable position, and $\tilde{v}_{i,t-1}$ is its value standardised within the model's sample. The rating itself and the foreign-currency share are the other two, and exposure to the global shock is E_i, mean trade openness over 1995–2004, standardised across economies. Groups follow the World Economic Outlook (WEO) split into advanced and emerging market and developing economies, with the International Monetary Fund's (IMF) three-way classification where a third group is informative.

The identification principle is the same in every model with a global shock. A shock to world uncertainty is exogenous to the fiscal position of any one economy, and its interaction with a predetermined position is identified under country and year fixed effects. I refer to the condition as Condition G: (i) $\varepsilon^{\text{g}}_{t}$ is uncorrelated with the disturbance of economy i in year t and after, conditional on the controls (exogeneity); (ii) the position it is interacted with, $d_{i,t-1}$, $v_{i,t-1}$ or E_i, is determined before the shock arrives (a predetermined position). Year effects absorb everything common to all economies in a year, the shock's own average effect included, so what is identified is the difference in response between positions, not the level of the response, which only the specification without year effects of Appendix C identifies. The domestic measures are different. A fiscal crisis raises domestic uncertainty as surely as domestic uncertainty raises the cost of a crisis. The coefficients on $U_{i,t}$ and $\varepsilon^{\text{d}}_{i,t}$ are therefore identified only from within-country variation under the year effects of equations (4) and (5) and read as patterns

rather than as causal effects.

2.2 Erosion and the conduct of fiscal policy

The first model asks how a global shock reaches the debt ratio. The change in debt decomposes exactly into five contributions, which I write as

$$\Delta d_{i,t} = c^{\text{int}}_{i,t} + c^{\text{gr}}_{i,t} + c^{\text{inf}}_{i,t} + c^{\text{pb}}_{i,t} + c^{\text{sfa}}_{i,t}, \tag{1}$$

where $c^{\text{int}}_{i,t} = r_{i,t}\, d_{i,t-1}/(1+g_{i,t})$ is the interest contribution, $c^{\text{gr}}_{i,t} = -\gamma_{i,t}\, d_{i,t-1}/(1+g_{i,t})$ the real-growth contribution, $c^{\text{inf}}_{i,t} = -\pi_{i,t}(1+\gamma_{i,t})\, d_{i,t-1}/(1+g_{i,t})$ the inflation contribution and $c^{\text{pb}}_{i,t} = -pb_{i,t}$ the primary-balance contribution, with $r_{i,t}$ the effective nominal interest rate on the debt stock, interest paid in t over the stock at the end of $t-1$, $\gamma_{i,t}$ real growth, $\pi_{i,t}$ deflator inflation, $g_{i,t}$ nominal growth and $pb_{i,t}$ the primary balance in percent of GDP, and where $c^{\text{sfa}}_{i,t}$ is the stock-flow adjustment that closes the identity and carries valuation effects and below-the-line operations.[1] For each contribution $c^{(\ell)}$, $\ell \in \{\text{int}, \text{gr}, \text{inf}, \text{pb}, \text{sfa}\}$, I estimate cumulative local projections in the sense of Jordà (2005), with k the summation index and h the horizon:

$$\sum_{k=0}^{h} c^{(\ell)}_{i,t+k} = \alpha^{(h,\ell)}_i + \delta^{(h,\ell)}_t + \theta^{(h,\ell)} \left(\varepsilon^{\text{g}}_t \times \tilde{d}_{i,t-1}\right) + \Gamma^{(h,\ell)\prime} X_{i,t-1} + u^{(h,\ell)}_{i,t+h}, \qquad h = 0, \ldots, 5, \tag{2}$$

where $\tilde{d}_{i,t-1}$ is lagged debt standardised within the sample, and $\alpha^{(h,\ell)}_i$ and $\delta^{(h,\ell)}_t$ are country and year effects. The vector $\Gamma^{(h,\ell)}$ holds the coefficients on the six lagged controls $X_{i,t-1}$ (real growth, inflation, the primary balance, the debt ratio, trade openness and commodity terms-of-trade growth). The error $u^{(h,\ell)}_{i,t+h}$ is the projection error at horizon h, serially correlated by construction because it accumulates the shocks that arrive between t and $t+h$ and correlated across economies within a year, which the Driscoll–Kraay standard errors of Section 2.3 allow for. The estimand is the differential response, the cumulative change in contribution ℓ at horizon h for the low-space economy minus that for the high-space economy, which is $\theta^{(h,\ell)}$ times the distance between the two positions in standard deviations of debt. Because the contributions add up, the differential debt response is the sum of the five, so equation (2) says through which channel the same global shock leaves the indebted economy with more debt. Figure 3 reports the responses at the two positions relative to the mean-debt economy, and Figure 4 reads the same slope at each economy's latest debt ratio.

The second model is the fiscal reaction function in the lineage of Bohn (1998), estimated for panels by Mendoza and Ostry (2008) and extended with a cubic in debt by Ghosh et al. (2013) to allow for fiscal fatigue, the weakening of the primary-balance response at high debt. I add uncertainty and its interaction with debt:

$$pb_{i,t} = \alpha_i + \rho\, pb_{i,t-1} + f(d_{i,t-1}) + \eta\, U_{i,t} + \theta_U\, d_{i,t-1} U_{i,t} + \varphi\, gap_{i,t} + \varphi_U\, gap_{i,t} U_{i,t} + \Gamma' X^{\text{rf}}_{i,t} + u_{i,t}, \tag{3}$$

where $f(d) = \psi_1 d + \psi_2 d^2 + \psi_3 d^3$ with debt in percent of GDP, α_i is a country effect, ρ the persistence of the balance, η the level effect of uncertainty, φ the response to the output gap $gap_{i,t}$ (in

[1] Nominal growth is $g = (1+\gamma)(1+\pi) - 1$, and the effective rate r is interest paid in year t over the debt stock at the end of $t-1$.

percent of trend GDP, from the regression filter of Hamilton (2018)), and φ_U its change with uncertainty. The vector Γ holds the coefficients on the controls $X^{\mathrm{rf}}_{i,t}$, which are lagged real growth, inflation and openness, contemporaneous commodity terms-of-trade growth, and election-year and fiscal-rule indicators with indicators for the years in which they are unavailable. The error $u_{i,t}$ has mean zero conditional on α_i, the instruments and the regressors dated $t-1$ and earlier, and is serially correlated within economies and correlated across economies within a year, which the Driscoll–Kraay standard errors allow for. The output gap and its product with uncertainty are correlated with it, which is why they are instrumented. The reaction function carries country effects only: year effects would absorb the common component of uncertainty to which a reaction function responds and the world-growth instrument with it, so $U_{i,t}$ enters with its global component, and the country-trend and system-GMM versions of Table C.20 are the checks on the lagged balance with country effects, the setting of Nickell (1981). The coefficient of interest is θ_U, the change in the marginal response of the primary balance to debt per standard deviation of uncertainty. The output gap is endogenous to the balance. The gap and its product with uncertainty are therefore instrumented by world real growth excluding the economy itself, by its product with exposure, and by the lagged gap, each alone and multiplied by $U_{i,t}$. The instruments move the economy's own cycle (relevance); world growth affects the primary balance only through that cycle once the controls and the country effect are held fixed, and the lagged gap is uncorrelated with the current disturbance given the lagged balance (exclusion), a strong assumption in a year such as 2009, which Section 3.5 checks by excluding that year. Model-based debt limits in the sense of Ghosh et al. (2013) are not identified in this panel, because the reaction function rarely crosses a positive interest–growth line from above (Appendix C.2.4).

The third model asks whether uncertainty changes the cyclical stance and whether fiscal space conditions the change. The dependent variable is the change in the primary balance, and the cyclical response is allowed to depend on uncertainty, on fiscal space and on their product:

$$\Delta pb_{i,t} = \alpha_i + \delta_t + gap_{i,t}(\phi + \phi_U U_{i,t} + \phi_F \tilde{d}_{i,t-1} + \phi_{UF} U_{i,t} \tilde{d}_{i,t-1}) + \boldsymbol{\lambda}' \mathbf{m}_{i,t} + \Gamma' X^{*}_{i,t-1} + u_{i,t}, \qquad (4)$$

where $\mathbf{m}_{i,t} = (U_{i,t},\ \tilde{d}_{i,t-1},\ U_{i,t}\tilde{d}_{i,t-1})'$ collects the lower-order terms and $\boldsymbol{\lambda}$ their coefficients, $X^{*}_{i,t-1}$ is $X_{i,t-1}$ without the debt ratio and Γ its coefficients, and α_i and δ_t are country and year effects. The cyclical response is ϕ, and ϕ_U, ϕ_F and ϕ_{UF} are its modifiers, the subscript F marking the fiscal-position term ($\tilde{d}_{i,t-1}$ here, $\tilde{v}_{i,t-1}$ in the vulnerability column of Table 3). The error $u_{i,t}$ has the properties of its counterpart in equation (3) under the additional year effects, and the four gap terms are correlated with it and are instrumented as described below. The estimand is the cyclical response $\phi+\phi_U U+\phi_F \tilde{d}+\phi_{UF} U \tilde{d}$ at the four corners defined by the 25th and 75th percentiles of uncertainty and of debt. The hypothesis under test is a sign switch, in which a government with space leans against a downturn when uncertainty is high and a government without space does not. The switch requires ϕ_{UF} to differ from zero by enough to reverse the sign of the response at one corner. The four gap terms are instrumented as in equation (3), which gives twelve instruments for four endogenous variables and an identification statistic too low for two-stage least squares inference. The inference of record is therefore a just-identified estimator that uses the fitted gap from the first stage and its products with $U_{i,t}$ and $\tilde{d}_{i,t-1}$ as the four instruments, the generated-instrument device of Wooldridge (2010). Confidence sets for ϕ_{UF} come from the S statistic of Stock and Wright (2000), with the other gap coefficients concentrated out at

each value in the subset form of Kleibergen and Mavroeidis (2009), whose size is asymptotically no larger than nominal whatever the strength of the instruments (Appendix C.2.5).

2.3 Regimes, insurance and the pandemic

The fourth model lets the response to a shock differ between a vulnerable and a safe fiscal regime without imposing where one ends and the other begins. Following the smooth-transition local projections of Auerbach and Gorodnichenko (2012), with the logistic transition of Teräsvirta (1994), the response of an outcome y at horizon h is a weighted average of two regime responses:

$$y_{i,t+h}-y_{i,t-1} = \alpha_i^{(h)}+\delta_t^{(h)}+F(z_{i,t-1})\,\mathbf{b}_{\mathrm{V}}^{(h)\prime}\mathbf{s}_{i,t}+\big[1-F(z_{i,t-1})\big]\,\mathbf{b}_{\mathrm{S}}^{(h)\prime}\mathbf{s}_{i,t}+\boldsymbol{\lambda}^{(h)\prime}\mathbf{m}_{i,t}+\Gamma^{(h)\prime}X_{i,t-1}+u_{i,t+h}^{(h)}, \quad (5)$$

$$\mathbf{s}_{i,t} = \big(\varepsilon_{i,t}^{\mathrm{d}},\ \varepsilon_t^{\mathrm{g}}E_i\big)', \qquad F(z) = \frac{\exp(\kappa z)}{1+\exp(\kappa z)}, \quad (6)$$

where $z_{i,t-1} = \tilde{v}_{i,t-1}$ is the standardised composite vulnerability, $F(\cdot)$ the logistic transition with speed κ, V and S the vulnerable and the safe regime, and $\mathbf{b}_{\mathrm{V}}^{(h)}$ and $\mathbf{b}_{\mathrm{S}}^{(h)}$ the responses to the two shocks in $\mathbf{s}_{i,t}$ (the domestic shock and the global shock times exposure). The vector $\mathbf{m}_{i,t}$ collects the state and the lower-order terms of the regime interactions, $z_{i,t-1}$, $F(z_{i,t-1})$, $F(z_{i,t-1})\varepsilon_t^{\mathrm{g}}$ and $F(z_{i,t-1})E_i$, with coefficients $\boldsymbol{\lambda}^{(h)}$, $\alpha_i^{(h)}$ and $\delta_t^{(h)}$ are country and year effects, and $\Gamma^{(h)}$ holds the coefficients on the controls $X_{i,t-1}$ of equation (2). The error $u_{i,t+h}^{(h)}$ is the projection error at horizon h, with the properties of its counterpart in equation (2). The outcomes are the primary balance, primary spending, the effective interest rate, debt and revenue. Under the year effects the regime responses to the global shock are exposure gradients, the difference between an economy one standard deviation more open than the mean and the mean-exposure economy in the same regime. The regime difference is the difference of the gradients, which Condition G identifies. The regime difference in the level response to the global shock is the coefficient on $F(z_{i,t-1})\varepsilon_t^{\mathrm{g}}$ in $\mathbf{m}_{i,t}$, the smooth-transition counterpart of $\theta^{(h,\ell)}$ in equation (2), and Appendix C.2.6 reports it. The domestic shock enters directly. The transition speed is a tuning constant, $\kappa = 1.5$, chosen so that the regimes blend rather than switch. Across the latest positions of the economies in the sample the weight on the vulnerable regime rises from 0.44 at the 25th percentile of vulnerability to 0.80 at the 75th, and the values 1 and 3 are reported as checks. The lagged composite is predetermined, which makes the regression well defined, but it depends on past shocks and moves after the current one, and for such a state the local projection recovers the population response only for a small shock (Gonçalves et al., 2024); I therefore read the estimates as responses to a one-standard-deviation shock conditional on the regime in which it arrives. The sharp alternative is a panel threshold model in the sense of Hansen (1999), with an estimated debt threshold and a bootstrap test of its existence (Appendix C.2.6).

The fifth model is the insurance question, and it uses the same form with the regime indexed by fiscal space rather than by vulnerability:

$$y_{i,t+h}-y_{i,t-1} = \alpha_i^{(h)}+\delta_t^{(h)}+\Big[F(z_{i,t-1})\,\theta_{\mathrm{H}}^{(h)}+(1-F(z_{i,t-1}))\,\theta_{\mathrm{L}}^{(h)}\Big]\varepsilon_t^{\mathrm{g}}E_i+\boldsymbol{\lambda}^{(h)\prime}\mathbf{m}_{i,t}+\Gamma^{(h)\prime}X_{i,t-1}+u_{i,t+h}^{(h)}, \quad (7)$$

where the regime variable is $z_{i,t-1} = -\tilde{d}_{i,t-1}$, so that F approaches one for the high-space economy, H and L denote high and low space, and $\theta_{\mathrm{H}}^{(h)}$ and $\theta_{\mathrm{L}}^{(h)}$ are the regime responses to the global shock times

exposure. The vector $\mathbf{m}_{i,t} = (F(z_{i,t-1}),\ F(z_{i,t-1})\varepsilon^{g}_{t},\ F(z_{i,t-1})E_i)'$ collects the lower-order terms (the level of debt is in $X_{i,t-1}$), the remaining symbols are those of equation (5), $u^{(h)}_{i,t+h}$ included, and the outcomes are log real GDP, in hundredths of a log point, the effective interest rate and the sovereign rating. Fiscal space insures against the shock if $|\theta^{(h)}_{\mathrm{H}}| < |\theta^{(h)}_{\mathrm{L}}|$, and I count the horizons at which the inequality holds.

The pandemic is arguably the one episode in the sample in which a large, common and exogenous shock hit every economy in the same year, and two cross-sectional designs treat it as an identified shock. The first regresses the fiscal response of 2020–2021 on the fiscal position of 2019:

$$y_i = \alpha + \beta_d\, d_{i,2019} + \beta_R\, R_{i,2019} + \beta_W\, W_{i,2020} + \beta_{AE}\, AE_i + \Gamma'\mathbf{q}_i + e_i, \tag{8}$$

where y_i is in turn the IMF's measures of above-the-line and of liquidity support, the change in the primary balance in 2020, its cumulative change over 2020–2021, and the change in debt from 2019 to 2021. On the right, α is the intercept, $d_{i,2019}$ the 2019 debt ratio, $R_{i,2019}$ the sovereign rating in notches, $W_{i,2020}$ the pandemic component of the 2020 WUI, AE_i the advanced-economy indicator, β_d, β_R, β_W and β_{AE} their coefficients, $\mathbf{q}_i$ the severity controls (deaths per million, the stringency of containment and the pandemic-discussion index) with log income per head and trade openness in 2019, and Γ its coefficients. The error e_i has mean zero given the regressors and a variance that differs across economies, which the heteroskedasticity-robust standard errors allow for. The coefficient β_{AE} is the test of status against space. The second design is an event study of the years after 2019,

$$y_{i,2019+h} - y_{i,2019} = \alpha^{(h)} - \theta^{(h)}\,\tilde{d}_{i,2019} + \beta^{(h)}_W\,\tilde{W}_{i,2020} + \beta^{(h)}_{AE}\,AE_i + \Gamma^{(h)\prime}\mathbf{q}_i + e^{(h)}_i, \qquad h = 1, \ldots, 6, \tag{9}$$

where $\tilde{d}_{i,2019}$ is the 2019 debt ratio standardised across the cross-section and enters with a negative sign, so that $\theta^{(h)}$ is the effect of one standard deviation more pre-pandemic space, that is, one standard deviation less debt, on debt, the primary balance, the effective interest rate and output h years on. The standardised pandemic index is $\tilde{W}_{i,2020}$, $\alpha^{(h)}$ the intercept, and $\beta^{(h)}_W$, $\beta^{(h)}_{AE}$ and $\Gamma^{(h)}$ the coefficients on the pandemic index, the advanced-economy indicator and the controls $\mathbf{q}_i$ of equation (8). The error $e^{(h)}_i$ is that of the cross-section at horizon h, with the properties of e_i, and the cross-section shrinks with h as the actuals run out.

Inference is the same throughout the panel models. Standard errors are those of Driscoll and Kraay (1998) with two lags, which allow for cross-sectional dependence within a year and for serial correlation, and the bands in the figures are the 68% and 90% bands they imply. The lag is set to $h + 1$ as a check on the projections (Table C.21). Where a global regressor enters without year effects the effective number of time observations is the number of years, and I add a wild cluster bootstrap by year (Cameron et al., 2008) with 999 Rademacher draws; the group medians of Figure 4 carry 90% bands from a block bootstrap over economies with 500 draws. Appendix C.1 gives the samples, the first-stage statistics and the bootstrap settings.

3 Empirical Findings

3.1 Data

The panel consists of the general government accounts of 139 economies, 4,327 country-years over 1990–2025, from the April 2026 World Economic Outlook actuals: gross debt, the primary and overall balances, revenue and expenditure, real growth and deflator inflation, from which the contributions of equation (1) are computed. The main sample keeps the country-years with debt, the primary balance, real growth and the WUI, in economies with at least ten such years. Fiscal space comes from the Spring 2026 database of Kose et al. (2022), the uncertainty measures from Ahir et al. (2022) and from Appendix B, and the controls from the sources Appendix A lists with their vintages. Throughout, fiscal variables are in percent of GDP, interest-rate responses in percentage points and output in log points. Table 1 gives the counts of economies and country-years by group and by model sample. Low-income developing countries supply 37% of the country-years, and the GDELT index covers 134 economies in its decade, 47 of the 52 low-income ones among them.

Three construction decisions bear on what the measures mean. I average the country WUI over each economy's fiscal year, from monthly data where the index has them and otherwise from the four quarters that best overlap the year. The uncertainty in a reaction function is then the uncertainty of the budget year. The global index is the GDP-weighted mean of the country indices over the economies of the panel, which tracks the published global series with a correlation of 0.999. The pandemic component, the World Pandemic Uncertainty Index of the same authors, is subtracted to form the purged series, since in 2020 the pandemic accounted for 51% of the global index. Figure 2 shows what the GDELT index adds, with the domestic and the foreign indices of six economies, monthly from January 2016 to September 2026. The correlation between the domestic and the foreign series is 0.17 in Türkiye against 0.85 in the United States, and April 2025 is a common peak in all six.

Three features of the construction bear on the results, and each has a check. The pandemic years carry the largest common shock in the sample, which is why the purged series exist and why Section 3.5 excludes 2020–2021. The identity attributes debt restructurings and valuation jumps to the stock-flow adjustment, so the decomposition sample drops the 425 country-years (11% of the complete ones) beyond the thresholds in the notes to Table 1, and the rule applies to the identity model only.[2]

The fiscal data are the April 2026 vintage, in which 2025 is an actual for a minority of economies and an estimate for the rest, so the last actual year differs across economies (Table A.2), and the 2026 projections are not used.

3.2 How uncertainty erodes fiscal space

Figure 3 plots the cumulative responses of the five contributions of equation (1) and of the change in debt, in panel (f), to a one-standard-deviation global uncertainty shock, from equation (2). I evaluate each response at two positions, the low-space economy with lagged debt at the 75th percentile of the sample, 66 percent of GDP, and the high-space economy at the 25th, 31 percent. Under the year effects each line is a differential from the economy with mean debt, and a positive value adds to debt. The differential between the two lines, which is the estimand, carries a standard error of roughly a fifth of

[2]The largest exclusions are debt restructurings and hyperinflations (Lebanon in 2021, Sudan in the 1990s, Argentina in 2002 and 2023).

Table 1. Sample coverage by country group

			WUI		GDELT domestic		Country-years in sample		
	Economies	Fiscal data (years)	Economies	Country-years	Economies	Country-years	M1	M2	COVID
AE	28	1990–2024	28	974	28	260	854	943	60
EM	59	1991–2024	59	2,046	59	513	1,543	1,837	81
LIDC	52	1995–2024	52	1,784	47	275	970	1,421	41
All	139	1991–2024	139	4,804	134	1,048	3,367	4,201	182

Notes: Entries are counts of economies and country-years in the estimation samples defined in Section 3.1. Years of fiscal data are the median across economies of the first and the last year with general government gross debt and primary balance (IMF WEO April 2026 actuals, 1990–2025). WUI columns count economies and country-years with the World Uncertainty Index (Ahir, Bloom and Furceri) among the economies of the main sample; GDELT columns count country-years with the domestic fiscal-policy-uncertainty index (2016–2025, thin cells excluded). The debt-decomposition sample requires the global uncertainty shock, the five debt-dynamics components and the lagged controls, and excludes country-years with $|pb| > 25$, $|\Delta d| > 40$ or a stock-flow adjustment above 10 percent of GDP in absolute value (425 country-years); the reaction-function sample requires the lagged primary balance and debt ratio, the Hamilton output gap and fiscal-year WUI; the COVID cross-section requires the IMF COVID-19 fiscal measures (October 2021) and the primary balance in 2019–2021. Groups are advanced economies (AE), emerging markets (EM) and low-income developing countries (LIDC) by the IMF income classification of the World Pandemic Uncertainty Index; membership is fixed at the current classification (IMF WEO April 2026).

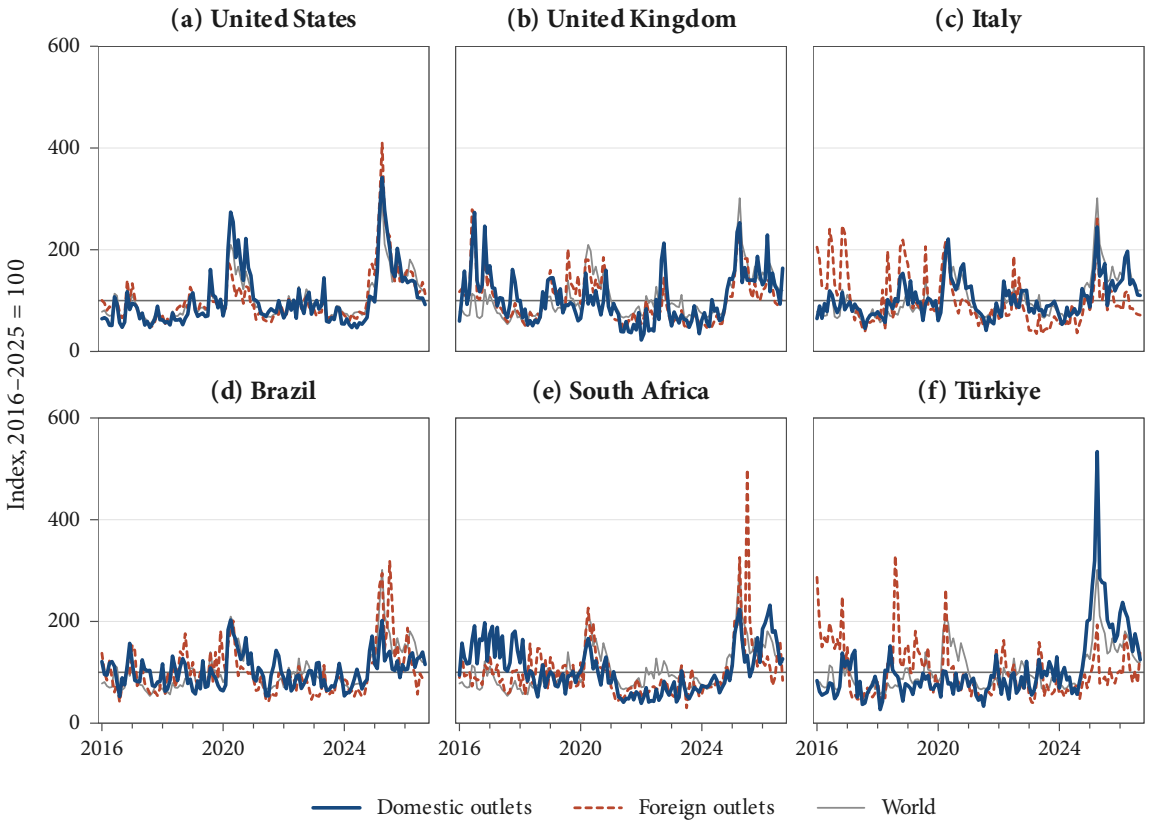


Figure 2. Fiscal-policy uncertainty in six economies, domestic and foreign outlets

Notes: Entries are monthly GDELT fiscal-policy-uncertainty indices, 2016–2025 mean = 100, from January 2016 to September 2026: domestic outlets of the economy (view D), foreign outlets writing about it (view X) and all articles worldwide (view W, the same line in every panel). The axis runs to 600; no value exceeds it.

its size two years after the shock and of the same order as its size after five. The figure suggests three conclusions.

First, a global uncertainty shock widens the debt gap between an indebted and a lightly indebted economy within two years, and the gap then narrows. Two years after the shock the low-space economy has added 1.82 (standard error [SE] of 0.39) percentage points of GDP more debt than the high-space economy, and after five years the gap has fallen back to 0.71 (SE = 0.69). The gap opens fast and closes slowly, so the cost to the indebted economy is a level shift in its debt while the shock is live, not a permanent divergence.

Second, early on the gap opens through growth and inflation, not through the budget. Of the two-year differential, the real-growth contribution accounts for 58% and the inflation contribution for 28%, against 14% for the primary balance and nothing of note for interest and stock-flow adjustments. The indebted economy loses more output and more deflator growth after a world shock, so the denominator of the debt ratio does most of the early work, and in this sense the erosion of fiscal space begins before any budget decision is taken.

Third, by the fifth year the primary balance has taken over and the interest channel works the other way. At that horizon the primary-balance contribution per standard deviation of debt (the two positions are 1.1 standard deviations apart) is 0.97 (SE = 0.59) percentage points of GDP and the interest contribution is −1.10 (SE = 0.57), while the stock-flow channel carries nothing. The indebted economy runs a weaker primary balance for years after the shock, the budgetary counterpart of the reaction function of Section 3.3, and a negative interest contribution partly offsets it, perhaps because the shock lowers the interest–growth differential of the economies that borrow most. The decomposition by country group, the responses to the domestic shock and the specification without year effects, which I read as an upper bound on the level because it credits the shock with the common surges of 2009 and 2020, are in Appendix C.2.1.

Figure 3 states the cost of a global shock per standard deviation of debt, and what the shock costs a particular economy depends on where that economy sits. Figure 4 reads the same estimates at each economy's own position in its latest sample year, one point per economy, with group medians as bars and the six economies of Figure 2 labelled. Panel (a) evaluates the slope of Figure 3 at the economy's lagged debt ratio, the debt added two years after the shock relative to the economy with mean debt. Panel (b) evaluates the response of the effective interest rate at the economy's regime position in the model of Section 3.4, for an economy one standard deviation more exposed than the mean relative to the mean-exposure economy in the same regime. The medians carry 90% bands from a block bootstrap over economies, and the advanced-economy band in panel (a) includes zero. The figure suggests four conclusions.

(i) Relative to the average economy, the median advanced economy adds more debt after a global shock than the median low-income economy. The median debt added after two years is 0.68 percentage points of GDP in the advanced economies, 0.06 in the emerging markets and −0.13 in the low-income developing countries, and the ordering of the first group above the third holds in 94% of the bootstrap draws. Who can still afford to act is, on this evidence, a question about where an economy sits on the debt distribution.

(ii) In levels the shock adds several points of GDP to debt everywhere, and the position decides the margin. The specification without year effects puts the median debt added after two years at 6.01

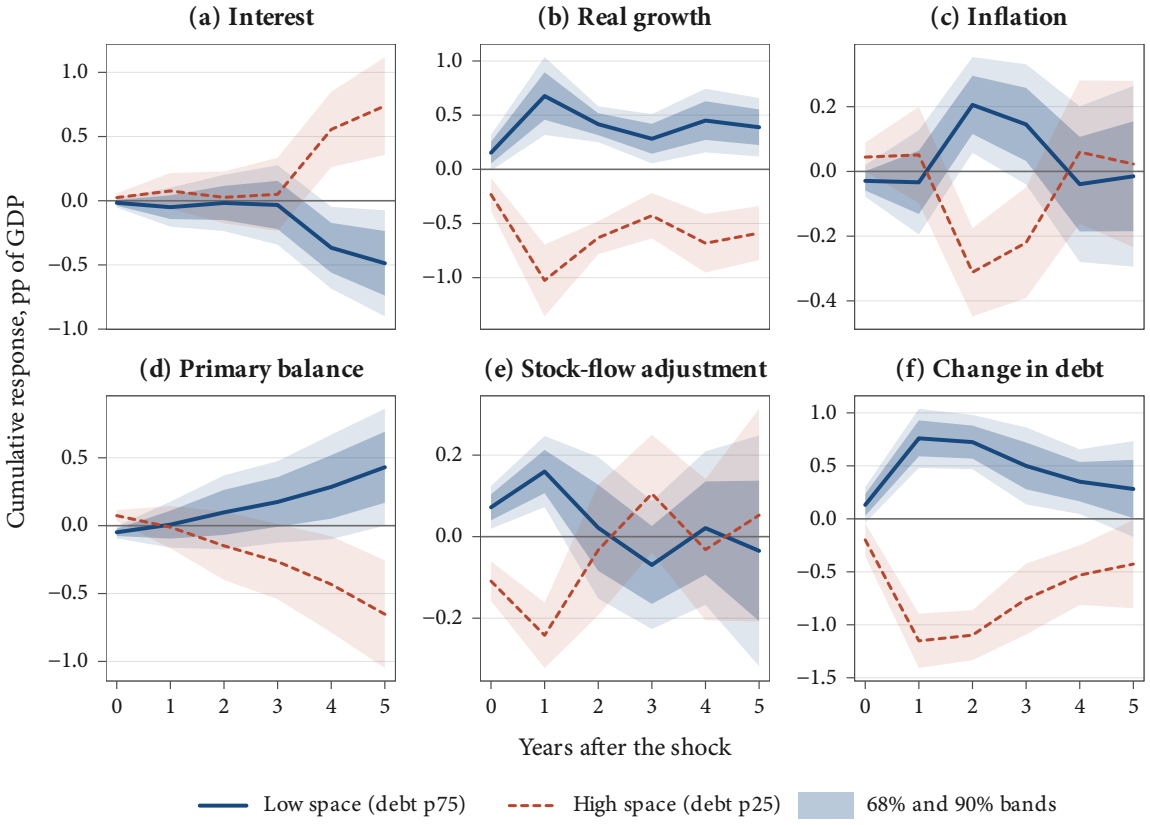


Figure 3. Responses of the debt-dynamics components to a global uncertainty shock

Notes: Cumulative responses of the debt-dynamics components to a one-sd global uncertainty shock, pp of GDP, from local projections with country and year effects and lagged controls, evaluated at lagged debt of 66 (low space) and 31% of GDP (high space) relative to the mean economy. Driscoll–Kraay bands (lag 2); the lighter band is the 68% band of the high-space response. 1992–2025, 132 economies.

percentage points of GDP in the advanced economies and 5.08 in the low-income economies (Table C.2 lists every economy). No economy escapes the level cost; position decides the margin, which is what panel (a) identifies.

(iii) Among the six economies the shock adds most debt in Italy and least in Türkiye. In levels Italy adds 10.10 (SE = 1.58) percentage points of GDP after two years and Türkiye 3.58 (SE = 1.58). Türkiye's low debt ratio is what buys it room in this exercise, not its status as an emerging market, and Italy's debt ratio is what removes it.

(iv) The price of the stance differs by regime position. In panel (b) the median low-income economy pays for the shock in its effective interest rate, 0.09 percentage points after two years, while the median advanced economy does not (−0.10). The cost of carrying debt through a global shock is the second dimension of affording to act, and it does not line up with the first: the advanced economies add the most debt and pay the least for it.

How much debt a shock leaves behind depends in the end on how governments respond to it, which the next subsection takes up.

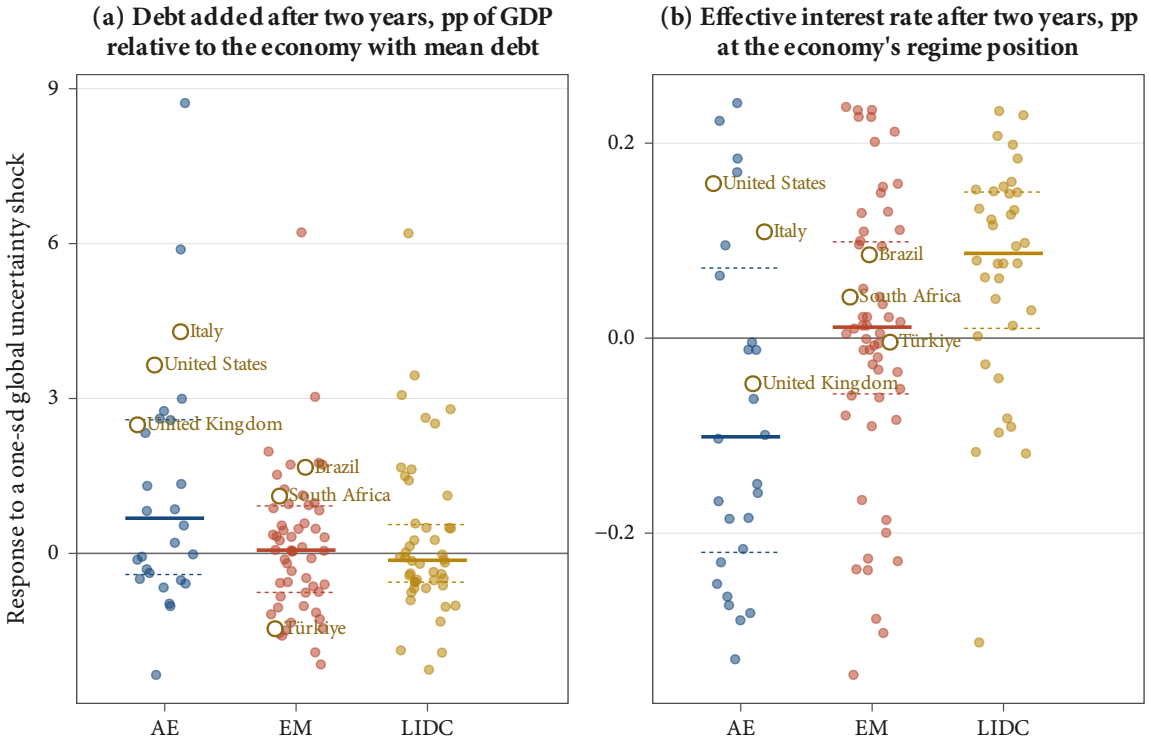


Figure 4. Fiscal space at risk: what a global uncertainty shock implies for each economy

Notes: One point per economy at its latest sample year; bars are group medians, dashed bars the interquartile range; the six economies of Figure 2 are labelled. (a) The Figure 3 slope evaluated at the economy's lagged debt: debt added two years after a one-sd global uncertainty shock relative to the economy with mean debt (the direct-effect level for that economy is 5.2 pp, standard error 1.5; Table C.2). (b) Effective-interest-rate response of an economy one sd more open than the mean, relative to mean exposure, at its regime weight F in the model of Figure 5 ($\kappa = 1.5$).

3.3 Uncertainty and the conduct of fiscal policy

Table 2 reports the reaction function of equation (3), with the primary balance in percent of GDP as the dependent variable, the cubic in lagged debt, and the output gap and its product with uncertainty instrumented as Section 2.2 describes. The columns vary the uncertainty measure: fiscal-year WUI, the same with the advanced-economy indicator interacted with the uncertainty block and the cubic, its pre-budget and pandemic-purged timings, and the GDELT index from domestic and from foreign outlets. The standard error of the debt–uncertainty interaction is half its estimate in the pooled column, and the hypothesis that it is zero is not rejected at the 5% level there. I therefore report the interaction with its standard error and do not lean on it. The table suggests four conclusions.

First, country-level uncertainty weakens the response of the primary balance to debt, by a margin the data do not pin down. The interaction is −0.0034 (standard error of 0.0017) percentage points of GDP per percentage point of debt per standard deviation of uncertainty. A move of uncertainty from its 10th to its 90th percentile changes the marginal response of the balance to debt by −0.074 (SE = 0.037) percentage points per 10 points of debt, 44% of the response at the median debt ratio in a quiet year, 0.17 per 10 points. The sign is the one fiscal fatigue predicts, and the precision says that the data cannot distinguish a modest weakening from none: the interaction is a direction rather than a

Table 2. Uncertainty and the fiscal reaction function

	Primary balance, % of GDP					
	(1)	(2)	(3)	(4)	(5)	(6)
Uncertainty U_t (sd)	0.107	0.168	0.0900	0.106	−0.0878	−0.514
	(0.104)	(0.130)	(0.145)	(0.0925)	(0.126)	(0.0799)
$\text{Debt}_{t-1} \times U_t$	−0.00343	−0.00386	−0.00334	−0.00309	−0.00119	0.00583
	(0.00172)	(0.00238)	(0.00237)	(0.00170)	(0.00344)	(0.00268)
Output gap (% of trend GDP)	0.595	0.512	0.603	0.511	0.439	0.449
	(0.0634)	(0.0645)	(0.0616)	(0.0625)	(0.0736)	(0.0795)
Output gap $\times U_t$	0.00646	−0.0120	−0.0123	−0.0120	0.105	0.0474
	(0.0325)	(0.0437)	(0.0260)	(0.0368)	(0.0385)	(0.0389)
Debt_{t-1} (% of GDP)	0.0263	0.0300	0.0253	0.0288	0.0590	0.0901
	(0.0123)	(0.0135)	(0.0126)	(0.0119)	(0.111)	(0.0933)
Debt^2_{t-1} ($\times 10^{-3}$)	−0.118	−0.115	−0.117	−0.124	−0.0993	−0.467
	(0.0935)	(0.115)	(0.0965)	(0.0902)	(1.00)	(0.832)
Debt^3_{t-1} ($\times 10^{-6}$)	0.235	0.209	0.241	0.233	−0.300	0.818
	(0.149)	(0.184)	(0.159)	(0.147)	(2.73)	(2.19)
$\text{Primary balance}_{t-1}$	0.508	0.507	0.509	0.500	0.551	0.509
	(0.0579)	(0.0578)	(0.0569)	(0.0565)	(0.0500)	(0.0614)
ΔAE: U_t		−0.534				
		(0.263)				
ΔAE: $\text{Debt}_{t-1} \times U_t$		0.00606				
		(0.00420)				
ΔAE: Output gap		0.229				
		(0.0649)				
ΔAE: Output gap $\times U_t$		0.0475				
		(0.0920)				
ΔAE: Debt_{t-1}		−0.0396				
		(0.0259)				
ΔAE: Debt^2_{t-1}		0.209				
		(0.311)				
ΔAE: Debt^3_{t-1}		−0.509				
		(0.916)				
Fixed effects	Country	Country	Country	Country	Country	Country
Instrumented output gap (first-stage F)	43.8	144.2	28.8	24.6	42.9	127.3
Cragg–Donald statistic	82.9	37.1	93.5	37.2	38.2	41.7
Wald test, AE = EMDE (p)		0.002				
Observations	3,834	3,834	3,833	3,594	954	1,110
Economies	127	127	127	127	114	124
Years	1990–2025	1990–2025	1990–2025	1990–2025	2016–2025	2016–2025

Columns: (1) pooled; (2) advanced-economy (AE) interactions; (3) pre-budget U; (4) purged U, excl. 2020–21; (5) GDELT domestic fiscal-policy uncertainty (FPU); (6) GDELT foreign FPU. Notes: Entries are coefficients of the fiscal reaction function, with the primary balance in percent of GDP as the dependent variable; Driscoll–Kraay standard errors (lag 2) in parentheses. The output gap (Hamilton filter) and its product with U_t are instrumented by world real growth excluding the own economy, its product with 1995–2004 trade openness (standardised) and the lagged gap, each alone and multiplied by U_t; the first-stage row reports the smallest Driscoll–Kraay-robust first-stage Wald F across the instrumented variables, and the Cragg–Donald statistic follows (the Kleibergen–Paap rk statistic is not defined under the Driscoll–Kraay covariance in the software used). Uncertainty is standardised within the main sample: fiscal-year WUI wui_fy (mean 0.180, sd 0.161) in columns (1)–(2), pre-budget WUI wui_pb (mean 0.176, sd 0.175) in (3), pandemic-purged WUI wui_np (mean 0.170, sd 0.162) in (4) with 2020–2021 excluded, GDELT domestic fpu_dom_z (mean -0.087, sd 0.895) and foreign fpu_for_z (mean -0.116, sd 0.871) fiscal-policy uncertainty (2016–2025) in (5)–(6). Column (2) interacts the advanced-economy indicator with the uncertainty block and the cubic; the rows report coefficients for emerging market and developing economies (EMDE) and the AE differences (ΔAE), with the Wald test of their joint equality to zero. Controls: lagged real growth and deflator inflation, commodity terms-of-trade growth, election-year and fiscal-rule indicators (zero-filled with missing indicators), lagged trade openness; country fixed effects. Debt in percent of GDP, so the cubic terms are scaled as shown.

magnitude. The cubic itself does not bend down at high debt, so fatigue in the sense of Ghosh et al. (2013) is absent from the pooled sample.

Second, uncertainty itself does not move the balance on average, and the cyclical response is strong. The level effect of uncertainty is 0.107 (SE = 0.104) percentage points of GDP per standard deviation, while the balance rises by 0.595 (SE = 0.063) percentage points per point of the instrumented gap, with a first-stage statistic of 43.8. Once the cycle is instrumented, uncertainty appears to act on the slope of the reaction function with respect to debt, not on the level of the balance. The rule bends rather than shifts.

Third, advanced and emerging economies run different reaction functions, but the debt–uncertainty interaction is not where they differ. The Wald test of equality of the uncertainty block and the cubic across the groups rejects at the 1% level, and the advanced-economy difference in the cyclical coefficient is 0.23 (SE = 0.06) percentage points per point of gap, while the difference in the interaction is 0.0061 (SE = 0.0042). Status changes how the balance responds to the cycle and does not absorb the response to debt under uncertainty, which is the first of several readings in this paper of space rather than status.

Fourth, the sign of the interaction depends on whose uncertainty is measured. The interaction is negative in 5 of the six columns, and the exception is the foreign-outlet GDELT column, where it is 0.0058 (SE = 0.0027) against −0.0012 (SE = 0.0034) for the domestic index. Table C.4 enters a country-specific and a global measure in the same equation, each interacted with debt. With the WUI the country-specific interaction is −0.0043 (SE = 0.0023) and the global one 0.0040 (SE = 0.0021), a difference rejected at the 5% level, and the GDELT pair gives the same signs with a difference also rejected at the 5% level. Global uncertainty disciplines, perhaps because it raises the price of debt, and country-specific uncertainty delays. Two qualifications hold the reading to its bands: the GDELT split does not survive the exclusion of 2020–2021, and the calendar-year decomposition of the WUI does not reproduce it, although the pattern holds in the point estimates of 4 of the five two-source columns. I read the split as the policy-rule counterpart of the regimes of Section 3.4, not as a headline.

Table 3 turns from the response to debt to the response to the cycle, from equation (4), with the change in the primary balance as the dependent variable. The top block reports the cyclical coefficient and its interactions with uncertainty, with fiscal space and with their product, and the middle block the implied cyclical response at the four corners, in percentage points of balance per point of gap, with uncertainty and lagged debt each at its 25th and 75th percentiles. The columns are the pooled sample, the two groups, the vulnerability composite in place of debt, the global shock in place of country uncertainty, and the pooled specification by ordinary least squares. The triple interaction carries a standard error three times its estimate, and the twelve-instrument estimator is weakly identified, so the weak-instrument-robust set of Table C.6 is the inference of record. The table suggests three conclusions.

(i) The primary balance leans against the cycle. The instrumented cyclical response is 0.236 (SE = 0.107) percentage points of GDP per point of gap in the pooled column, with a first-stage statistic of 24.5. A one-point wider gap raises the balance by roughly a quarter of a point of GDP, the stance the hypothesis expected uncertainty to bend.

(ii) Uncertainty does not change the cyclical stance, with or without fiscal space, so the hypothesised sign switch is not found. Under high uncertainty the response is 0.222 (SE = 0.104) with high space and 0.245 (SE = 0.104) with low space, and the triple interaction behind them is 0.010 (SE =

Table 3. Uncertainty, fiscal space and the cyclical response of the primary balance

	Change in the primary balance, pp of GDP					
	Pooled	AE	EMDE	Vulnerability	Global shock	Pooled, OLS
Coefficients						
Output gap ϕ	0.236	0.169	0.0769	0.378	0.213	0.0743
	(0.107)	(0.132)	(0.104)	(0.0757)	(0.113)	(0.0314)
Output gap × U (ϕ_U)	0.00114	−0.0462	−0.00459	−0.0393	0.0155	0.0120
	(0.0251)	(0.0572)	(0.0372)	(0.0218)	(0.0382)	(0.0144)
Output gap × $\tilde{d}$ (ϕ_F)	0.0174	0.0326	0.0126	−0.00761	0.0211	0.0244
	(0.0246)	(0.0351)	(0.0319)	(0.0443)	(0.0276)	(0.00943)
Output gap × U × $\tilde{d}$ (ϕ_{UF})	0.00972	0.0323	0.0155	0.0232	−0.0390	−0.00827
	(0.0294)	(0.0264)	(0.0255)	(0.0295)	(0.0387)	(0.0116)
U	0.0341	−0.0700	0.0339	0.110		0.00211
	(0.0500)	(0.0907)	(0.0569)	(0.0513)		(0.0499)
$\tilde{d}$	0.553	0.458	0.599	0.470	0.564	0.577
	(0.212)	(0.219)	(0.196)	(0.186)	(0.222)	(0.143)
U × $\tilde{d}$	−0.0811	−0.00924	−0.0724	−0.0658	0.149	−0.0373
	(0.0668)	(0.0848)	(0.0933)	(0.0639)	(0.0956)	(0.0528)
Implied cyclical response at the corners (pp of GDP per pp of gap)						
Low U, high space	0.228	0.187	0.0792	0.416	0.182	0.0470
	(0.111)	(0.136)	(0.123)	(0.0804)	(0.119)	(0.0365)
Low U, low space	0.239	0.203	0.0807	0.390	0.221	0.0776
	(0.118)	(0.120)	(0.119)	(0.0918)	(0.116)	(0.0379)
High U, high space	0.222	0.118	0.0627	0.353	0.205	0.0665
	(0.104)	(0.149)	(0.102)	(0.0698)	(0.117)	(0.0286)
High U, low space	0.245	0.172	0.0819	0.359	0.221	0.0876
	(0.104)	(0.139)	(0.0962)	(0.0836)	(0.112)	(0.0300)
Fixed effects	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year
Instrumented output gap (first-stage F)	24.5	31.7	23.0	13.8	18.4	
Cragg–Donald statistic	9.5	9.1	9.3	10.4	11.1	
Observations	3,832	1,011	2,821	2,839	3,721	3,832
Economies	127	31	96	121	127	127
Years	1990–2025	1990–2025	1990–2025	1991–2025	1992–2025	1990–2025

Notes: Entries are coefficients of the regression of the change in the primary balance (pp of GDP) on the output gap (percent of trend GDP, Hamilton filter) interacted with uncertainty U and lagged standardised debt $\tilde{d}$, with country and year fixed effects; Driscoll–Kraay standard errors (lag 2) in parentheses. The gap terms are instrumented by world growth excluding the own economy, its product with 1995–2004 openness and the lagged gap, each multiplied by $(1, U, \tilde{d}, U\tilde{d})$; column (6) is the pooled specification by ordinary least squares. U is fiscal-year WUI standardised (mean 0.180, sd 0.161) in columns (1)–(4), (6) and the global uncertainty shock (sd units, no main effect under year effects) in column (5). $\tilde{d}$ is lagged debt standardised within each column's sample (pooled mean 54.5, sd 37.1% of GDP), higher being more debt; in column (4) the standardised composite fiscal-vulnerability index $\tilde{v}$ (higher = more vulnerable; 121 economies) replaces $\tilde{d}$, with lagged debt as a control. The middle block reports the implied cyclical response $\phi + \phi_U U + \phi_F \tilde{d} + \phi_{UF} U\tilde{d}$ at the 25th and 75th percentiles of U and of $\tilde{d}$ (high space = 25th percentile of debt or of vulnerability), delta-method standard errors. Controls: lagged real growth, deflator inflation, primary balance, trade openness and commodity terms-of-trade growth.

0.029). The corners under low uncertainty are the same to the second decimal, and the picture is the same within each country group, with the vulnerability composite in place of debt, and with the global shock in place of country uncertainty. A government that was countercyclical stays countercyclical whatever its space and whatever the year's uncertainty. What uncertainty changes, on the evidence of Figures 3 and 4, is the price of the stance and the debt it leaves behind, not the stance itself.

(iii) The null is not an artefact of weak instruments. With the fitted-gap instruments the just-identified estimate of the triple interaction is −0.039 (SE = 0.031), and its 90% S set is 0.11 units wide and contains zero. The sign switch would need a value far outside the range over which the set was searched, and with the twelve instruments the set is unbounded.

The two tables do not deliver model-based debt limits in the sense of Ghosh et al. (2013), the natural complement to Table 2, because the interest–growth differential is non-positive over 2010–2025 for 122

of 127 economies and the emerging-market reaction function has no fatigue bend (Appendix C.2.4).

If uncertainty does not change the stance, the regimes in which governments act differently must be found elsewhere, and that is where the next subsection looks.

3.4 Regimes of vulnerability, insurance and the pandemic

Figure 5 plots the responses of the primary balance, primary spending and the effective interest rate in the vulnerable and the safe regime of equation (5), to the global shock scaled by exposure in the top row and to the domestic shock in the bottom row. The global shock is one standard deviation for an economy one standard deviation more open than the mean, relative to the mean-exposure economy in the same regime, the domestic shock one standard deviation, and the vulnerable regime is the focal line. The regime differences in the budget variables are imprecise and change sign across horizons, and only the interest-rate difference is estimated with a standard error below a quarter of its value, so I weight the conclusions accordingly. The figure suggests three.

First, after a global shock the vulnerable economy pays in its effective interest rate through its exposure, and the safe economy does not. Two years after the shock the effective rate of an economy one standard deviation more open than the mean has risen by 0.25 (standard error of 0.06) percentage points relative to the mean-exposure economy in the vulnerable regime and changed by −0.35 (SE = 0.10) in the safe one, a difference of 0.60 (SE = 0.13) that reaches 0.86 in the third year. At mean exposure the level difference between the regimes, reported in Table C.8, is −0.80 with a standard error of 0.32 at two years and within 1.645 standard errors of zero at the other horizons. The price of the stance is where the regimes separate, and that asymmetry is the main result of the figure. For an exposed economy a global shock lowers the cost of debt for a safe sovereign, the flight to quality, and raises it for a vulnerable one, so the same shock widens the gap in what the two must pay to carry the debt that Figure 3 says they accumulate.

Second, the budget does not separate the regimes as the price does, and its differences change sign across horizons. The primary-balance difference between the regimes is −0.01 (SE = 0.41) percentage points of GDP two years after the global shock and within two standard errors of zero at every horizon, and after the domestic shock the vulnerable economy's balance falls relative to the safe one's by −0.62 (SE = 0.45), within 1.645 standard errors of zero. The hypothesised loosening of the vulnerable regime is not in the exposure gradients, the regime counterpart of the null of Table 3. The domestic pattern is identified only within economies and is read as a pattern.

Third, the sharp alternative finds a regime at a low debt ratio in the level response to the global shock, and none in the exposure gradient. The threshold tests of Table C.7 reject the absence of a threshold at the 5% level in 4 of 6 outcome–horizon equations, with thresholds between 20 and 25 percent of GDP. The rejection comes from the regime-level terms: in the year of the shock the primary balance below the threshold moves by –1.13 (SE = 0.41) percentage points of GDP relative to the balance above it, while the below-minus-above differences in the exposure gradient are within 1.645 standard errors of zero in 6 of the 6 equations. The threshold model's regime is a level effect at low debt, the sharp counterpart of the slope of Figure 3, not a gap in the exposure gradient, which is why the smooth transition rather than the threshold carries the main result.

Fiscal space measured by debt alone does not detectably insure against the average global shock, in output or in its price. In the model of equation (7), with the regime indexed by space, shown in

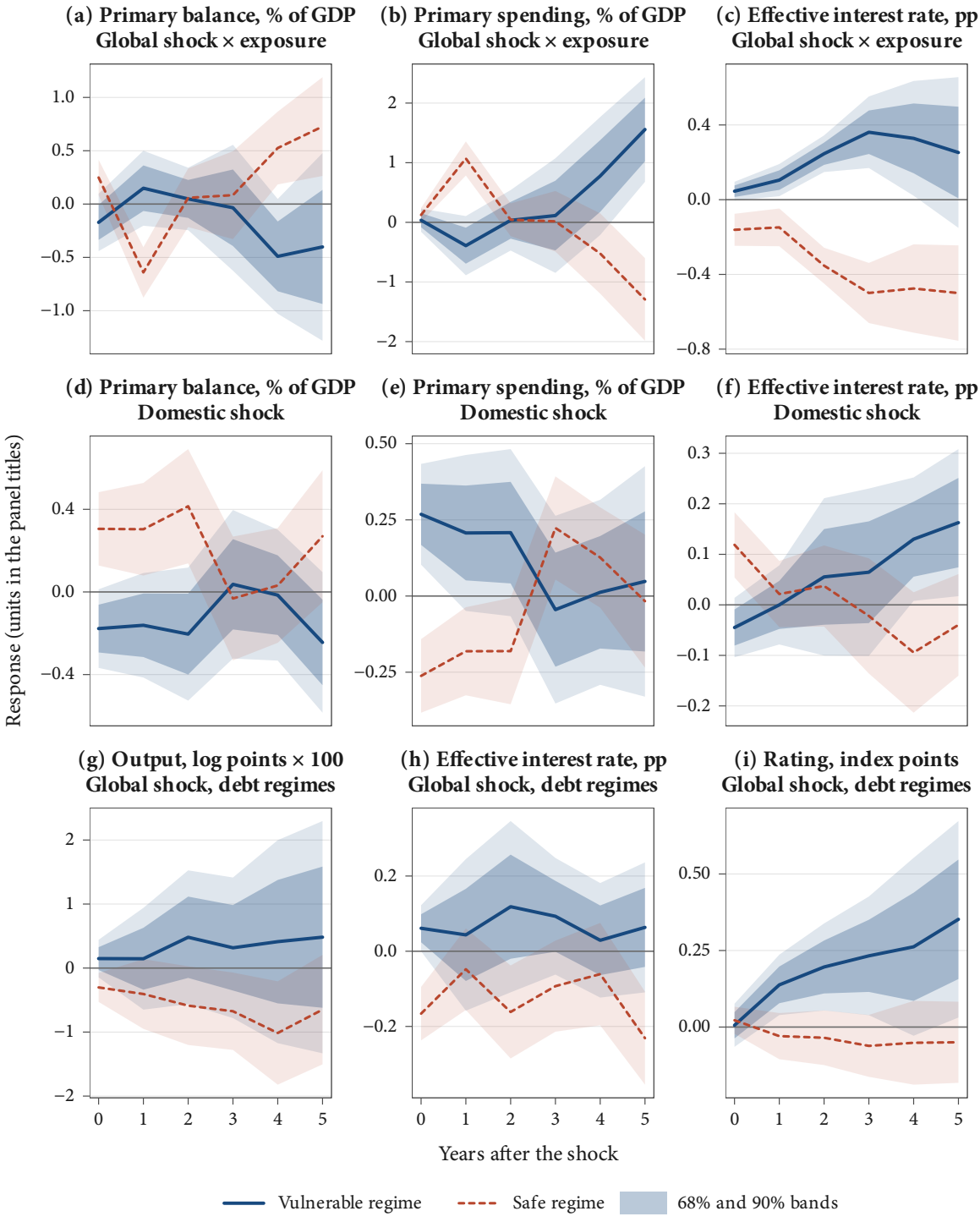


Figure 5. Responses in vulnerable and safe fiscal regimes: fiscal outcomes and insurance

Notes: Smooth-transition local projections of $y_{i,t+h} - y_{i,t-1}$ with country and year effects, lagged controls and the lower-order regime terms; Driscoll–Kraay bands (lag 2). Rows 1–2: weight $F(z)$, z lagged standardised vulnerability, $\kappa = 1.5$ (122 economies). Row 3: regimes by lagged debt (low space p75 = 68%, high space p25 = 31%; 129 economies). Global shock: one sd, exposure gradient (one sd of openness).

Figure C.9, the insurance inequality holds at 0 of the six horizons for output, at 1 of six for the effective interest rate and at 5 of six for the sovereign rating.

The output responses in the two regimes are too imprecise to support or refute the insurance hypothesis. The rating count rests on a gradient that is steeper in the low-space regime, 0.20 index points two years on against −0.03 with space (standard errors 0.09 and 0.09), a difference within 1.645 standard errors of zero at every horizon. What separates economies after an ordinary global shock is the vulnerability of the regime above, not debt on its own.

The pandemic is where the insurance question can be asked of a visible shock. Table 4 reports the cross-section of equation (8), with the IMF's above-the-line and liquidity support, the change in the primary balance in 2020, its cumulative change over 2020–2021 and the change in debt from 2019 to 2021 as the five outcomes. Panel A carries the 2019 debt ratio and sovereign rating, Panel B the composite vulnerability index. The standard errors are heteroskedasticity-robust on cross-sections of 57 to 109 economies. The table suggests three conclusions.

Table 4. Fiscal space and the fiscal response to the pandemic, 2020–2021

	% of 2020 GDP		pp of GDP		
	Above-the-line measures	Liquidity support	Δ primary balance 2020	Cumulative Δ pb 2020–21	Δ debt 2019–21
Panel A: fiscal-space variables					
Debt 2019 (% of GDP)	0.0417	0.0610	−0.00733	−0.0330	0.240
	(0.0194)	(0.0328)	(0.0109)	(0.0165)	(0.161)
Sovereign rating 2019 (1–21)	0.512	−0.186	0.0589	0.00624	0.647
	(0.249)	(0.525)	(0.148)	(0.201)	(0.493)
Pandemic uncertainty 2020 (WPUI)	−2.84	−1.45	2.06	5.05	−7.68
	(4.47)	(5.42)	(2.26)	(3.57)	(8.41)
log(1 + deaths per million)	−0.0386	0.787	−0.0873	−0.379	−0.152
	(0.386)	(0.531)	(0.195)	(0.300)	(0.999)
Stringency index 2020	0.0390	−0.00814	−0.0656	−0.0730	−0.0121
	(0.0538)	(0.105)	(0.0254)	(0.0378)	(0.104)
Pandemic discussion 2020	2.64	2.18	0.241	−0.410	−10.1
	(1.89)	(2.68)	(0.779)	(1.24)	(10.4)
Advanced economy	1.80	6.53	−1.14	−2.52	−8.43
	(2.14)	(3.93)	(1.21)	(1.66)	(7.49)
Observations	57	53	109	109	110
R^2	0.55	0.47	0.25	0.25	0.21
Panel B: composite vulnerability					
Fiscal vulnerability 2019 (sd)	1.77	3.62	0.575	0.136	13.0
	(1.22)	(2.37)	(0.950)	(1.30)	(8.93)
Pandemic uncertainty 2020 (WPUI)	0.544	−0.118	2.43	4.19	3.30
	(4.55)	(4.96)	(2.42)	(3.83)	(7.34)
log(1 + deaths per million)	−0.273	0.887	−0.113	−0.351	−0.637
	(0.385)	(0.575)	(0.187)	(0.303)	(1.07)
Stringency index 2020	0.0653	0.00905	−0.0625	−0.0745	0.0298
	(0.0566)	(0.102)	(0.0251)	(0.0374)	(0.0971)
Pandemic discussion 2020	3.53	2.59	0.411	−0.164	−8.91
	(1.70)	(2.58)	(0.774)	(1.21)	(10.3)
Advanced economy	4.81	8.19	−1.10	−3.40	1.23
	(1.78)	(2.91)	(0.878)	(1.38)	(2.86)
Observations	57	53	115	115	116
R^2	0.51	0.44	0.27	0.22	0.16

Notes: Entries are ordinary least squares coefficients of cross-country regressions of the pandemic fiscal response on pre-pandemic fiscal space, with heteroskedasticity-robust (HC1) standard errors in parentheses. Outcomes: IMF above-the-line measures (additional spending and forgone revenue, January 2020–September 2021) and liquidity support (below-the-line measures and guarantees), percent of 2020 GDP; the change in the primary balance in 2020 and cumulated over 2020–2021, and the change in gross debt from 2019 to 2021, pp of GDP. Panel A uses the 2019 debt ratio and sovereign rating (the foreign-currency share, available for few economies, is omitted); Panel B the composite vulnerability index (equal-weight average of standardised debt, deficit, rating, foreign-currency share and short-term debt; higher = more vulnerable). Pandemic uncertainty is the 2020 World Pandemic Uncertainty Index (WUI scale), $W_{i,2020}$ of equation (8); severity controls are cumulative COVID-19 deaths per million at end-2020 (log), the stringency index of the Oxford COVID-19 Government Response Tracker (2020 mean) and the pandemic-discussion index; controls also include log GDP per capita at purchasing-power parity and trade openness in 2019; together these form $\mathbf{q}_i$, and the advanced-economy row is β_{AE}. Every WEO economy with the variables.

(i) Economies that entered the pandemic with less space spent more on support, not less. Ten percentage points more debt in 2019 go with 0.42 (SE = 0.19) percent of GDP more above-the-line support, and a better rating goes with more support too, 0.51 (SE = 0.25) percent of GDP per notch. Market access, not low debt, is what bought support in 2020, and since a high debt ratio and a good rating together are the advanced-economy configuration, the configuration rather than the label carried the response.

(ii) The pandemic deficits and the debt they left do not line up with pre-pandemic debt, and pandemic uncertainty does not sort them either. Ten points more 2019 debt change the primary balance in 2020 by −0.07 (SE = 0.11) percentage points of GDP and the cumulative two-year change by −0.33 (SE = 0.17), and the coefficients on the 2020 pandemic-uncertainty index are within one standard error of zero in every column. The 2020 balance change was a wash across debt levels, the two-year deterioration leans weakly against the indebted economies, and the size of the health shock, as the news measured it, was not what sorted fiscal responses.

(iii) Advanced-economy status loses its role once space is controlled for. Without the space variables the advanced-economy indicator adds 5.06 (SE = 1.81) percent of GDP of above-the-line support, with them 1.80 (SE = 2.14), and the indicator differs from zero at the 5% level in 0 of the five outcome regressions that include space. In the cross-section, status is space by another name, and Tables C.10 and C.11 run the same race in the panel.

Figure 6 follows the economies through the years after 2019. It plots the coefficient on 2019 fiscal space from equation (9) for debt, the primary balance, the effective interest rate and output, per standard deviation of space, with 68% and 90% bands, on a cross-section that shrinks from 122 economies in 2020 to 34 in 2025 as the actuals run out, so I read the last point of every panel for its sign only. The figure suggests three conclusions.

(i) Economies with more pre-pandemic space accumulated less debt in the pandemic years and then caught up. One standard deviation more space changed debt by −3.37 (SE = 1.51) percentage points of GDP in 2020 and by −7.49 (SE = 5.48) in 2021, and by 2025 the sign has reversed. Space bought a smaller debt surge while the shock was live, and the reversal says that the gap was timing, not a permanent saving.

(ii) Their primary balances held up and their effective interest rates rose. In 2021 one standard deviation more space meant a primary balance 0.86 (SE = 0.28) percentage points of GDP higher and an effective rate 0.49 (SE = 0.16) percentage points higher. The economies with space did not run the larger deficits. Their interest bill per unit of debt rose, perhaps because they had borrowed less at the low rates of 2020.

(iii) Output was higher in the economies with space. One standard deviation more space meant output 2.14 (SE = 0.95) log points higher in 2021, and the gap had not closed by 2023. For the identified shock space protected output, where for the average global uncertainty shock the output responses do not separate. The insurance that fiscal space provides is insurance against crises, not against turbulence.

Whether these readings survive the choices made along the way is the question of the next subsection.

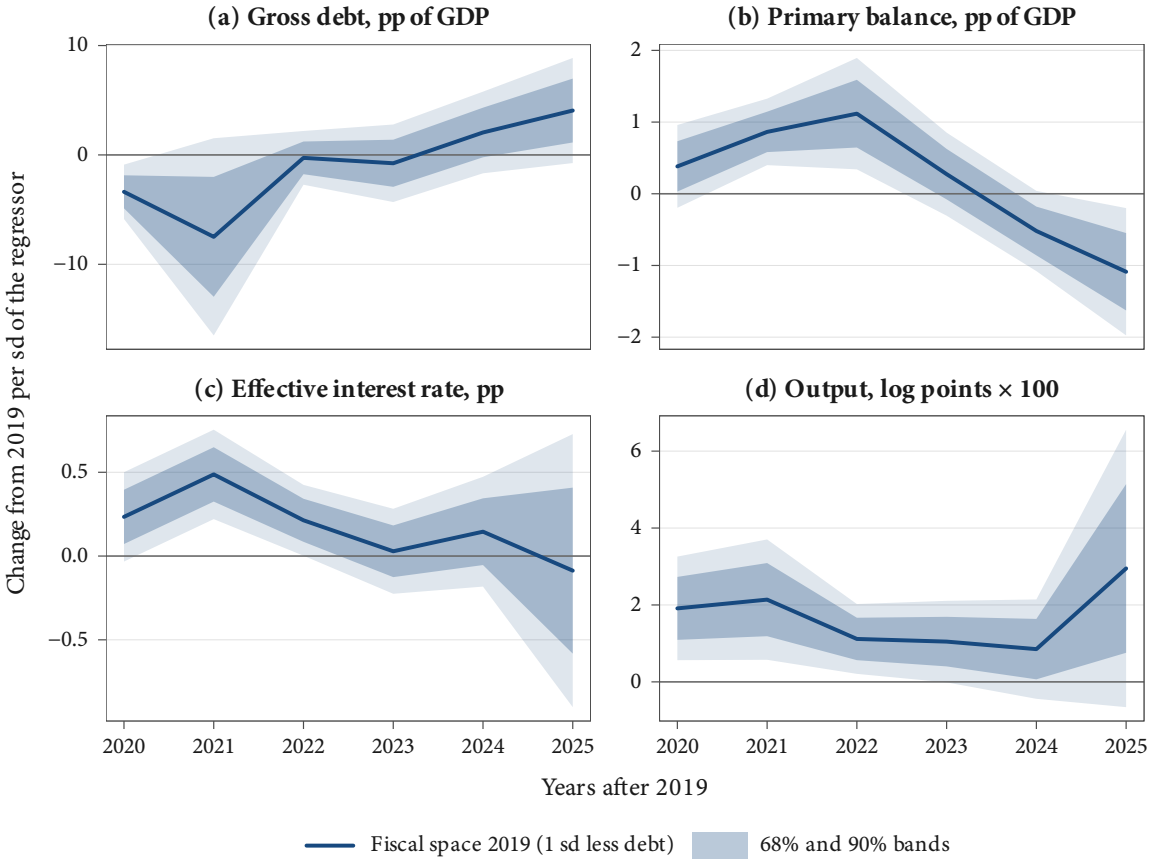


Figure 6. Pre-pandemic fiscal space and outcomes after 2020

Notes: Coefficients $\theta^{(h)}$ of equation (9), one standard deviation less 2019 debt (sd 36 pp), in cross-country regressions of $y_{i,2019+h} - y_{i,2019}$ on the regressor, the pandemic-uncertainty index, severity controls (deaths, stringency, pandemic discussion), the advanced-economy indicator, log income per capita at purchasing-power parity and openness; heteroskedasticity-robust (HC1) bands (68% and 90%). Economies: 122 in 2020 to 34 in 2025.

3.5 Sensitivity analysis

I summarize ten sets of sensitivity checks, 78 re-estimations in all, each changing one thing in a baseline and reporting the objects under test in the baseline's units: the debt–uncertainty interaction of Table 2, the triple interaction of Table 3 by the fitted-gap estimator, the two-year debt differential of Figure 3 with the interest-rate regime difference of Figure 5, and the group medians of Figure 4. Table C.12 collects every number. Across the checks the interaction keeps its sign in 40 of 59, the triple interaction stays within 1.645 standard errors of zero in 49 of 63, the debt differential keeps its sign in 26 of 32, and the group ordering holds in 28 of 31.

(i) The global shock is the innovation of one world index, so I re-estimated the decomposition with six other global series and the reaction function with other country measures (Table C.13, Figure C.13). The debt differential keeps its sign for 6 of the 7 global series, the exception being geopolitical risk, while the interaction is negative for 3 of the 11 other measures and indistinguishable from zero for an index of non-fiscal economic policy uncertainty (EPU) built from the same GDELT news articles (0.0004, SE = 0.0017). The erosion result is a property of uncertainty indices broadly. The interaction

appears to be specific to the measures built, as the WUI is, from an economy's own narrative.

(ii) Under the calendar-year, pre-budget, lagged and fourth-quarter timings of the WUI (Table C.14) the interaction is negative in all four and within one standard error of the baseline.

(iii) The pandemic years hold the sample's largest common shock, so the three panel models were re-estimated with the pandemic-purged measures, without 2020–2021, and without 2009 (Table C.15). The two-year debt differential is 1.27 (SE = 0.34) percentage points of GDP purged and 0.92 without the pandemic years, against 1.82 in the baseline, and the interaction and the interest-rate difference keep their signs under every treatment. The pandemic years contribute to the size of the erosion and not to its sign.

(iv) The pooled sample mixes groups and decades, and one of them could drive the results. Re-estimating by group, by decade, without commodity exporters and on a balanced panel (Table C.16) leaves the debt differential in the advanced and the low-income economies and removes it in the emerging markets, where it is −0.30 (SE = 1.02), and moves the interaction from positive in the 1990s to negative in the 2020s. The erosion result does not rest on one group, and the interaction seems to be a feature of the last two decades.

(v) With the cyclically adjusted or the WEO structural balance in place of the headline balance (Table C.17) the interaction keeps its sign, and the overall-balance contribution reproduces the sum of the primary-balance and interest contributions.

(vi) With six other measures of space in place of debt (Table C.19) the triple interaction differs from zero at the 10% level for 1 of the six, while the debt differential keeps its sign with 2 of them. The cyclicality null does not depend on how space is measured, and the erosion result is a result about debt.

(vii) The estimators were varied next: system generalized method of moments (GMM), other output gaps, ordinary least squares (OLS) and country trends for the reaction function, other lag structures, no outlier rule and winsorised components for the projections, and $\kappa = 1$, $\kappa = 3$ and the estimated threshold for the regimes (Table C.20). The interaction keeps its sign in every variant but the WEO gap, which exists for 23 advanced economies. The debt differential is largest without the outlier rule, which binds on size, not on sign.

(viii) The interaction sits close to a conventional threshold, so the inference was redone with one and four lags, two-way clustering, and wild bootstraps by year and by economy (Table C.21). The interaction is rejected at the 10% level under every covariance and at the 5% level under the four-lag covariance alone, and the triple interaction is not rejected at the 10% level under any. These results leave the interaction where Section 3.3 placed it.

(ix) With the GDELT index rebuilt on the stable outlet panel, in native languages and without the composition-flagged economies (Table C.22), the domestic interaction stays within one standard error of zero, and the domestic–foreign split keeps its signs in all 6 pairs. The split is a feature of the narrative, not of one corpus.

(x) The horse race of Table 4 was run in the panel, with uncertainty interacted with status, the rating, the currency share, government effectiveness, the exchange-rate regime and fiscal rules (Table C.11). The debt interaction keeps its sign in 5 of 7 columns and loses it when all six enter at once, while the interactions with government effectiveness and with a floating regime carry the sign of discipline. Institutions are the other candidate the panel points to. Appendix C.4 re-estimates the local projections under a Bayesian prior and changes no conclusion.

4 Discussion

Who can still afford to act when global uncertainty rises is a question about fiscal space, not about income group or status. The analysis leads to four major conclusions. First, a global uncertainty shock erodes fiscal space through the debt ratio before any budget decision is taken: within two years the economy at the 75th percentile of debt has accumulated roughly 2 points of GDP more than its counterpart at the 25th, through growth and inflation rather than the balance. Second, the cyclical stance is invariant to uncertainty, with or without space. Governments that leaned against the cycle in quiet years lean against it in turbulent ones, and the null carries the same weight as the erosion result. Third, what uncertainty changes is the price of the stance: after the same global shock the effective interest rate of an exposed economy rises in the vulnerable regime and falls in the safe one, a difference of roughly 0.6 percentage points. Fourth, in the one identified crisis of the sample, space bought a smaller debt surge and a smaller output loss, and the advantage of advanced-economy status disappears once the fiscal position is held fixed.

These findings share a mechanism. The identity of Section 2.2 says that debt rises after a shock through the growth and inflation terms, which scale with the debt stock, so the same shock costs the indebted economy more. The primary-balance channel that follows is the budgetary expression of a reaction function that uncertainty bends, and the source of the uncertainty seems to decide the direction of the bend. Global uncertainty strengthens the response to debt, arguably because it raises the price of debt for the vulnerable sovereign, while country-specific uncertainty weakens it, perhaps because it delays decisions whose cost is domestic.

Ostry et al. (2010) fit the cubic response of the primary balance to lagged debt for 23 advanced economies over 1985–2007, and report a marginal response of 0.53 percentage points of GDP per 10 points of debt at debt ratios of 50 to 60 percent and model-based limits between 150 and 260 percent of GDP. The marginal response at the median debt ratio of this panel, 0.17 per 10 points, is roughly a third of theirs, on a sample that is mostly emerging and developing economies over years in which the interest–growth differential was negative for most of them, which is also why their limits are not identified here for those economies, while the advanced-economy limits sit above their range with wide bands (Appendix C.2.4). The gap is one of sample as much as of specification. Nguyen et al. (2023) estimate the effect of the World Uncertainty Index on the fiscal balance of 143 countries over 1990–2019 and find that a 0.1-point rise in the index lowers the balance by 0.15 percentage points of GDP. Their object is the overall balance unconditional on the cycle, so the comparison is with the unconditional counterpart here, the direct-effect projection of Appendix C.2.1, and on sign alone, on which the two agree, since there too uncertainty worsens the primary balance over the years after a shock.

The regime result on the price of debt agrees on its qualitative property with the spread studies. Hong et al. (2024) find in monthly panel vector autoregressions on 189 countries that a global fiscal-uncertainty shock raises sovereign spreads by more in emerging markets than in advanced economies, and Lian et al. (2020) find larger interest-rate increases after global volatility shocks where debt is high. The property here is the same, with vulnerability in place of income group or debt; basis points of monthly spreads and percentage points of the annual effective rate are different objects, so no magnitudes are compared. Geopolitical risk is the exception on one side only, since Gamboa-Estrada and Romero (2026) find with panel local projections that it raises emerging-market credit-default-swap

spreads, while it is the one global series in Section 3.5 that does not reproduce the erosion result, perhaps because such a shock moves spreads for months while the identity moves over years.

The pandemic findings agree with Augustin et al. (2022), who find in daily data for 30 developed countries in 2020 that the credit-default-swap spreads of fiscally constrained sovereigns were the more sensitive to the spread of infections, and with Romer and Romer (2019), who find in OECD panel data over 1980–2017 that economies with lower debt entered financial distress with more support and suffered milder aftermaths. The objects are spreads and crisis aftermaths rather than fiscal flows, so the agreement is on the property that pre-crisis space governs the response. In this panel space protects output in the identified crisis and not detectably after the average global shock, which suggests that the insurance Romer and Romer document is insurance against crises rather than against turbulence. The cyclicality null bears on Bianchi et al. (2023), who show in a model with sovereign risk and in panel data for 70 countries over 1980–2016 that fiscal policy turns procyclical where default risk is high. Their state is sovereign risk and mine is uncertainty interacted with space, but the null is consistent with their mechanism in that the stance is set by the fiscal position itself rather than by the uncertainty of the year.

Four things the results do not settle. The absence of a sign switch in the cyclical response could reflect the coarseness of an annual headline balance as a measure of discretionary policy, although the cyclically adjusted balance gives the same null. The debt limits might be identified in a period of positive interest–growth differentials. The GDELT evidence covers one decade dominated by the pandemic, so the domestic–foreign split it supports is a result for that decade. And the domestic measures are identified only within economies, so their patterns could be the reverse causation they are flagged as. Of the four, the measurement of discretionary policy is the one I would pursue first.

The implications are scoped by these limits. For a government deciding how much to lean against a downturn in an uncertain year, the results say that the stance it takes is not what will distinguish it from its peers. The debt it carries into the year, and the regime position that debt implies, will decide how much the shock costs and what it pays to carry the cost. I would not read the results as a ranking of creditworthiness, since they say nothing about the quality of the room, nor as a debt limit, since the limits are the one object the panel does not identify.

The room to act is measured in debt. Uncertainty removes it through growth and prices before the budget moves, and the economies with the least room pay the most to use it.

5 Conclusion

This paper has asked how global uncertainty reaches the public finances of an economy and whether the room to respond depends on the fiscal position the economy brings to the shock. It has done so in an annual panel of advanced, emerging and low-income economies, with a world shock interacted with each economy's own position, a news-based index of fiscal-policy uncertainty, and the pandemic as a shock every government faced in the same year.

The analysis leads to four conclusions. The first is that uncertainty erodes fiscal space through the debt ratio, by way of growth and prices before the budget, and that the erosion is larger where debt is higher, roughly 2 percentage points of GDP within two years between the upper and the lower quartile of debt. The second is that the cyclical stance does not change with uncertainty, whatever the space behind it, so what a government does in a turbulent year is not what separates it from its peers. The

third is that the price of carrying debt through the shock is what separates them, rising with exposure in the vulnerable regime and falling in the safe one. The fourth is that in the one crisis the sample identifies, the economies with space came through with less debt and more output.

I see the main use of these results as a way of reading an economy's exposure to the next global shock from two numbers it already knows, its debt ratio and its regime position, rather than from its income group. It is not a measure of how much more an economy could borrow.

Four problems remain open. Model-based debt limits were computed and set aside, because the interest–growth differential over 2010–2025 leaves the reaction function without a crossing for most economies, and a panel with positive differentials could revisit them. The domestic–foreign split of fiscal-policy uncertainty did not survive the exclusion of the pandemic years in the news data, and a longer news sample would say whether the pattern is general. A dynamic-panel estimator for the reaction function gave standard errors too wide to be informative, and an estimator free of the dynamic-panel bias that keeps the cross-section of debt merits further work. The smooth transition carries the price-of-debt result, and the threshold model finds a regime only in the level of the response, at a low debt ratio, which leaves the vulnerability at which the price of debt turns as a problem for a model of sovereign risk.

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Online Appendix to "Global Uncertainty and Fiscal Space: Who Can Still Afford to Act?"

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This online appendix accompanies the paper and has three parts. Appendix A documents the data: sources and vintages, the construction of the uncertainty measures and the fiscal-space measures, the estimation sample economy by economy, and the descriptive exhibits. Appendix B documents the news-based fiscal-policy-uncertainty index built from the Global Knowledge Graph of the Global Database of Events, Language, and Tone (GDELT). Appendix C gives the estimation details that Section 2 of the paper summarises, the additional results for the main models, the robustness checks family by family, and a Bayesian re-estimation of the local projections. The main text refers to the exhibits here by their appendix numbers (Table C.4, for example), and the appendix refers to the main text's exhibits by their numbers there (Figure 3, for example).

A Data

This appendix records the sources of the panel with their vintages, the construction of every derived series the main text uses, the estimation sample, and the descriptive exhibits that Section 3.1 of the main text refers to. The appendix contains no results.

A.1 Sources and construction

The fiscal and macroeconomic series are the April 2026 vintage of the International Monetary Fund (IMF) World Economic Outlook (WEO), read from the IMF's Statistical Data and Metadata eXchange (SDMX) service for every economy from 1980. They are general government revenue, expenditure, the overall and the primary balance and gross debt, each in domestic currency and in percent of GDP, together with nominal and real GDP, the GDP deflator, consumer-price inflation, the current account, unemployment and population. The WEO flags for each series the last year that is an actual rather than a staff estimate. Every observation after that year is a projection and is excluded from estimation. Where a WEO fiscal series starts late, the Public Finances in Modern History database of Mauro et al. (2015), in its December 2025 release, supplies the earlier years of debt, the primary balance, revenue and expenditure for the same economy. The overall balance of those years is the primary balance less interest paid. The cyclically adjusted balances of the April 2026 Fiscal Monitor are carried as alternative outcomes.

The contributions of the debt identity in equation (1) are computed from these series. Nominal growth is $(1+\gamma)(1+\pi)-1$. The effective interest rate is net interest paid in year t, defined as the primary less the overall balance, over the debt stock at the end of $t-1$. The stock-flow adjustment closes the

identity. The effective rate is set to missing where lagged debt is below one half of one percent of GDP. In the models where it is an outcome it is trimmed at the 1st and 99th percentiles of the main sample, –16.4 and 18.9 percent. The output gap is the residual of the regression filter of Hamilton (2018) applied to log real GDP, in percent of trend. A Hodrick–Prescott gap with the annual smoothing parameter and the WEO staff gap, available for a small set of advanced economies, serve as alternatives. The cyclically adjusted primary balance subtracts the revenue ratio times the gap from the primary balance, which assumes a unit revenue elasticity and a zero expenditure elasticity.

The World Uncertainty Index (WUI) of Ahir et al. (2022) is read from the July and September 2026 releases of the quarterly country index, the monthly country index, the World Pandemic Uncertainty Index (WPUI) and the sentiment index. The calendar-year WUI is the mean of the four quarters of the year, and a year with fewer than four quarters is missing.

The fiscal-year index $U_{i,t}$ aligns the measure with the budget year of each economy as the WEO metadata define it. The WEO reports, for each economy with a non-calendar fiscal year, the start month and whether the year is labelled by its first or its last calendar year. In the sample, 16 of the 139 economies have such a year. For a calendar-year economy the fiscal-year index equals the calendar-year index. For the others it is the mean of the monthly index over the months of the fiscal year where the monthly file covers the economy. Otherwise it is the mean of the four quarters that best overlap the fiscal year. The first of them is the quarter that contains the start month when that month is the first or second month of the quarter, and the next quarter otherwise. The pre-budget index is the mean of the two quarters before the first fiscal quarter, labelled with the fiscal year that follows them. The WPUI, published as a share of words at a different scale, is rescaled to the units of the WUI. The purged country index is the WUI less the WPUI, floored at zero.

The global index is the GDP-weighted mean of the country WUI over the economies with a value in the year, with each economy's nominal dollar GDP of the previous year as its weight. Its purged counterpart applies the same weights to the purged country index. The global shock $\varepsilon_t^{\mathrm{g}}$ is the standardised residual of an autoregression of order two with a constant, fitted to the annual global index over 34 years. Its autoregressive coefficients are 0.37 and 0.41. The purged global shock is the residual of the same autoregression fitted to the purged global index. The domestic shock $\varepsilon_{i,t}^{\mathrm{d}}$ is the standardised residual of a country autoregression of order two of the WUI with the contemporaneous global index as a regressor, estimated for every economy with at least fifteen observations. The domestic component used in the two-source reaction functions of Appendix C.2.3 is the residual of a country regression of the WUI on the global index.

The GDELT fiscal-policy-uncertainty indices are described in Appendix B. The annual domestic index, the foreign-outlet index, the world index and the index of non-fiscal economic policy uncertainty enter the panel as z-scores of the log article share over 2016–2025, on the calendar, the fiscal-year and the pre-budget basis. Country-years in which the domestic index rests on too few articles are flagged as thin and dropped from the domestic samples. Economies in which the set of domestic outlets changes materially within the decade carry a composition flag, which the robustness checks of Appendix C.3.9 use. Three further measures enter the checks. The global economic policy uncertainty index of Davis (2016), in the lineage of Baker et al. (2016), is the annual mean of its non-imputed months. The global and the country geopolitical risk indices of Caldara and Iacoviello (2022) are annual means of the monthly series. The Cboe Volatility Index (VIX) is the annual mean of the daily close.

Every uncertainty regressor is standardised with the mean and standard deviation of the main estimation sample, which Table A.1 reports. The global and domestic shocks are autoregressive innovations already in standard-deviation units and are not rescaled.

Fiscal space is the lagged gross debt ratio, in percent of GDP. The alternative measures come from the Spring 2026 release of the fiscal-space database of Kose et al. (2022). They are the overall and the primary balance, the foreign-currency share of government debt, the share of debt held by non-residents, debt maturing within a year and the average maturity of sovereign debt, the external debt ratios, and the foreign-currency long-term sovereign rating on a scale from 1 to 21. The composite vulnerability index $v_{i,t}$ is the equal-weight mean of five standardised inputs: the debt ratio, the fiscal deficit, the foreign-currency share, the share of debt maturing within a year, and the rating with its sign reversed. Each input is standardised over the pooled sample of the database, and the weights are renormalised over the inputs an economy reports. The index is oriented so that a higher value is a more vulnerable position. The first principal component of the same five inputs has loadings of mixed sign, with debt, the deficit and the rollover need loading against the rating and the foreign-currency share. It cannot be read as a scale of vulnerability, and it appears only in the robustness checks of Appendix C.3.6.

Groups follow two classifications that are fixed over the sample. The WEO split into advanced economies and emerging market and developing economies is the two-way group, and the IMF income classification published with the WPUI, into advanced, emerging market and low-income developing economies, is the three-way group. Exposure to the global shock, E_i, is the mean of trade openness, exports plus imports in percent of GDP from the World Development Indicators, over 1995–2004 for economies with at least three years in that window. It is standardised across the economies of the sample, where its mean is 70.8 percent of GDP and its standard deviation 35.6. Exposure through the foreign-currency share of debt is built in the same way for the economies that report it, with the first five years after 2004 where the share starts later. An economy is a commodity exporter when fuel, ores and metals, food and agricultural raw materials together exceed one half of its merchandise exports on average over 1995–2004.

The controls come from seven sources. Lagged real growth and inflation are WEO series, and trade openness comes from the World Development Indicators. The commodity terms-of-trade control is the annual change in the log of the index of Gruss and Kebhaj (2019), weighted by the economy's commodity exports and imports relative to GDP. Government effectiveness is the estimate of the Worldwide Governance Indicators of Kaufmann et al. (2011), which end in 2022, and capital-account openness is the index of Chinn and Ito (2006), which ends in 2023. The exchange-rate regime is the coarse classification of Ilzetzki et al. (2019), taken as the modal monthly code within the year and grouped into pegs, floats and freely falling regimes. The fiscal-rule indicator records a numerical rule in force in the IMF fiscal rules dataset of Davoodi et al. (2022), which covers the economies in that database to 2024. The election-year indicator marks an executive or legislative election in the Database of Political Institutions of Scartascini et al. (2021), which ends in 2020. In the reaction function the election-year and fiscal-rule indicators enter zero-filled with an indicator for a missing value, so that the years after the end of each source and the economies outside the fiscal-rules database stay in the sample.

The pandemic module has three parts. The fiscal response is the IMF database of country fiscal measures in response to the pandemic in its October 2021 release. It gives, for the economies it covers,

Table A.1. Standardisation constants of the uncertainty measures

Measure	Mean	SD	Country-years	Economies	Years
WUI, calendar year	0.181	0.165	4,327	139	1990–2025
WUI, fiscal year	0.180	0.161	4,327	139	1990–2025
WUI, pre-budget	0.176	0.175	4,326	139	1990–2025
WUI, pandemic-purged	0.170	0.162	4,327	139	1990–2025
WUI, domestic component (residual)	0.004	0.133	4,327	139	1990–2025
Global WUI, GDP-weighted	0.193	0.094	4,327	139	1990–2025
Global WUI, pandemic-purged	0.183	0.090	4,327	139	1990–2025
GDELT fiscal-policy uncertainty, domestic outlets (z)	−0.087	0.895	1,237	138	2016–2025
GDELT fiscal-policy uncertainty, foreign outlets (z)	−0.116	0.871	1,245	139	2016–2025
GDELT fiscal-policy uncertainty, world (z)	−0.165	0.775	1,245	139	2016–2025
GDELT non-fiscal economic policy uncertainty, domestic (z)	−0.093	0.892	1,237	138	2016–2025
World Pandemic Uncertainty Index	0.012	0.049	3,854	139	1996–2025

Notes: Entries are the mean and standard deviation of each uncertainty measure over the country-years of the main estimation sample (Section 3.1), in the units of the measure (mentions per thousand words for the World Uncertainty Index series; z-scores of the log article share over 2016–2025 for the GDELT indices; share of words for the pandemic index), with the number of country-years and economies with a non-missing value and the years covered. Every standardised regressor is $(x - \text{mean})/\text{SD}$ with these constants; the global and domestic shocks are AR(2) innovations already in standard-deviation units and are not rescaled. Source: `output/estimates/standardisation.csv`.

the above-the-line measures and the liquidity support announced between January 2020 and September 2021, in percent of 2020 GDP, as time-invariant attributes of the economy. The European Union row is dropped because its measures finance member states. The severity controls are three. Cumulative confirmed deaths per million at the end of each year come from the Johns Hopkins compilation of Dong et al. (2020). The stringency and the economic-support indices of the Oxford COVID-19 Government Response Tracker of Hale et al. (2021) enter as annual means over 2020–2022. The pandemic-discussion index is the one published with the WPUI. The pandemic component of uncertainty in 2020 is the WPUI of that year, and its share of the WUI is reported in Table C.9.

A.2 The estimation sample

The estimation window is 1990–2025. It starts in 1990, the first year of the fiscal-space database, and it ends with the last WEO actual, so that no projection enters. Within the window the WEO actuals give 6,888 country-years for 197 economies. The main sample keeps the country-years with the four core variables, gross debt, the primary balance, real growth and the WUI, in economies with at least ten such country-years. The rule removes 58 economies, none of which has a single complete country-year, because none is covered by the WUI. A further 4 economies have a WUI but no WEO debt or primary balance. The main sample is 4,327 country-years in 139 economies.

The decomposition sample of the identity model applies the outlier rule of Section 3.1. The rule drops a country-year when the primary balance exceeds 25 percent of GDP in absolute value, the change in the debt ratio exceeds 40 points, or the stock-flow adjustment exceeds 10 points. It must hold in every year of a cumulative window, so the sample shrinks with the horizon. The reaction-function sample is the main sample less the country-years without an output gap or a lagged primary balance. The pandemic cross-section is the set of economies with IMF fiscal measures and a 2019–2021 primary balance. Table A.2 lists every economy of the main sample by group with its first and last year of fiscal data, its number of complete country-years and its membership of the model samples. The counts by group are those of Table 1.

Table A.2. Economies by group: years of fiscal data and model samples

Economy	First year	Last year	Country-years	M1	M2	Pandemic
Advanced economies (28 economies)						
Australia	1990	2024	35	33	35	3
Austria	1990	2024	35	33	35	0
Belgium	1990	2024	35	24	35	3
Canada	1990	2024	35	32	35	3
Czech Republic	1996	2024	29	28	27	3
Denmark	1990	2024	35	32	35	3
Finland	1990	2024	35	32	35	3
France	1990	2024	35	33	35	3
Germany	1990	2025	36	34	36	3
Greece	1990	2024	35	30	35	0
Hong Kong SAR	2005	2024	20	13	19	0
Ireland	1990	2024	35	30	35	0
Israel	1990	2025	36	34	36	0
Italy	1990	2025	36	34	36	3
Japan	1990	2024	35	33	35	3
Korea	1990	2024	35	33	35	3
Latvia	1998	2024	27	26	26	0
Netherlands	1990	2024	35	32	35	3
New Zealand	1990	2025	36	34	36	3
Norway	1990	2024	35	14	35	3
Portugal	1990	2024	35	33	35	3
Slovak Republic	1995	2024	30	29	29	0
Slovenia	1995	2024	30	29	29	0
Spain	1990	2025	36	34	36	3
Sweden	1990	2024	35	33	35	3
Switzerland	1990	2025	36	34	36	3
United Kingdom	1990	2025	36	34	36	3
United States	1990	2025	36	34	36	3
Emerging markets (59 economies)						
Albania	1997	2024	28	27	27	3
Algeria	1990	2024	35	26	34	0
Angola	1996	2024	29	12	28	0
Argentina	1990	2025	36	22	36	3
Armenia	2005	2025	21	19	20	0
Azerbaijan	1994	2024	31	21	30	0
Belarus	2004	2024	21	18	20	0
Bosnia and Herzegovina	1998	2024	27	24	26	0
Botswana	1999	2023	25	21	24	0
Brazil	1996	2024	29	26	28	3
Bulgaria	1995	2025	31	28	30	3
Chile	1990	2024	35	33	35	3
China	1995	2024	30	29	29	3
Colombia	1990	2025	36	34	36	3
Costa Rica	1990	2024	35	32	35	0
Croatia	1997	2024	28	27	27	0

Table A.2 (continued)

Economy	First year	Last year	Country-years	M1	M2	Pandemic
Dominican Republic	1990	2025	36	33	36	0
Ecuador	1995	2025	31	29	30	0
Egypt	1999	2025	27	20	26	3
El Salvador	1991	2024	34	33	33	0
Gabon	1990	2024	35	28	34	0
Georgia	1995	2025	31	25	29	3
Guatemala	1995	2024	30	29	29	3
Hungary	1995	2025	31	30	30	0
India	1990	2024	35	33	35	3
Indonesia	1990	2025	36	32	36	3
Iran	1990	2024	31	24	28	0
Iraq	2004	2025	22	15	21	0
Jamaica	1990	2024	35	0	34	0
Jordan	1990	2024	35	16	35	0
Kazakhstan	2002	2024	23	22	22	3
Kuwait	1991	2024	34	4	33	0
Lebanon	1990	2021	32	29	31	0
Lithuania	2000	2024	25	24	24	0
Malaysia	1990	2025	36	34	35	0
Mexico	1990	2024	35	32	35	3
Morocco	1990	2024	35	33	34	0
Namibia	1991	2024	34	33	32	0
North Macedonia	1997	2025	29	27	28	3
Oman	1990	2024	35	30	34	0
Pakistan	1990	2025	36	33	36	3
Panama	1990	2025	36	33	36	0
Paraguay	1990	2024	35	30	35	0
Peru	1990	2025	36	32	36	3
Philippines	1990	2025	36	33	36	3
Poland	1995	2024	30	29	29	3
Qatar	1991	2023	33	22	32	0
Romania	1995	2024	30	28	29	3
Russian Federation	1998	2025	28	26	27	3
Saudi Arabia	1991	2025	35	30	34	3
South Africa	1990	2024	35	33	35	3
Sri Lanka	1990	2024	35	25	34	0
Thailand	1990	2024	35	31	35	3
Tunisia	1991	2025	35	34	34	3
Türkiye	1990	2025	36	30	36	3
Ukraine	1995	2024	30	23	29	0
United Arab Emirates	1991	2024	34	15	33	3
Uruguay	1990	2024	35	29	34	0
Venezuela	1990	2017	28	13	28	0
Low-income developing countries (52 economies)						
Afghanistan	2004	2024	21	4	20	0
Bangladesh	1990	2025	36	34	35	3
Benin	1990	2024	35	29	35	0

Table A.2 (continued)

Economy	First year	Last year	Country-years	M1	M2	Pandemic
Bolivia	1990	2025	36	33	36	0
Burkina Faso	1990	2024	35	32	35	0
Burundi	1992	2024	33	0	32	0
Cambodia	1996	2024	29	28	28	0
Cameroon	2000	2024	25	23	24	0
Central African Republic	1990	2019	30	22	30	0
Chad	1995	2024	30	29	29	3
Congo	2000	2023	24	14	23	0
Congo	1990	2022	33	14	33	0
Côte d'Ivoire	1997	2024	28	26	27	0
Eritrea	2000	2019	20	3	19	0
Ethiopia	1990	2025	36	12	36	3
Gambia	2000	2023	24	17	23	0
Ghana	1990	2024	35	26	35	3
Guinea	1990	2024	35	25	34	0
Guinea-Bissau	1991	2023	33	22	32	3
Haiti	1997	2025	29	26	28	0
Honduras	1990	2024	35	28	35	3
Kenya	1990	2024	35	32	35	3
Kyrgyz Republic	1995	2024	30	21	29	0
Lao P.D.R.	2001	2024	24	15	23	0
Lesotho	1990	2024	35	13	35	0
Liberia	2001	2025	25	0	23	0
Madagascar	1990	2024	35	28	35	0
Malawi	2002	2024	23	3	22	0
Mali	2000	2024	25	23	24	0
Mauritania	2004	2023	20	15	19	0
Moldova	1995	2025	31	26	30	0
Mongolia	1993	2025	33	21	33	0
Mozambique	1999	2023	25	13	24	0
Myanmar	1999	2020	22	0	20	2
Nepal	2000	2025	26	25	25	0
Nicaragua	1997	2024	28	23	27	0
Niger	1995	2022	28	25	27	3
Nigeria	1991	2024	34	0	32	3
Papua New Guinea	1992	2024	33	12	32	0
Rwanda	1995	2024	30	25	29	0
Senegal	1994	2024	31	24	30	3
Sierra Leone	2001	2024	24	18	23	0
Sudan	1992	2019	28	7	27	0
Tajikistan	1998	2023	26	20	25	0
Tanzania	1991	2023	33	27	32	0
Togo	1990	2023	34	29	34	0
Uganda	1997	2024	28	27	27	0
Uzbekistan	1997	2024	28	0	27	3
Vietnam	2000	2024	25	24	0	3
Yemen	2015	2024	10	0	0	0

Table A.2 (continued)

Economy	First year	Last year	Country-years	M1	M2	Pandemic
Zambia	2000	2024	25	14	24	3
Zimbabwe	2005	2024	20	13	19	0

Notes: Entries are, for each of the 139 economies of the main sample, the first and last year with the four core variables (gross debt, primary balance, real growth and the World Uncertainty Index; IMF WEO April 2026 actuals), the number of such country-years, and the number of country-years in the decomposition sample (M1, after the outlier rule), the reaction-function sample (M2) and the pandemic cross-section (the 2019–2021 rows of economies with IMF COVID-19 fiscal measures). Groups follow the IMF income classification of the World Pandemic Uncertainty Index. Source: `output/estimates/sample_annual.parquet`, `economy_groups.csv`.

A.3 Descriptive exhibits

Table A.3 reports, for each of the 44 variables of the estimation, the mean, the standard deviation, the 10th percentile, the median and the 90th percentile over the country-years of the main sample. The entries are raw and unstandardised, in the units of the variable, with the number of country-years and of economies that report it. The three panels are the pooled sample, the advanced economies and the emerging market and developing economies. The pooled median debt ratio is 46.9 percent of GDP against a mean of 56.2, and the change in the debt ratio has a standard deviation of 13.1 points of GDP, of which the stock-flow adjustment alone has 12.8.

Table A.3. Summary statistics of the estimation variables

Variable	Units	Mean	SD	p10	Median	p90	N	Economies
Panel A. All economies								
Gross debt	% of GDP	56.2	42.5	18.8	46.9	100.4	4,327	139
Primary balance	% of GDP	−0.3	5.2	−4.6	−0.6	4.1	4,327	139
Overall balance	% of GDP	−2.3	5.7	−7.2	−2.5	2.1	4,327	139
Revenue	% of GDP	27.1	13.0	12.2	24.8	45.3	4,327	139
Expenditure	% of GDP	29.4	13.0	14.3	27.6	48.1	4,327	139
Primary expenditure	% of GDP	27.5	13.0	12.7	25.1	45.6	4,327	139
Net interest payments	% of GDP	2.0	2.5	0.2	1.5	4.5	4,327	139
Effective interest rate	fraction	0.0257	0.297	0.00766	0.0346	0.0824	4,256	139
Real GDP growth	fraction	0.0367	0.0474	−0.00845	0.0381	0.0811	4,327	139
GDP deflator inflation	fraction	0.136	1.29	0.00289	0.0465	0.197	4,327	139
Output gap (Hamilton)	% of trend GDP	0.5	6.1	−5.7	0.8	6.5	4,277	137
Change in debt ratio	pp of GDP	−0.1	13.1	−7.8	0.0	8.5	4,263	139
Interest contribution	pp of GDP	2.0	2.5	0.2	1.5	4.5	4,256	139
Real growth contribution	pp of GDP	−1.6	4.4	−4.0	−1.4	0.3	4,263	139
Inflation contribution	pp of GDP	−4.1	10.5	−9.5	−1.8	−0.1	4,263	139
Primary balance contribution	pp of GDP	0.3	5.2	−4.1	0.6	4.6	4,327	139
Stock-flow adjustment	pp of GDP	3.3	12.8	−2.4	1.4	10.3	4,256	139
WUI, calendar year	per 1,000 words	0.181	0.165	0.0280	0.140	0.375	4,327	139
WUI, fiscal year	per 1,000 words	0.180	0.161	0.0282	0.139	0.377	4,327	139
WUI, pre-budget	per 1,000 words	0.176	0.175	0	0.132	0.397	4,326	139
Domestic WUI	per 1,000 words	0.00379	0.133	−0.128	−0.0191	0.165	4,327	139
Global WUI	per 1,000 words	0.193	0.0937	0.100	0.188	0.298	4,327	139
Non-pandemic WUI	per 1,000 words	0.170	0.162	0.0238	0.130	0.359	4,327	139
WPUI (pandemic)	per 1,000 words	0.0123	0.0495	0	0	0.0187	3,854	139
Global uncertainty shock	sd units	−0.0882	0.672	−0.573	−0.167	0.640	4,188	139
Domestic uncertainty shock	sd units	0.00594	0.951	−1.06	−0.126	1.29	4,184	139
GDELT FPU, domestic	z-score	−0.0871	0.895	−1.20	−0.108	1.12	1,237	138

Table A.3 (continued)

Variable	Units	Mean	SD	p10	Median	p90	N	Economies
GDELT FPU, foreign	z-score	−0.116	0.871	−1.16	−0.219	1.16	1,245	139
GDELT FPU, world	z-score	−0.165	0.775	−0.907	−0.440	1.47	1,245	139
Fiscal vulnerability composite	sd units	−0.0342	0.554	−0.685	−0.0158	0.582	3,191	132
Sovereign rating	index, 1–21	12.5	5.31	6.07	11.9	21.0	3,119	124
Foreign-currency debt share	% of total	30.2	30.8	0.0	20.0	76.1	818	55
Debt due within a year	% of GDP	7.0	8.3	0.5	4.5	16.4	1,601	120
Trade openness	% of GDP	75.4	42.5	35.3	66.2	128.8	4,022	134
Commodity export share	% of exports	49.2	29.9	12.1	46.2	91.1	3,715	138
Chinn–Ito openness	index	0.342	1.56	−1.25	−0.0571	2.28	4,102	137
Government effectiveness	index	−0.0704	0.971	−1.19	−0.256	1.54	3,211	139
Country geopolitical risk	index	0.231	0.466	0.0190	0.0732	0.523	1,447	43
Election year	0/1	0.283	0.450	0	0	1.00	3,750	138
Numerical fiscal rule	0/1	0.668	0.471	0	1.00	1.00	2,840	89
COVID-19 fiscal support	% of 2020 GDP	6.8	5.3	2.0	5.3	15.3	61	61
COVID-19 liquidity support	% of 2020 GDP	5.8	7.5	0.3	2.9	15.3	56	56
OxCGRT stringency	index 0–100	43.2	17.4	16.8	46.1	64.0	393	132
COVID-19 deaths	per million	908.6	1,189.5	12.2	378.3	2,733.5	400	135
Panel B. Advanced economies								
Gross debt	% of GDP	64.3	39.0	23.4	56.4	113.1	1,033	31
Primary balance	% of GDP	−0.4	3.8	−4.6	−0.4	3.7	1,033	31
Overall balance	% of GDP	−2.3	4.3	−7.2	−2.4	1.9	1,033	31
Revenue	% of GDP	40.6	9.1	29.9	40.6	52.3	1,033	31
Expenditure	% of GDP	42.9	9.5	32.2	44.0	54.3	1,033	31
Primary expenditure	% of GDP	41.0	9.1	31.1	41.7	51.9	1,033	31
Net interest payments	% of GDP	1.9	2.3	0.0	1.5	4.6	1,033	31
Effective interest rate	fraction	0.00587	0.258	−0.000735	0.0287	0.0674	1,022	31
Real GDP growth	fraction	0.0237	0.0323	−0.00920	0.0244	0.0575	1,033	31
GDP deflator inflation	fraction	0.0385	0.286	0.00202	0.0209	0.0659	1,033	31
Output gap (Hamilton)	% of trend GDP	0.1	4.3	−4.8	0.5	4.2	1,030	31
Change in debt ratio	pp of GDP	0.8	5.8	−4.5	0.0	7.0	1,028	31
Interest contribution	pp of GDP	1.9	2.3	0.0	1.5	4.6	1,022	31
Real growth contribution	pp of GDP	−1.2	2.4	−3.1	−1.1	0.4	1,028	31
Inflation contribution	pp of GDP	−1.7	3.9	−3.4	−1.1	0.0	1,028	31
Primary balance contribution	pp of GDP	0.4	3.8	−3.7	0.4	4.6	1,033	31
Stock-flow adjustment	pp of GDP	1.4	4.9	−1.9	0.7	4.9	1,022	31
WUI, calendar year	per 1,000 words	0.184	0.150	0.0433	0.149	0.354	1,033	31
WUI, fiscal year	per 1,000 words	0.183	0.150	0.0433	0.149	0.353	1,033	31
WUI, pre-budget	per 1,000 words	0.176	0.156	0.0280	0.139	0.361	1,033	31
Domestic WUI	per 1,000 words	0.000843	0.104	−0.112	−0.0162	0.134	1,033	31
Global WUI	per 1,000 words	0.189	0.0944	0.0968	0.181	0.267	1,033	31
Non-pandemic WUI	per 1,000 words	0.174	0.143	0.0416	0.144	0.336	1,033	31
WPUI (pandemic)	per 1,000 words	0.0112	0.0474	0	0	0.0221	892	31
Global uncertainty shock	sd units	−0.0939	0.673	−0.573	−0.180	0.640	987	31
Domestic uncertainty shock	sd units	−0.00263	0.947	−1.16	−0.0896	1.25	986	31
GDELT FPU, domestic	z-score	−0.105	0.885	−1.17	−0.259	1.25	288	31
GDELT FPU, foreign	z-score	−0.100	0.891	−1.09	−0.245	1.20	288	31
GDELT FPU, world	z-score	−0.161	0.778	−0.907	−0.440	1.47	288	31
Fiscal vulnerability composite	sd units	−0.261	0.535	−0.802	−0.368	0.386	1,027	31
Sovereign rating	index, 1–21	18.1	3.41	12.7	19.4	21.0	1,031	31
Foreign-currency debt share	% of total	3.7	6.7	0.0	0.5	14.7	369	16
Debt due within a year	% of GDP	11.2	11.1	2.2	8.0	20.3	443	31
Trade openness	% of GDP	90.1	58.3	41.1	74.6	150.2	1,031	31
Commodity export share	% of exports	26.6	20.4	7.8	20.3	65.7	1,018	31
Chinn–Ito openness	index	1.85	0.857	0.698	2.28	2.28	985	31

Table A.3 (continued)

Variable	Units	Mean	SD	p10	Median	p90	N	Economies
Government effectiveness	index	1.31	0.554	0.497	1.47	1.92	734	31
Country geopolitical risk	index	0.333	0.589	0.0270	0.123	0.820	692	20
Election year	0/1	0.320	0.467	0	0	1.00	884	30
Numerical fiscal rule	0/1	0.838	0.368	0	1.00	1.00	989	30
COVID-19 fiscal support	% of 2020 GDP	11.1	6.1	4.8	9.2	19.3	21	21
COVID-19 liquidity support	% of 2020 GDP	11.3	9.5	2.4	7.4	27.8	21	21
OxCGRT stringency	index 0–100	41.8	16.5	16.3	46.8	58.5	93	31
COVID-19 deaths	per million	1,530.6	1,232.4	107.2	1,273.9	3,290.8	93	31
Panel C. Emerging market and developing economies								
Gross debt	% of GDP	53.6	43.2	18.0	44.4	91.6	3,294	108
Primary balance	% of GDP	−0.3	5.6	−4.6	−0.7	4.3	3,294	108
Overall balance	% of GDP	−2.3	6.1	−7.1	−2.5	2.2	3,294	108
Revenue	% of GDP	22.9	11.0	11.3	20.9	37.6	3,294	108
Expenditure	% of GDP	25.2	10.8	13.4	23.2	39.0	3,294	108
Primary expenditure	% of GDP	23.2	11.0	12.0	21.2	37.0	3,294	108
Net interest payments	% of GDP	2.0	2.6	0.3	1.5	4.5	3,294	108
Effective interest rate	fraction	0.0320	0.308	0.0101	0.0375	0.0864	3,234	108
Real GDP growth	fraction	0.0408	0.0505	−0.00819	0.0431	0.0855	3,294	108
GDP deflator inflation	fraction	0.166	1.47	0.00378	0.0636	0.228	3,294	108
Output gap (Hamilton)	% of trend GDP	0.7	6.5	−5.9	0.8	7.0	3,247	106
Change in debt ratio	pp of GDP	−0.4	14.7	−9.2	0.1	8.8	3,235	108
Interest contribution	pp of GDP	2.0	2.6	0.3	1.5	4.5	3,234	108
Real growth contribution	pp of GDP	−1.7	4.8	−4.2	−1.5	0.2	3,235	108
Inflation contribution	pp of GDP	−4.9	11.8	−11.4	−2.3	−0.1	3,235	108
Primary balance contribution	pp of GDP	0.3	5.6	−4.3	0.7	4.6	3,294	108
Stock-flow adjustment	pp of GDP	3.9	14.3	−2.7	1.8	12.4	3,234	108
WUI, calendar year	per 1,000 words	0.180	0.170	0.0233	0.137	0.384	3,294	108
WUI, fiscal year	per 1,000 words	0.179	0.164	0.0230	0.136	0.386	3,294	108
WUI, pre-budget	per 1,000 words	0.177	0.180	0	0.129	0.410	3,293	108
Domestic WUI	per 1,000 words	0.00472	0.141	−0.134	−0.0205	0.181	3,294	108
Global WUI	per 1,000 words	0.194	0.0934	0.100	0.188	0.298	3,294	108
Non-pandemic WUI	per 1,000 words	0.169	0.167	0.0174	0.127	0.373	3,294	108
WPUI (pandemic)	per 1,000 words	0.0126	0.0501	0	0	0.0182	2,962	108
Global uncertainty shock	sd units	−0.0864	0.672	−0.573	−0.167	0.640	3,201	108
Domestic uncertainty shock	sd units	0.00859	0.952	−1.05	−0.132	1.31	3,198	108
GDELT FPU, domestic	z-score	−0.0818	0.898	−1.20	−0.0792	1.08	949	107
GDELT FPU, foreign	z-score	−0.121	0.865	−1.16	−0.212	1.11	957	108
GDELT FPU, world	z-score	−0.166	0.774	−0.907	−0.440	1.47	957	108
Fiscal vulnerability composite	sd units	0.0735	0.530	−0.496	0.0881	0.624	2,164	101
Sovereign rating	index, 1–21	9.70	3.60	5.96	9.00	15.0	2,088	93
Foreign-currency debt share	% of total	52.0	25.3	16.0	49.8	87.6	449	39
Debt due within a year	% of GDP	5.4	6.3	0.2	3.5	13.6	1,158	89
Trade openness	% of GDP	70.3	34.1	34.4	62.5	120.0	2,991	103
Commodity export share	% of exports	57.8	28.4	18.8	61.2	93.2	2,697	107
Chinn–Ito openness	index	−0.136	1.43	−1.25	−0.750	2.28	3,117	106
Government effectiveness	index	−0.480	0.632	−1.26	−0.538	0.354	2,477	108
Country geopolitical risk	index	0.136	0.282	0.0154	0.0483	0.352	755	23
Election year	0/1	0.271	0.445	0	0	1.00	2,866	108
Numerical fiscal rule	0/1	0.578	0.494	0	1.00	1.00	1,851	59
COVID-19 fiscal support	% of 2020 GDP	4.6	3.1	1.8	3.8	9.2	40	40
COVID-19 liquidity support	% of 2020 GDP	2.4	2.5	0.2	1.4	5.8	35	35
OxCGRT stringency	index 0–100	43.6	17.6	17.9	45.2	64.8	300	101
COVID-19 deaths	per million	720.2	1,111.4	10.8	204.4	2,441.7	307	104

Notes: Entries are the mean, standard deviation, 10th percentile, median and 90th percentile of each variable over the country-years of the main estimation sample (Section 3.1; 4,327 country-years, 139 economies, 1990–2025, IMF WEO April 2026 actuals), raw and unstandardised, with the number of country-years (N) and economies with a non-missing value. Advanced economies (AE) and emerging market and developing economies (EMDE) follow the WEO classification. Debt-dynamics contributions sum to the change in the debt ratio. The uncertainty shocks are standardised AR(2) residuals; the GDELT indices are z-scores of the log article share over 2016–2025. COVID-19 fiscal measures (IMF, October 2021, percent of 2020 GDP) are economy-level and counted once. Sources: IMF WEO and Fiscal Monitor, World Bank Fiscal Space database and WDI/WGI, Ahir, Bloom and Furceri (WUI, WPUI), Caldara and Iacoviello (GPR), Chinn and Ito, DPI, OxCGRT, Our World in Data, GDELT (own construction).

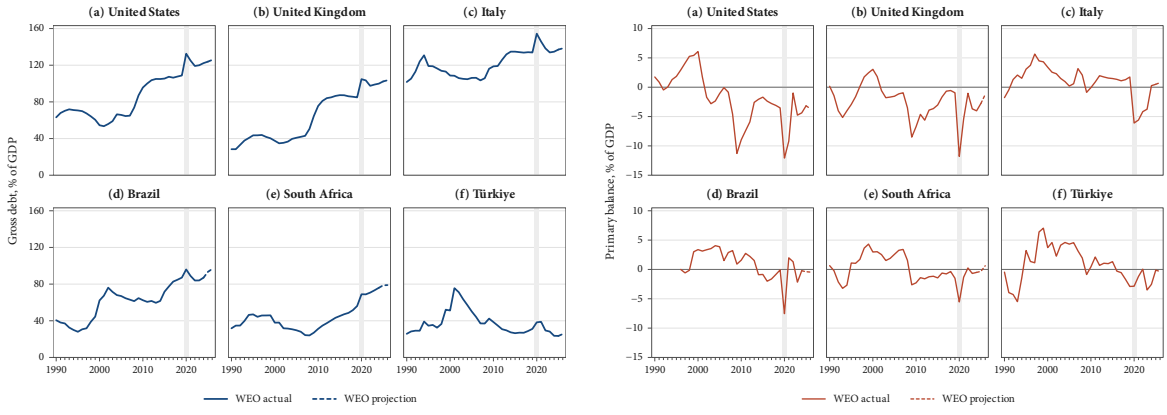
Table A.4 reports the correlations among the thirteen uncertainty measures, in levels below the diagonal and in first differences above it, on the common sample of the eleven panel measures: 399 country-years in 43 economies over 2016–2025, the decade that the GDELT and the country geopolitical risk indices bound. The two columns of the Baker–Bloom–Davis country index are computed on the 150 country-years of the 16 economies that publish it. The 343 imputed monthly cells of its workbook are dropped first. The GDELT world index and the global economic policy uncertainty index correlate at 0.93 on this sample, and the domestic GDELT index and the WUI at 0.36.

Figure A.1 plots gross debt (panel a) and the primary balance (panel b), in percent of GDP, for the six economies of Figure 2 over 1990–2026, with the WEO actuals as solid lines and the projections from each economy's last actual year as dashed segments. Figure A.2 plots the cross-country median and the interquartile range of the debt ratio, the primary balance and the WUI by year for the two WEO groups. A debt or primary-balance cell with fewer than 20 reporting economies is omitted. The rule removes the 2 cells of the advanced economies in 2025, where 9 economies have an actual, and the WUI panel covers every sample economy to 2025. Figure A.3 plots, for the same six economies over 2016–2025, the annual WUI, its pandemic component on the same scale, and the GDELT domestic index rescaled by the ratio of the economy's 2016–2019 mean WUI to its 2016–2019 mean index. The two series share a scale in the base years by construction only. The smallest rescaling constant among the six is 0.375 and the largest 0.887.

Table A.4. Correlations among the uncertainty measures, levels and first differences

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. WUI		0.89	0.53	0.49	0.05	0.15	0.16	0.42	0.44	0.37	0.43	0.43	0.44
2. Non-pandemic WUI	0.94		0.53	0.60	0.08	0.27	−0.04	0.26	0.32	0.20	0.29	0.32	0.30
3. Global WUI	0.41	0.43		0.92	0.07	0.19	0.23	0.52	0.60	0.71	0.53	0.53	0.82
4. Global non-pandemic WUI	0.38	0.48	0.94		0.09	0.25	−0.08	0.32	0.38	0.40	0.35	0.37	0.55
5. Country GPR	0.08	0.11	0.08	0.10		0.39	0.12	0.11	0.18	0.06	0.14	0.35	0.09
6. Global GPR	0.09	0.20	0.17	0.30	0.21		0.24	0.16	0.21	0.16	0.17	0.25	0.21
7. VIX	0.09	−0.10	0.16	−0.15	0.03	0.00		0.54	0.62	0.75	0.52	0.43	0.69
8. GDELT FPU, domestic	0.36	0.30	0.51	0.38	0.06	0.15	0.38		0.70	0.70	0.84	0.67	0.68
9. GDELT FPU, foreign	0.40	0.33	0.58	0.43	0.10	0.00	0.39	0.61		0.80	0.68	0.65	0.77
10. GDELT FPU, world	0.32	0.20	0.77	0.52	0.05	0.05	0.66	0.62	0.69		0.69	0.61	0.96
11. GDELT non-fiscal EPU, domestic	0.37	0.30	0.46	0.32	0.04	0.16	0.42	0.87	0.57	0.60		0.64	0.65
12. BBD country EPU	0.31	0.28	0.32	0.26	0.32	0.17	0.17	0.39	0.32	0.33	0.36		0.62
13. BBD global EPU	0.32	0.22	0.80	0.58	0.10	0.21	0.63	0.60	0.58	0.93	0.58	0.35	

Notes: Entries are Pearson correlations among the uncertainty measures, in levels below the diagonal and in first differences above it, on the common sample of the eleven panel measures (country-years with all eleven non-missing: 399 country-years, 43 economies, 2016–2025; 356 differences). The Baker–Bloom–Davis (BBD) country EPU, annual means of the monthly indices with imputed months dropped, is available for 150 of these country-years (16 economies); the BBD global EPU (GEPU, current-price GDP weights) varies by year only. WUI measures are mentions per thousand words; GPR and VIX are annual means; GDELT measures are z-scores of the log article share over 2016–2025.



(a) (b)

Figure A.1. Gross debt and primary balance in six economies, 1990–2026

Notes: General government gross debt (panel a) and primary balance (panel b), percent of GDP, IMF WEO April 2026, 1990–2026. Solid lines are WEO actuals; dashed segments are WEO projections from the last actual year (USA 2025, GBR 2025, ITA 2025, BRA 2024, ZAF 2024, TUR 2025) to 2026. The y axis is shared within each figure; 2020 is shaded.

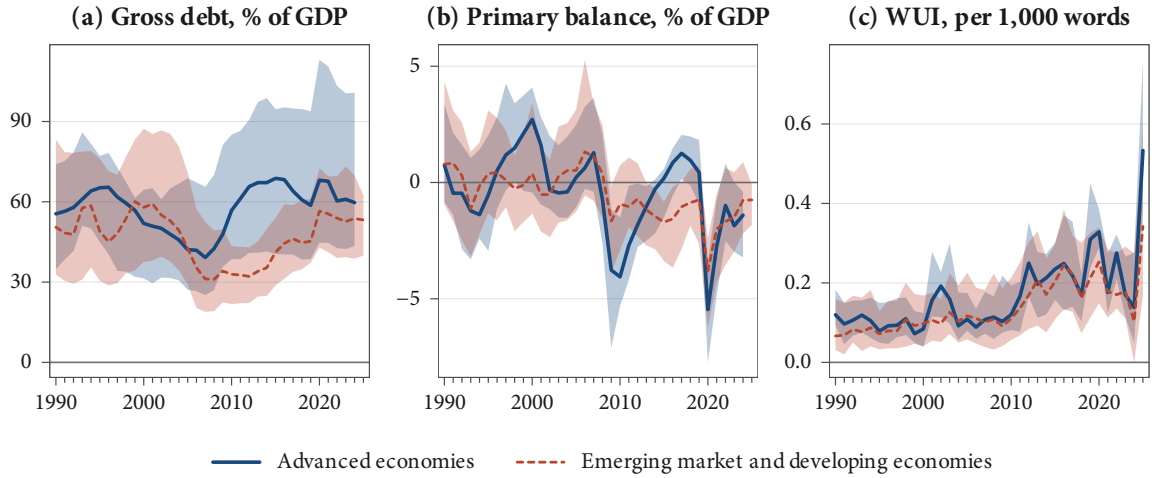


Figure A.2. Cross-country distributions of debt, the primary balance and uncertainty by group, 1990–2025

Notes: Lines are cross-country medians by year; shaded bands are the interquartile range (25th to 75th percentile) in the colour of the series. Debt and the primary balance are WEO actuals over the main estimation sample, shown where at least 20 economies of the group report (AE 2025 omitted); WUI covers all sample economies, 1990–2025. Groups follow the WEO classification.

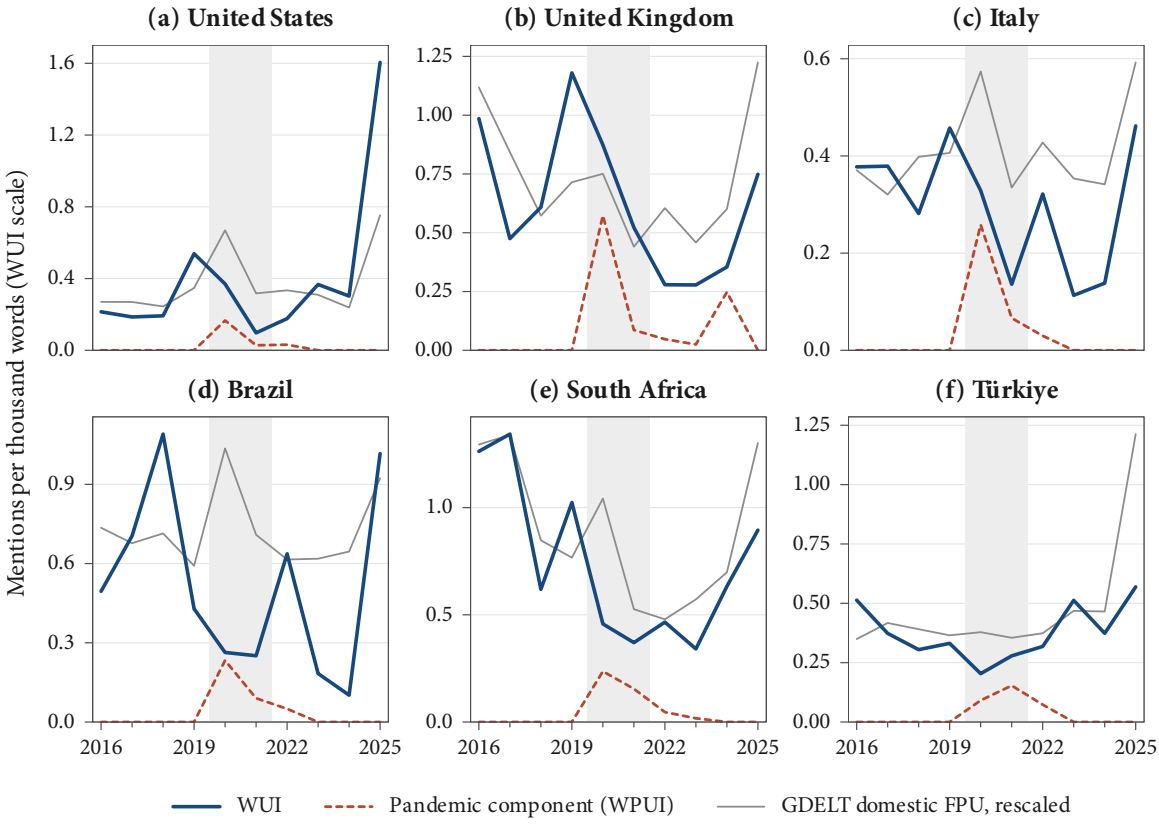


Figure A.3. WUI, pandemic uncertainty and GDELT domestic fiscal-policy uncertainty in six economies, 2016–2025

Notes: Annual WUI (mentions of uncertainty per thousand words), its pandemic component WPUI on the same scale, and the GDELT domestic fiscal-policy-uncertainty index divided by 100 and multiplied by the ratio of the economy's 2016–2019 mean WUI to its 2016–2019 mean index, 2016–2025; 2020–2021 shaded; y axes differ across panels.

B The GDELT fiscal-policy-uncertainty index

B.1 Overview

This appendix describes the construction of the news-based fiscal policy uncertainty (FPU) index used in the paper, the variants of the index used in the robustness checks, and the checks made on the index before use. The index is built from the Global Knowledge Graph (GKG) of the GDELT Project, which records every online news article that GDELT monitors, in English and in roughly 65 other languages, together with the themes the article is tagged with and the locations it mentions. Section B.2 describes the corpus. Section B.3 describes how articles are classified. Section B.4 describes how articles are attributed to countries and defines the domestic, foreign, and world views of each country. Sections B.5 and B.6 define the counts, the annual aggregation, the shrinkage, and the scaling of the index. Section B.7 lists the variables that enter the paper, Section B.8 reports the validation checks, and Section B.9 describes the computation.

Newspaper-based measurement of policy uncertainty dates to Baker et al. (2016), who measure economic policy uncertainty (EPU) by the share of articles in a fixed set of national newspapers that contain at least one term from each of three lists (economy, policy, and uncertainty). I make two departures from that design. First, the policy list is replaced by a fiscal list, so that the index counts articles about taxes, budgets, government spending, deficits, and public debt rather than about policy in general. Second, the fixed set of newspapers is replaced by the GDELT corpus, so that one dictionary and one procedure produce comparable indices for roughly 200 economies, from the national press of each economy and from the press of other countries writing about it. The first departure follows the categorical indices of Baker et al. (2016) and the fiscal volatility literature of Fernández-Villaverde et al. (2015). The second follows Davis (2016), who aggregates national EPU indices into a global one, and Caldara and Iacoviello (2022), who measure country risk from articles in the press of other countries. The World Uncertainty Index (WUI) of Ahir et al. (2022), which counts the word "uncertainty" in Economist Intelligence Unit country reports, is the long-sample uncertainty measure in the paper. The GDELT index covers 2016 onward. It adds three things the WUI does not have (a fiscal focus, a within-year resolution, and a distinction between the domestic and the foreign narrative), and it is built for the same economies and the same fiscal years as the panel.

B.2 The corpus

The data set consists of the GKG records of the online news articles processed by GDELT from 1 January 2016 to 30 September 2026, 1,649,845,652 articles in total, of which 648,021,199 (39.3 percent) are in the English feed and 1,001,824,453 in the machine-translated feed (Leetaru and Schrodt, 2013; GDELT Project, 2015).[1] One record is one article identified by its URL. GDELT processes English articles directly and articles in roughly 65 other languages through machine translation, of which 63 occur in the sample. It tags every article with themes from a fixed vocabulary (the current GDELT theme inventory lists 59,315 distinct themes), including the Baker, Bloom, and Davis term lists, which GDELT applies to the translated text with one dictionary. I therefore classify articles by GDELT's theme tags rather

[1]The volume processed in June 2025 is 45 percent of the average of May and July 2025. Every series of this appendix is a ratio of counts from the same articles, so the shortfall lowers the precision of the 2025 shares slightly but does not shift them in expectation.

than by a search of the article text, which the GKG does not contain. No keyword list for any language is needed.

Only records from the web news collection are kept. Broadcast transcripts and the other GDELT collections are excluded. Seven fields of each record are read: the date, the collection identifier, the outlet domain, the translation field (which gives the source language), the theme list, the list of geocoded locations, and the average tone.[2]

The sample starts in 2016 because GDELT began applying the Baker, Bloom, and Davis term lists on 14 December 2015. No record before that date carries the tags, which was checked file by file, so 2016 is the first complete year. The base period used for scaling is 2016–2025, the ten complete years. Calendar year 2026 is partial and is flagged as such wherever it appears.

B.3 Classification of articles

Let $\Theta(a)$ denote the set of themes tagged to article a, and let a lexicon $\mathcal{L}$ be a set of themes. An article carries a lexicon if at least one of its themes belongs to it:

$$L_a = \mathbb{1}\{\Theta(a) \cap \mathcal{L} \neq \varnothing\}, \qquad L \in \{U, E, P, F, M, T, S\}, \tag{B.1}$$

where the indicators U_a, E_a, P_a, F_a, M_a, T_a, and S_a correspond to the uncertainty, economy, policy, fiscal, monetary, trade, and sovereign-debt lexicons listed in Table B.1. An article is a fiscal policy uncertainty article if it carries the uncertainty, economy, and fiscal lexicons at once:

$$\mathrm{FPU}_a = U_a\, E_a\, F_a. \tag{B.2}$$

The companion classifications used in the paper are $\mathrm{EPU}_a = U_a E_a P_a$ (the Baker, Bloom, and Davis triple), $\mathrm{EPU}_a^{-F} = \mathrm{EPU}_a(1 - F_a)$ (non-fiscal EPU), $\mathrm{MPU}_a = U_a E_a M_a$ (monetary policy uncertainty, as in Husted et al., 2020), $\mathrm{TPU}_a = U_a E_a T_a$ (trade policy uncertainty, as in Caldara et al., 2020), $\mathrm{SDU}_a = U_a S_a$ (sovereign-debt uncertainty), and $\mathrm{FA}_a = E_a F_a$ (fiscal attention, an article about fiscal matters whether or not it mentions uncertainty).

The uncertainty lexicon $\mathcal{L}_U$ is the single theme under which GDELT records the Baker, Bloom, and Davis uncertainty terms ("uncertain" and "uncertainty"), $\mathcal{L}_E$ is the theme recording their economy terms, and $\mathcal{L}_P$ is the family of themes recording their policy terms. The fiscal lexicon is the design choice specific to this paper. I use three nested versions of it, so that the results in the paper can be bracketed. The narrow lexicon $\mathcal{L}_{F^-}$ (4 themes) consists of the categorical EPU terms of Baker et al. (2016) for taxes, government spending, entitlement programs, and debt-ceiling and government-shutdown episodes, as GDELT records them. The headline lexicon $\mathcal{L}_F$ (20 themes) adds three groups of themes: their sovereign-debt category, GDELT's own fiscal policy themes (tax, budget, spending, deficit, fiscal policy, ministry of finance, and public investment), and the taxation, budget-deficit, sovereign-debt, and austerity topics of GDELT's economic taxonomy. The broad lexicon $\mathcal{L}_{F^+}$ (40 themes) adds 20 public-finance topics from the World Bank taxonomy that GDELT also applies (fiscal consolidation, fiscal sustainability, government expenditure, tax policy, debt management, and

[2]The remaining fields of the record (the extended theme and location lists with character offsets, the Global Content Analysis Measures, and the extras) are never read. They are an order of magnitude larger than the fields used, and reading them would raise the cost of the computation in Section B.9 by the same factor without changing any count.

subsidies among them). Because $\mathcal{L}_{F^-} \subset \mathcal{L}_F \subset \mathcal{L}_{F^+}$, every article counted by the narrow lexicon is counted by the headline, and every article counted by the headline is counted by the broad lexicon.

Two further variants widen the other two lists. The lexicon $\mathcal{L}_{U^+}$ adds GDELT's broader theme for speculative and uncertain language, and $\mathcal{L}_{E^+}$ adds GDELT's broader economy theme. The headline index uses U, E, and F. Themes that mention debt without a public-sector qualifier are excluded from every fiscal lexicon, because household and corporate debt dominate their counts.[3]

B.4 Attribution to countries and views

Two attributions are computed for every article and kept separately. The outlet country $s(a)$ is the country in which the publishing outlet is located. It is obtained from the outlet's domain by applying GDELT's list of 189,545 outlet domains with their countries, and then the country-code top-level domain (244 codes) for outlets not on the list.[4] The mention set $\mathcal{M}(a)$ is the set of countries of the geocoded locations in the record. An article mentioning three cities in one country contributes once to that country.

Let $\mathcal{A}_\tau$ denote the articles dated in month τ. For country c, four views of the press are defined:

$$\begin{aligned} \mathcal{A}^D_{c\tau} &= \{a \in \mathcal{A}_\tau : s(a) = c\}, \qquad \mathcal{A}^X_{c\tau} = \{a \in \mathcal{A}_\tau : c \in \mathcal{M}(a),\ s(a) \notin \{c, \varnothing\}\}, \\ \mathcal{A}^A_{c\tau} &= \{a \in \mathcal{A}_\tau : c \in \mathcal{M}(a)\}, \qquad \mathcal{A}^W_\tau = \mathcal{A}_\tau, \end{aligned} \tag{B.3}$$

where D (domestic) is every article published by an outlet located in c, whatever it mentions; X (foreign, or external) is every article published elsewhere that mentions c; A (all sources) is every article that mentions c; and W (world) is every article.

The domestic view is the GDELT counterpart of a national newspaper index in the sense of Baker et al. (2016). It measures fiscal uncertainty as it is discussed in the country's own press, and I take it as the domestic uncertainty measure in the paper. The foreign view measures the external narrative about the country, which is what a foreign lender or a rating agency reads. I use it for the foreign-minus-domestic gap. The all-sources view is the widest and is the only one with usable counts for the smallest economies. The world view gives the global component. Table B.2 summarizes the views. The difference $\mathcal{A}^A_{c\tau} \setminus (\mathcal{A}^X_{c\tau} \cup \{a \in \mathcal{A}^D_{c\tau} : c \in \mathcal{M}(a)\})$ consists of articles mentioning c from outlets with no mapped country. Its share is reported in Section B.8.

Three restrictions of the views are used in the robustness checks. First, the language of the article $\ell(a)$ (English, or the source language recorded in the translation field) splits the domestic view into a native-language and an English-language press. Second, a stable outlet panel $\mathcal{O}^P$, the set of outlets with at least 50 articles in every year from 2016 to 2025, restricts every view to outlets that were present throughout the sample, so that changes in GDELT's source list do not enter the index. Third, a focus indicator $\text{foc}_a = \mathbb{1}\{1 \le |\mathcal{M}(a)| \le 2\}$ restricts the all-sources view to articles that mention at most two

[3] The excluded themes are the generic debt topics of the economic and World Bank taxonomies and the macroeconomic-vulnerability-and-debt topic. Each was inspected in GDELT's theme inventory, which reports the number of articles ever tagged with each theme.

[4] A generic country-code domain (`.co`, `.me`, `.tv`, and 11 others) is accepted only when the second-level label is one of 15 national conventions (`com`, `gov`, `gob`, `gouv`, `org`, `edu`, `ac`, and the like), so that `eltiempo.com.co` is Colombian but `t.co` is not. Outlets that neither step maps have $s(a) = \varnothing$. In a sample of 70,559 web articles from the raw GKG files (one fifteen-minute file per year from 2016 to 2026 in each feed), 96% to 100% of the articles in every major source language have a mapped outlet country. The share is 98.4% in English, 98.8% in Spanish, 99.9% in Turkish, 99.2% in Chinese, and 95.7% in French.

countries, which separates articles about a country from articles that list it.

B.5 Counts and annual aggregation

For each view $v \in \{D, X, A, W\}$, the monthly counts are the number of articles in the view and the number of fiscal policy uncertainty articles among them:

$$N_{c\tau}^{v} = |\mathcal{A}_{c\tau}^{v}|, \qquad K_{c\tau}^{v} = \sum_{a \in \mathcal{A}_{c\tau}^{v}} \mathrm{FPU}_{a}. \tag{B.4}$$

The same counts are formed for every companion classification and every lexicon variant of Section B.3, and for every restriction of Section B.4, in the same pass over the data.

Three annual windows are used. Let $\mathcal{T}_{c}^{b}(t)$ denote the set of months assigned to year t of country c under basis $b \in \{\mathrm{cal}, \mathrm{fy}, \mathrm{pb}\}$. Under the calendar basis, $\mathcal{T}_{c}^{\mathrm{cal}}(t)$ is the twelve months of calendar year t. Under the fiscal-year basis, $\mathcal{T}_{c}^{\mathrm{fy}}(t)$ is the twelve months of the fiscal year that the IMF World Economic Outlook (WEO) labels t for country c. The window begins in the first month of the fiscal year, m_c, of calendar year t when the WEO labels fiscal years by their starting year and of calendar year $t-1$ when it labels them by their ending year. India's fiscal year April t to March $t+1$ is labelled t, for example, and Egypt's fiscal year July $t-1$ to June t is labelled t. The starting month and the labelling convention were taken for each of the 197 WEO economies from the series metadata of the WEO database (the reporting-year months and the fiscal-year notes of the fiscal series). Thirty-two economies have a non-calendar fiscal year.

Under the pre-budget basis, $\mathcal{T}_{c}^{\mathrm{pb}}(t)$ is the six months before fiscal year t begins, which is the season in which the budget for year t is prepared and debated. For a calendar-year country it is July to December of $t-1$. The annual counts and the raw share are sums over the window and their ratio:

$$N_{ct}^{v,b} = \sum_{\tau \in \mathcal{T}_{c}^{b}(t)} N_{c\tau}^{v}, \qquad K_{ct}^{v,b} = \sum_{\tau \in \mathcal{T}_{c}^{b}(t)} K_{c\tau}^{v}, \qquad \hat{\pi}_{ct}^{v,b} = \frac{K_{ct}^{v,b}}{N_{ct}^{v,b}}. \tag{B.5}$$

The share is a ratio of sums rather than a mean of monthly shares, so that months are weighted by the volume of articles and a month with few articles cannot dominate the year of a small economy. A window is complete when every one of its months lies inside the data window. Incomplete windows (the fiscal years that begin before January 2016 and the windows that extend into the current year) are retained with a flag and excluded from the base period. The fiscal-year and pre-budget series carry the WEO fiscal-year label. They therefore align with the fiscal variables of the panel without further adjustment.

B.6 Shrinkage and scaling

Fiscal policy uncertainty articles are rare. In the full sample, 2.2 articles per 1,000 carry the classification in equation (B.2), against 4.4 per 1,000 for EPU and 12.9 per 1,000 for any mention of uncertainty. For an economy whose domestic press contributes a few thousand articles a year, the raw share in equation (B.5) is a count of a few articles divided by a few thousand. Its sampling variation is then of the same order as the variation the index is meant to measure. Following the empirical Bayes approach of Efron and Morris (1975), I therefore treat the annual share as a draw from a beta-binomial model

Table B.1. Theme lexicons used to classify articles

Lexicon	Indicator	GDELT themes (exact tags)
Uncertainty	U_a	EPU_UNCERTAINTY
Uncertainty, broad	U_a^+	EPU_UNCERTAINTY, USPEC_UNCERTAINTY1
Economy	E_a	EPU_ECONOMY
Economy, broad	E_a^+	EPU_ECONOMY, EPU_ECONOMY_HISTORIC
Policy	P_a	EPU_POLICY and every EPU_POLICY_* theme
Fiscal, narrow	F_a^-	EPU_CATS_TAXES, EPU_CATS_FISCAL_POLICY, EPU_CATS_ENTITLEMENT_PROGRAMS, EPU_CATS_DEBT_CEILING_GOV_SHUTDOWN
Fiscal (headline)	F_a	the narrow lexicon and EPU_CATS_SOVEREIGN_DEBT_CURRENCY_CRISES, EPU_POLICY_TAX, EPU_POLICY_BUDGET, EPU_POLICY_BUDGET_OUTFLOW(S), EPU_POLICY_SPENDING, EPU_POLICY_NATIONAL_SPENDING, EPU_POLICY_DEFICIT, EPU_POLICY_FISCAL_POLICY, EPU_POLICY_MINISTRY_OF_FINANCE, EPU_POLICY_PUBLIC_INVESTMENT, ECON_TAXATION, ECON_BUDGET_DEFICIT, ECON_SOVEREIGN_DEBT, AUSTERITY, WB_1121_TAXATION
Fiscal, broad	F_a^+	the headline lexicon and WB_713_PUBLIC_FINANCE, WB_445_FISCAL_POLICY, WB_1072_FISCAL_POLICY_AND_GROWTH, WB_1074_FISCAL_CONTRACTION, WB_716_MANAGING_PUBLIC_FINANCES, WB_983_TAX_POLICY, WB_720_TAX_AND_REVENUE_POLICY_AND_ADMINISTRATION, WB_1122_TAX_EXPENDITURES, WB_1073_FISCAL_STIMULUS_AND_FISCAL_RULES, WB_1116_FISCAL_CONSOLIDATION, WB_1097_FISCAL_SUSTAINABILITY, WB_449_FISCAL_RISKS, WB_1112_GOVERNMENT_EXPENDITURE, WB_1115_EXPENDITURE_POLICY, WB_718_PUBLIC_INVESTMENT_MANAGEMENT, WB_451_DEBT_MANAGEMENT, WB_452_DEBT_RELIEF, WB_454_DEBT_RESTRUCTURING, WB_722_FISCAL_TRANSPARENCY, ECON_SUBSIDIES
Monetary	M_a	EPU_CATS_MONETARY_POLICY, EPU_POLICY_CENTRAL_BANK, EPU_POLICY_INTEREST_RATE(S), EPU_POLICY_MONETARY_POLICY, EPU_POLICY_FEDERAL_RESERVE, EPU_POLICY_BANK_OF_JAPAN, EPU_POLICY_PEOPLE(S)_BANK_OF_CHINA, EPU_POLICY_BANK_OF_KOREA
Trade	T_a	EPU_CATS_TRADE_POLICY, EPU_POLICY_WORLD_TRADE_ORGANIZATION
Sovereign debt	S_a	EPU_CATS_SOVEREIGN_DEBT_CURRENCY_CRISES, ECON_SOVEREIGN_DEBT

Notes: Entries are the GDELT GKG theme tags (V1 `Themes` field) whose presence sets the indicator in equation (B.1); a tag in parentheses marks a spelling variant that GDELT also uses. EPU_* tags record the Baker, Bloom, and Davis term lists as applied by GDELT, ECON_* tags are GDELT's economic taxonomy, and WB_* tags are the World Bank topical taxonomy. The headline index uses U_a, E_a, and F_a.

Table B.2. Views of the press for country c

View	Symbol	Articles counted	Interpretation and use
Domestic	D	published by outlets located in c	national press; domestic uncertainty
Domestic, own	D°	published in c and mentioning c	national press about the country itself; robustness
Foreign	X	published outside c, mentioning c	external narrative; foreign-minus-domestic gap
All sources	A	every article mentioning c	widest coverage; small economies
World	W	every article	global component

Notes: Views are defined in equation (B.3). Each view is also computed on the stable outlet panel (outlets with at least 50 articles in each year 2016–2025); the domestic views are also split by the language of the article. Outlets without a mapped country enter view A only.

with a country-specific prior mean:

$$K_{ct}^{v,b} \mid \pi_{ct}^{v,b} \sim \text{Binomial}(N_{ct}^{v,b}, \pi_{ct}^{v,b}), \qquad \pi_{ct}^{v,b} \sim \text{Beta}(\kappa\, \bar{\pi}_{c}^{v,b},\ \kappa\,(1-\bar{\pi}_{c}^{v,b})), \tag{B.6}$$

where $\bar{\pi}_{c}^{v,b} = \sum_{t\in\mathcal{B}} K_{ct}^{v,b} / \sum_{t\in\mathcal{B}} N_{ct}^{v,b}$ is the country's pooled share over the base period $\mathcal{B}$ (the complete years from 2016 to 2025) and κ is the precision of the prior, measured in articles. The posterior mean of the share is

$$\tilde{\pi}_{ct}^{v,b} = \frac{K_{ct}^{v,b} + \kappa\, \bar{\pi}_{c}^{v,b}}{N_{ct}^{v,b} + \kappa}, \tag{B.7}$$

a weighted average of the raw share and the country's own mean with weight $N_{ct}^{v,b}/(N_{ct}^{v,b}+\kappa)$ on the raw share.

The precision is estimated once per series by the method of moments,

$$\hat{\kappa} = \frac{\overline{\bar{\pi}_c(1-\bar{\pi}_c)}}{\overline{(\hat{\pi}_{ct}-\bar{\pi}_c)^2} - \overline{\bar{\pi}_c(1-\bar{\pi}_c)/N_{ct}}} - 1, \tag{B.8}$$

where the overbars denote averages over the country-years of the series with at least 100 articles in the denominator, the view and basis superscripts are suppressed, and $\hat{\kappa}$ is restricted to the interval $[1, 10^7]$. The numerator is the average binomial variance of a share at the prior mean, and the denominator is the dispersion of the raw shares around their country means net of the part that binomial sampling alone would produce. For the headline domestic series, $\hat{\kappa} = 488$ articles. An economy with 2,000 domestic articles in a year, the threshold below which a country-year is flagged thin, therefore keeps a weight of 0.80 on its own share, and the shrinkage is negligible for the large economies, whose domestic press contributes hundreds of thousands of articles a year. The world series is not shrunk. It has no cross-section.

The prior mean in equation (B.6) is the country's own mean rather than the cross-country mean of the year. The choice matters for the paper, which separates a global and a domestic component of uncertainty. Shrinking toward the year's cross-country mean would move the domestic series of the small economies toward the global one and manufacture comovement where the data are merely thin. Shrinking toward the country's own mean leaves the cross-country differences in level untouched and only damps the year-to-year noise of the thin series. Shares shrunk toward the year's mean and unshrunk shares are produced as well and are used in the robustness checks.

Two scalings of the shrunk share are used:

$$\text{FPU}_{ct}^{v,b} = 100\, \frac{\tilde{\pi}_{ct}^{v,b}}{\bar{\tilde{\pi}}_{c}^{v,b}}, \qquad \text{fpu}_{ct}^{v,b} = \frac{\ln \tilde{\pi}_{ct}^{v,b} - \mu_{c}^{v,b}}{\sigma_{c}^{v,b}}, \tag{B.9}$$

where $\bar{\tilde{\pi}}_{c}^{v,b}$ is the base-period mean of the shrunk share and $\mu_{c}^{v,b}$ and $\sigma_{c}^{v,b}$ are the base-period mean and standard deviation of its logarithm within the country. The index FPU equals 100 on average over 2016–2025 in every country. It is the form used in the figures. The standardized form fpu is the form used in the regressions, where one unit is one within-country standard deviation of the log share, so that coefficients are comparable across countries whose press differs in size and in the base rate

of fiscal coverage. The foreign-minus-domestic gap is the difference of the two standardized series, $\Delta^{XD}_{ct} = \text{fpu}^{X}_{ct} - \text{fpu}^{D}_{ct}$, on the calendar basis. Throughout, a country-year is flagged thin when the view contains fewer than 2,000 articles or fewer than 20 fiscal policy uncertainty articles. The flag is a sample filter, not an adjustment.

B.7 Variables used in the paper

Table B.3 lists the series that enter the paper, with the symbol of this appendix, its variable name, and the role of the series. The headline series are the domestic, foreign, and all-sources indices and the world index, on the calendar basis. Every other series is obtained by one substitution. A lexicon variant replaces F_a, U_a, or E_a in equation (B.2). A construction variant replaces FPU_a by $U_a F_a$ (no economy requirement), replaces N in equation (B.4) by the count of fiscal-economic articles (the intensity series, FPU_a over FA_a at the level of counts), or restricts the view by foc_a. A composition variant restricts the view to the stable panel, to native-language articles, or to the own-country subset of the domestic view. A timing variant changes the basis b, and a companion series replaces FPU_a by one of the classifications of Section B.3. Each substitution then passes through equations (B.4) to (B.9) unchanged, with its own $\hat{\kappa}$. The mean GDELT tone of fiscal-economic articles is carried as a sentiment control. An increase in fiscal uncertainty and a turn toward negative coverage are distinct events, and the paper does not want to confound them.

B.8 Validation

Five checks were made on the index before use, and Table B.4 reports them. First, the monthly domestic EPU and FPU shares were compared with the 20 published national EPU indices that cover the sample, those of Baker et al. (2016) and of the studies that apply their method to other economies (the notes to Table B.4 list the sources). Months that the publishers impute are excluded, which leaves 2,492 country-months. The correlation of the monthly EPU share with the published index has a median of 0.74 across the 20 economies. It is positive in every economy and ranges from 0.33 for Greece to 0.93 for the United States. The agreement is not a shared trend, because the correlation of 12-month changes in the logarithms of the two series has a median of 0.68. The correlation of the FPU share with the published index, which measures a narrower object, has a median of 0.73. At the world level, the EPU share of view W correlates at 0.91 with the global EPU index of Davis (2016) over the 127 months from January 2016 to July 2026. Second, the annual domestic series was compared with the WUI in the panel of the paper, for the 1,421 country-years of 143 economies in which both exist. The pooled correlation is 0.28, the median within-country correlation is 0.36, and the within-country correlation is positive in 76 percent of the 142 economies for which it can be computed.

Third, the largest monthly values of the domestic series were read against the fiscal calendar for nine economies (Turkey, the United States, the United Kingdom, Brazil, Argentina, South Africa, India, Germany, and Egypt). The largest months of the series are those of the episodes that the news coverage of each economy would lead one to expect. For the United States, Germany, and Brazil, March to May 2020 is among the five largest months. April 2025, the month of the United States tariff announcements, is the largest or second-largest month in seven of the nine economies (all but Argentina and Egypt) and is among the five largest for Argentina. For the United Kingdom, July 2016 follows the referendum on membership of the European Union, and November 2016 coincides with the United States election

Table B.3. Series used in the paper and their names

Series	Symbol	Variable	Definition and role
Headline (calendar basis)			
Domestic FPU, index	FPU_{ct}^{D}	`fpu_dom_idx`	view D, equation (B.9); figures
Domestic FPU, standardized	fpu_{ct}^{D}	`fpu_dom_z`	regressions, domestic uncertainty
Foreign FPU	FPU_{ct}^{X}, fpu_{ct}^{X}	`fpu_for_idx`, `fpu_for_z`	view X; external narrative
All-sources FPU	FPU_{ct}^{A}, fpu_{ct}^{A}	`fpu_all_idx`, `fpu_all_z`	view A; small economies
World FPU	FPU_{t}^{W}, fpu_{t}^{W}	`fpu_glob_idx`, `fpu_glob_z`	view W; global component
Narrative gap	Δ_{ct}^{XD}	`fpu_gap_z`	$\mathrm{fpu}_{ct}^{X} - \mathrm{fpu}_{ct}^{D}$
Timing variants (headline, views D, X, A)			
Fiscal-year basis	$\mathrm{fpu}_{ct}^{v,\mathrm{fy}}$	`fpu_*_z_fy`	WEO fiscal year t
Pre-budget basis	$\mathrm{fpu}_{ct}^{v,\mathrm{pb}}$	`fpu_*_z_pb`	six months before fiscal year t
Lexicon variants (views D and A)			
Broad uncertainty	$\mathrm{fpu}_{ct}^{v}(U^{+})$	`fpu_ub_*_z`	$U_a^{+}E_aF_a$
Broad economy	$\mathrm{fpu}_{ct}^{v}(E^{+})$	`fpu_eb_*_z`	$U_aE_a^{+}F_a$
Narrow fiscal	$\mathrm{fpu}_{ct}^{v}(F^{-})$	`fpu_fn_*_z`	$U_aE_aF_a^{-}$
Broad fiscal	$\mathrm{fpu}_{ct}^{v}(F^{+})$	`fpu_fb_*_z`	$U_aE_aF_a^{+}$
Construction variants			
No economy requirement	$\mathrm{fpu}_{ct}^{D}(\neg E)$	`fpu_noe_dom_z`	U_aF_a
Focused coverage	$\mathrm{fpu}_{ct}^{A}(\mathrm{foc})$	`fpu_foc_all_z`	articles mentioning at most two countries
Intensity	$\mathrm{fpu}_{ct}^{v}(\mathrm{FA})$	`fpu_intensity_*_z`	FPU articles over fiscal-economic articles
Fiscal attention	FA_{ct}^{v}	`fiscal_attention_*_z`	fiscal-economic articles over all articles
Composition and language variants			
Stable outlet panel	$\mathrm{fpu}_{ct}^{v,P}$	`fpu_*_pnl_z`	outlets present in every year 2016–2025
Native-language press	$\mathrm{fpu}_{ct}^{D,\ell}$	`fpu_dom_nat_z`	domestic, non-English articles
Own-country domestic	$\mathrm{fpu}_{ct}^{D^{\circ}}$	`fpu_dom_own_z`	domestic articles mentioning the country
Companion series (views D, X, A, W as available)			
Economic policy uncertainty	EPU_{ct}^{v}	`epu_*_z`	$U_aE_aP_a$; horse race
Non-fiscal EPU	$\mathrm{EPU}_{ct}^{-F,v}$	`epu_nf_*_z`	$\mathrm{EPU}_a(1 - F_a)$; falsification check
Monetary, trade, sovereign debt	MPU, TPU, SDU	`mpu_*_z`, `tpu_*_z`, `sdu_*_z`	controls
Tone of fiscal coverage	tone_{ct}^{v}	`tone_fe_*`	mean GDELT tone of $\mathrm{FA}_a = 1$ articles
Diagnostics			
Counts and flags	N_{ct}^{v}, K_{ct}^{v}	`n_all_*`, `n_fpu_*`, `thin_*`, `complete_*`	sample filters

Notes: In variable names, `*` stands for the view tag (`dom`, `for`, `all`, `glob`), and the suffixes `_share`, `_idx`, and `_z` give the shrunk share, the index, and the standardized log share of equation (B.9). Every series is also available as a share and as an index. The counts, the raw share, the shrunk share, and $\hat{\kappa}$ are available for every basis, view, sample, language, and classification.

and the Autumn Statement. For Argentina, September 2018, August and September 2019, and July 2022 coincide with the agreement with the International Monetary Fund, the loss of the primary election and the capital controls that followed it, and the change of finance minister. For South Africa, March and April 2025 coincide with the postponed 2025 budget and the withdrawal of its increase in value-added tax. The high months of 2026 in India and South Africa are not an artifact of outlet composition. The stable-panel series has the same peaks, and the domestic share rises by more than the world share over the same months, to about twice its usual ratio to the world share in India and about 1.4 times in South Africa. Both fall in the budget season. In India the run of high months begins in January 2026, the month before the Union Budget, and January has the highest average value of the Indian series over 2016–2024. In South Africa the 2026 peaks are March and April, after the February budget, and February has the highest average value of the South African series over the same years. For Egypt, by contrast, the high months from October 2025 to April 2026 are an artifact of composition. English-language outlets outside the stable panel rose from about 1,000 articles a month in the third quarter of 2025 to between 23,000 and 29,000 a month from October 2025 to April 2026, and they account for most of the Egyptian FPU articles of those months. The stable-panel series for Egypt does not rise. Fourth, the variants of Table B.3 were correlated with the headline series over the complete country-years. The lexicon variants correlate with the headline at 0.77 to 0.98, the stable-panel series at 0.78, and the native-language series at 0.81. The non-fiscal EPU index, which shares the uncertainty and economy components of the headline but not the fiscal one, correlates at 0.68. Fifth, coverage and composition were tabulated. In the domestic view, 222 economies have at least eight complete years, the median economy contributes 41,017 domestic articles a year, 36.5 percent of the country-years in the domestic view are flagged thin, and 1.0 percent of the articles that mention a country come from outlets without a mapped country. The all-outlet and stable-panel domestic indices differ by more than a factor of two in 222 of the 2,042 country-years (10.9 percent), in 69 economies. Egypt in 2025 and 2026 is the only such case among the nine economies above. For an economy whose outlets change in this way, the stable-panel series $\mathrm{FPU}_{ct}^{D,P}$ is the one to use. Figure B.1 plots the domestic, foreign, and stable-panel domestic indices for the nine economies.

B.9 Computation

The counts in equation (B.4) are computed on Google BigQuery, where the GDELT Project publishes the GKG as a public table partitioned by day. One query per calendar year reads the seven fields of every record in the year. It classifies each article by the lexicons of Table B.1, attributes it to its outlet country and its mentioned countries, and writes a compact table of daily counts by outlet country, source language, panel membership, and mentioned country. Every lexicon variant and companion classification is one more counting column of the same query. The robustness series therefore cost nothing beyond the headline.

The monthly views of equation (B.3) and the annual series of Sections B.5 to B.6 are then computed from the compact table. A second, cheaper query over the outlet domain alone defines the stable outlet panel. The main pass reads 1.16 TiB, because only the seven fields are read, the outlet pass 35.5 GiB, and the monthly views 31 GiB, 1.23 TiB in all. The whole computation ran in less than four minutes at a cost of about US$8 at BigQuery's on-demand price of US$6.25 per TiB, before the free monthly allowance of 1 TiB.

Table B.4. Validation of the GDELT fiscal policy uncertainty index

Check	Comparison	Statistic	Value
Published EPU (monthly)	domestic EPU share vs. national EPU index	median correlation, 20 economies	0.74
		range of correlations	0.33–0.93
	12-month log changes of the same	median correlation	0.68
	domestic FPU share vs. national EPU index	median correlation	0.73
	world EPU share vs. global EPU index	correlation, 127 months	0.91
WUI (annual)	fpu_{ct}^{D} vs. WUI	pooled correlation, 1,421 country-years	0.28
		share of economies with positive within-country correlation	0.76
Variants	lexicon variants vs. headline	range of correlations	0.77–0.98
	stable panel vs. headline	correlation	0.78
	native-language vs. headline	correlation	0.81
	non-fiscal EPU vs. headline	correlation	0.68
Coverage	economies with ≥ 8 complete years, view *D*	count	222
	domestic articles per economy-year	median	41,017
	thin country-years, view *D*	share	0.365
	all-outlet and stable-panel index differ by a factor above 2, view *D*	share of country-years	0.109
	articles from unmapped outlets, view *A*	share	0.010

Notes: Entries are correlations, counts, and shares. The monthly comparisons run from January 2016 to the last month of each published index (July 2026 at the latest) and exclude the months that the publishers impute. The annual comparisons use the complete country-years of the calendar basis. The national EPU indices, as distributed at policyuncertainty.com, are those of Baker et al. (2016) for Australia, Brazil, Canada, France, Germany, India, Italy, Korea, Mexico, Russia, the United Kingdom, and the United States, Cerda et al. (2016) for Chile, Davis et al. (2019) for China, Hardouvelis et al. (2018) for Greece, Zalla (2016) for Ireland, Arbatli et al. (2017) for Japan, Choudhary et al. (2020) for Pakistan, Ghirelli et al. (2019) for Spain, and Davis (2016) for Singapore. The global EPU index is that of Davis (2016), with current-price GDP weights. The WUI is the quarterly index of Ahir et al. (2022), averaged to the calendar year.

Two properties of the classification were checked on a sample of raw GKG records, one fifteen-minute file per year from 2016 to 2026 in each of the two feeds (70,559 web articles). Every article that mentions a country falls in exactly one of the domestic-own, foreign, and unmapped subsets of view *A*, and every article counted under a narrower fiscal lexicon is also counted under the broader ones.

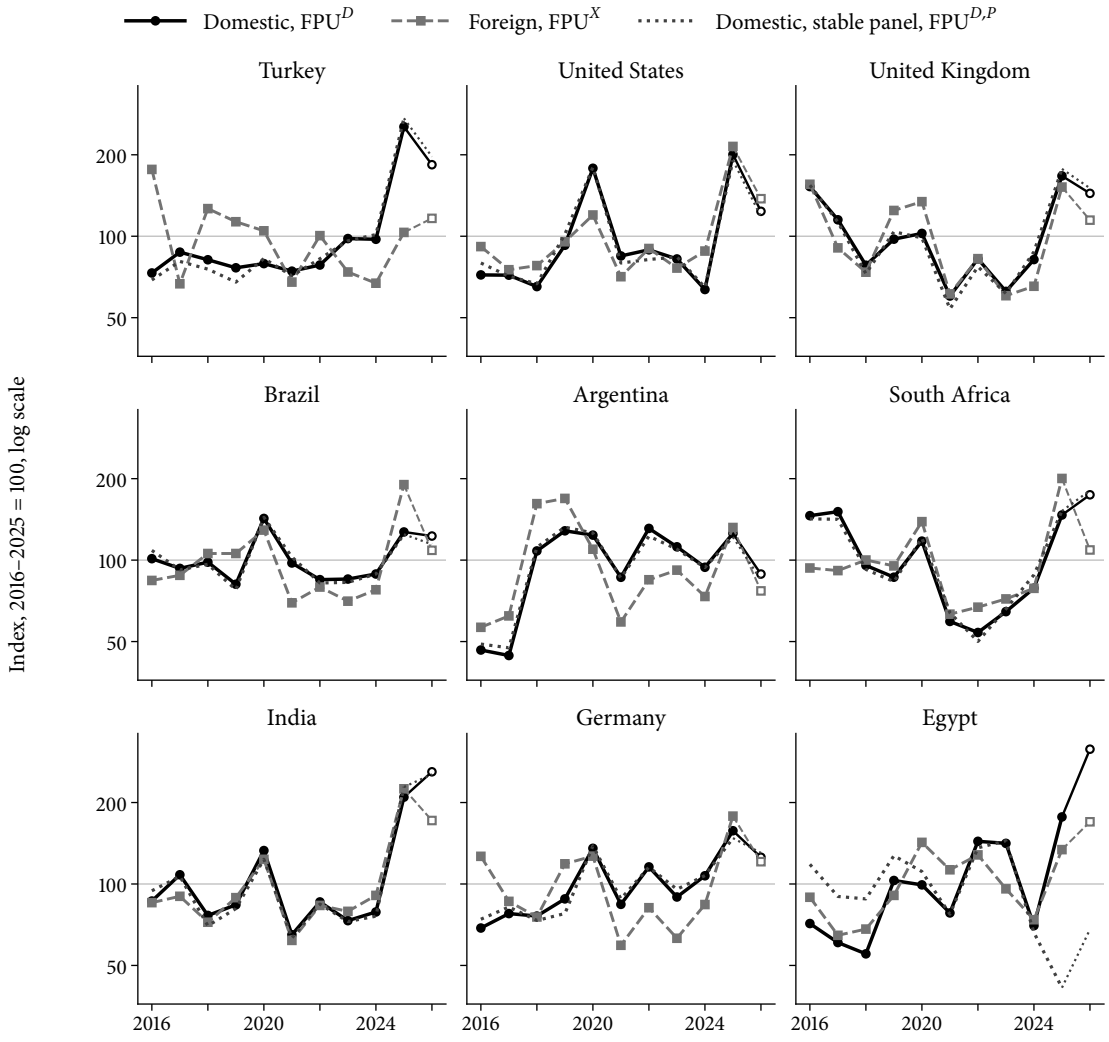


Figure B.1. Domestic and foreign fiscal policy uncertainty, nine economies, 2016–2026

Notes: Each panel plots FPU^D_{ct} (solid), FPU^X_{ct} (dashed), and the domestic index on the stable outlet panel, $\text{FPU}^{D,P}_{ct}$ (dotted), on the calendar basis of equation (B.9), with the 2016–2025 mean equal to 100 and a logarithmic vertical scale. The 2026 value (hollow marker) is computed on January to September and is flagged as incomplete in the data. For Egypt, the rise of FPU^D_{ct} in 2025 and 2026 comes from outlets outside the stable panel (Section B.8).

C Estimation details, additional results and robustness

C.1 Estimation details

This section records the choices that Section 2 refers to this appendix for. The conventions are the same in every panel model: the fixed effects the equation lists, Driscoll–Kraay standard errors with a Bartlett kernel and two lags, 68% and 90% bands from those standard errors, uncertainty regressors standardised within the main sample, and fiscal space standardised within the sample of the model in which it is interacted.

The global shock is the standardised innovation of the autoregression of Appendix A.1, fitted once over 34 years and held fixed across models. In the decomposition model the cumulative sums run over years that all belong to the decomposition sample, so the outlier rule applies to every year of a window, and the sample is 3,367 country-years at $h = 0$ and 2,178 at $h = 5$, in 132 economies. Fiscal space in that model is lagged debt standardised with a mean of 52 and a standard deviation of 32 percent of GDP, read at its 25th and 75th percentiles, 31 and 66 percent of GDP. The five component responses add to the debt response at every horizon, because the samples are identical component by component.

The reaction function was estimated by two-stage least squares with the output gap and its product with uncertainty instrumented by world real growth excluding the economy itself, by its product with standardised exposure, and by the lagged gap, each alone and multiplied by $U_{i,t}$. The first-stage statistics reported are the smallest Driscoll–Kraay-robust Wald F across the instrumented variables and the Cragg–Donald statistic of Cragg and Donald (1993), since the rank statistic of Kleibergen and Paap is not computable under the Driscoll–Kraay covariance in the software used. In the pooled column the robust F is 43.8 and the Cragg–Donald statistic 82.9. The debt limits of Appendix C.2.4 were computed from the group-specific cubic of the second column, with the country effect, the controls at country means and the gap at zero, as the largest positive root at which the long-run reaction crosses the interest–growth line from above, with bands from a block bootstrap over economies with 500 draws.

The cyclicality model has four endogenous gap terms and twelve instruments, which gives a Cragg–Donald statistic of 9.5 in the pooled column. The inference of record therefore uses the generated instruments of Wooldridge (2010): the fitted gap from the first stage on the three base instruments, the exogenous regressors and the fixed effects, multiplied by $(1, U, \tilde{d}, U\tilde{d})$, $\tilde{d}$ the lagged debt ratio standardised within the sample. The estimator is just identified, with a robust first-stage F of 35.5 and a Cragg–Donald statistic of 24.2. Each base instrument alone has a robust first-stage F between 4.0 and 4.7 for the four terms, so a single base instrument is weak, and the lagged gap is the strongest of the three. The S sets for ϕ_{UF} invert the S statistic of Stock and Wright (2000), the objective of a two-step generalized method of moments (GMM) estimator with the Driscoll–Kraay weighting matrix, over a grid from -0.6 to 0.6 in steps of 0.0025, with the other three gap coefficients concentrated out at each grid point in the subset form of Kleibergen and Mavroeidis (2009), whose size is asymptotically no larger than nominal whatever the strength of the instruments; the robustness checks use steps of 0.005.

The smooth-transition model uses the logistic transition $F(z) = \exp(\kappa z)/(1 + \exp(\kappa z))$ with $\kappa = 1.5$ and the state $z_{i,t-1} = \tilde{v}_{i,t-1}$, the composite vulnerability index of the previous year standardised within the model's sample, so that F approaches one in the vulnerable regime. The weight on the vulnerable regime exceeds one half for 51% of the country-years. The global shock enters as its product

with standardised exposure and the domestic shock directly, and the controls are the state variable, the lower-order terms $F(z_{i,t-1})$, $F(z_{i,t-1})\varepsilon_t^{\mathrm{g}}$ and $F(z_{i,t-1})E_i$ of the regime interactions, and $X_{i,t-1}$. The sample is 2,845 country-years in 122 economies at $h = 0$. The insurance model uses the same transition with $z_{i,t-1} = -\tilde{d}_{i,t-1}$, the sign of standardised lagged debt reversed, so that F approaches one for the high-space economy, with $F(z_{i,t-1})$, $F(z_{i,t-1})\varepsilon_t^{\mathrm{g}}$ and $F(z_{i,t-1})E_i$ as controls, on 3,757 country-years in 129 economies. The panel threshold model of Hansen (1999) searches the threshold on lagged debt over a grid from the 10th to the 90th percentile in steps of one point of GDP, with the regime indicator and its products with the global shock and with exposure as the regime-level terms. The test of no threshold covers the two regime differences and the three regime-level terms and uses a fixed-regressor wild bootstrap clustered by economy with 300 draws.

The direct-effect specification of Table C.1 drops the year effects and adds world real growth excluding the economy, the global geopolitical risk index and the VIX as global controls. Its inference adds a wild cluster bootstrap by year with 999 Rademacher draws and the null imposed, since a regressor that varies by year only has as many independent observations as there are years. The pandemic cross-sections use heteroskedasticity-robust standard errors of the HC1 form, and in the event study the number of economies falls from 122 in the first year to 34 in 2025, because 2025 is an actual for few economies. Singleton observations within a fixed effect were dropped; the zero-filled indicators and the trimming of the effective rate are as stated in Appendix A.1.

C.2 Additional results for the main models

C.2.1 The debt-dynamics decomposition (M1). Table C.1 reports the direct-effect specification behind Figures 3 and 4, which drops the year effects so that the level of the response to a global shock is identified, at the price of attributing the common debt surges of 2009 and 2020 to the shock. Without year effects a one-standard-deviation global shock raises the debt ratio by 4.70 (standard error [SE] of 1.33) points of GDP after five years, and the interaction with space keeps its sign, 0.87 per standard deviation of debt against 0.64 with year effects. The primary-balance component carries most of the five-year level effect and the interest component offsets a little of it, the channel pattern of the main text. At the two-year horizon the wild bootstrap by year gives $p = 0.010$ for the level, so the level is rejected at the 5% level under the inference that counts years.

Figure C.1 repeats Figure 3 for the advanced economies and for the emerging market and developing economies separately, and Figure C.2 shows the debt response alone by income group. The two-year interaction is 1.81 (SE 0.48) points of GDP per standard deviation of debt in the advanced economies and 0.79 (SE 0.59) in the others, so the erosion result holds within each group; at five years the groups diverge within their standard errors, and the low-income economies show the widest and least precise gap.

Figure C.3 replaces the global shock by the domestic shock, which enters directly under the year effects. A one-standard-deviation domestic shock changes the debt ratio by 0.06 (SE 0.12) points of GDP after two years and by a similar amount after five, so domestic uncertainty leaves no mark on the debt ratio that the identity can attribute. Figure C.4 repeats Figure 3 with the pandemic-purged global shock. The two-year interaction is 1.15 (SE 0.31) against 1.64 with the headline shock, so the pandemic reading of 2020 accounts for part of the erosion but not for its sign or its timing.

Table C.1. Direct effect of the global uncertainty shock on the debt-dynamics components, no year fixed effects

	Cumulative response, pp of GDP					
	Interest	Real growth	Inflation	Primary balance	Stock-flow adj.	Change in debt
Horizon $h = 0$						
Global shock $\varepsilon_t^{\mathrm{g}}$ (sd), $\beta^{(h,\ell)}$	−0.0104	0.0355	0.288	−0.00148	−0.0246	0.287
	(0.0403)	(0.0854)	(0.118)	(0.112)	(0.0681)	(0.250)
	[0.82]	[0.65]	[0.03]	[0.99]	[0.76]	[0.17]
$\varepsilon_t^{\mathrm{g}} \times \tilde{d}_{i,t-1}$ (sd), $\theta^{(h,\ell)}$	−0.0276	0.347	−0.0525	−0.136	0.143	0.274
	(0.0521)	(0.225)	(0.0702)	(0.0752)	(0.0658)	(0.221)
	[0.68]	[0.05]	[0.51]	[0.10]	[0.16]	[0.16]
Observations	3,367	3,367	3,367	3,367	3,367	3,367
Economies	132	132	132	132	132	132
Horizon $h = 2$						
Global shock $\varepsilon_t^{\mathrm{g}}$ (sd), $\beta^{(h,\ell)}$	−0.0354	1.29	1.42	2.43	0.127	5.23
	(0.217)	(0.246)	(0.501)	(1.07)	(0.254)	(1.49)
	[0.87]	[0.00]	[0.04]	[0.14]	[0.61]	[0.01]
$\varepsilon_t^{\mathrm{g}} \times \tilde{d}_{i,t-1}$ (sd), $\theta^{(h,\ell)}$	0.00770	1.00	0.617	0.205	0.0252	1.86
	(0.311)	(0.205)	(0.202)	(0.439)	(0.236)	(0.337)
	[0.99]	[0.02]	[0.07]	[0.72]	[0.89]	[0.03]
Observations	2,766	2,766	2,766	2,766	2,766	2,766
Economies	129	129	129	129	129	129
Horizon $h = 5$						
Global shock $\varepsilon_t^{\mathrm{g}}$ (sd), $\beta^{(h,\ell)}$	−0.339	0.750	−0.225	4.39	0.125	4.70
	(0.363)	(0.455)	(0.920)	(1.14)	(0.183)	(1.33)
	[0.42]	[0.07]	[0.86]	[0.02]	[0.60]	[0.00]
$\varepsilon_t^{\mathrm{g}} \times \tilde{d}_{i,t-1}$ (sd), $\theta^{(h,\ell)}$	−1.11	0.882	−0.132	1.28	−0.0391	0.875
	(0.592)	(0.330)	(0.379)	(0.592)	(0.368)	(0.550)
	[0.24]	[0.07]	[0.69]	[0.16]	[0.91]	[0.03]
Observations	2,178	2,178	2,178	2,178	2,178	2,178
Economies	120	120	120	120	120	120

Notes: Entries are coefficients of the cumulative local projection $\sum_{k=0}^{h} c_{i,t+k}^{(\ell)} = \alpha_i^{(h,\ell)} + \beta^{(h,\ell)}\,\varepsilon_t^{\mathrm{g}} + \theta^{(h,\ell)}\,\varepsilon_t^{\mathrm{g}}\tilde{d}_{i,t-1} + \Gamma^{(h,\ell)\prime} X_{i,t-1} + \Psi^{(h,\ell)\prime} G_t + u_{i,t+h}^{(h,\ell)}$, in percentage points of GDP per standard deviation of the global uncertainty shock, without year fixed effects, where $c_{i,t}^{(\ell)}$ is contribution ℓ of the debt identity (the last column is their sum, the change in debt), k the summation index, $\alpha_i^{(h,\ell)}$ a country effect, $\varepsilon_t^{\mathrm{g}}$ the global uncertainty shock, $\beta^{(h,\ell)}$ its direct effect, $\theta^{(h,\ell)}$ the coefficient on its product with $\tilde{d}_{i,t-1}$, $\Gamma^{(h,\ell)}$ and $\Psi^{(h,\ell)}$ the coefficients on the lagged controls $X_{i,t-1}$ and on the global controls G_t, and $u_{i,t+h}^{(h,\ell)}$ the projection error at horizon h, serially correlated by construction and correlated across economies within a year. Driscoll–Kraay standard errors (lag 2) in parentheses; wild cluster bootstrap p-values by year (Rademacher, 999 draws, null imposed) in brackets. $\tilde{d}_{i,t-1}$ is lagged debt standardised within the sample (mean 51.8, sd 31.7% of GDP). G_t contains world real growth excluding the own economy, the geopolitical risk index and the VIX; the US three-month rate is not in the panel. Controls $X_{i,t-1}$: lagged real growth, deflator inflation, primary balance, debt, trade openness and commodity terms-of-trade growth. Sample 1992–2025.

C.2.2 Fiscal space at risk. Table C.2 gives the numbers behind Figure 4 for the six economies of Figure 2 and for the group medians: the debt a global shock adds at the economy's latest position, as a level and relative to the average economy, and the effective-rate response at its regime position, the exposure gradient of Figure 5 weighted by the economy's regime weight. The level of debt added after two years has a median of 6.01 points of GDP among the advanced economies against 5.08 among the low-income economies, and its 90% band excludes zero for 98% of economies. Among the six, Italy's level is the largest, 10.10 (SE 1.58) points of GDP, and Türkiye's the smallest, 3.58. The main text reads the relative column, since the level column inherits the attribution of the common surges to the shock.

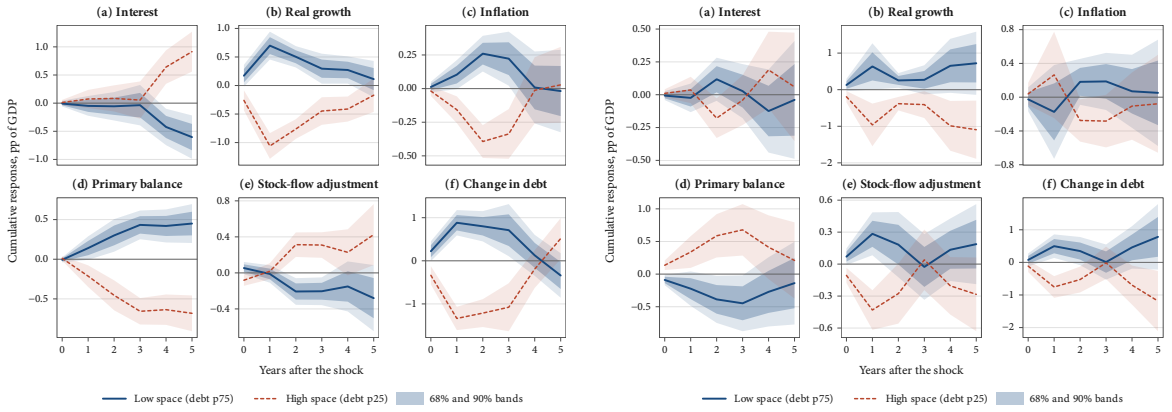


Figure C.1. Responses of the debt-dynamics components by country group

Notes: As Figure 3, estimated separately for advanced economies (panel set a, 31 economies, 933 country-years) and emerging market and developing economies (panel set b, 101 economies, 2,434 country-years); lagged debt $\tilde{d}_{i,t-1}$ standardised with the pooled sample's moments and evaluated at its p25 (31) and p75 (66% of GDP).

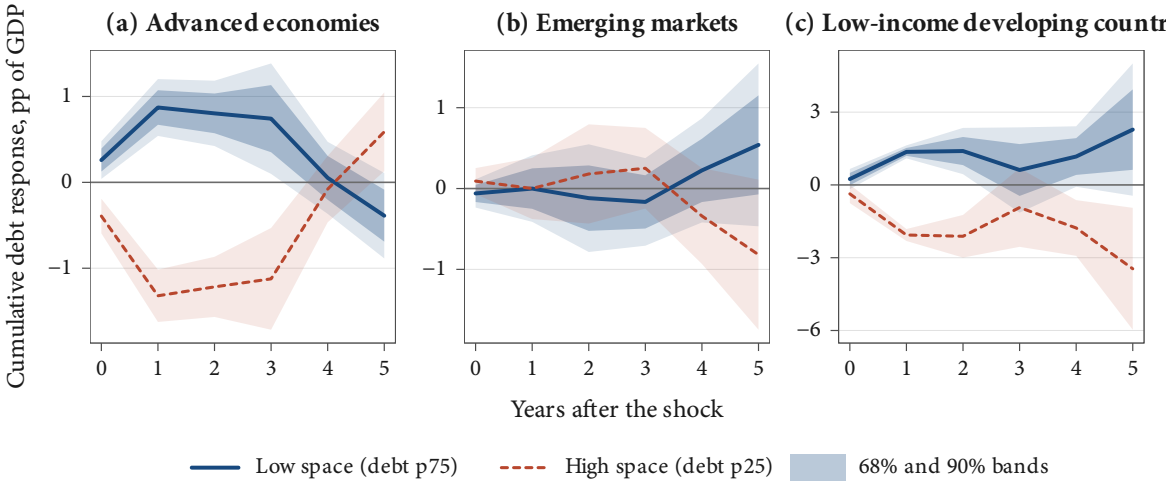


Figure C.2. Debt response to a global uncertainty shock by income group

Notes: The change-in-debt panel of Figure 3 estimated separately for AE, EM and LIDC (IMF WPUI classification); lagged debt $\tilde{d}_{i,t-1}$ standardised with the pooled moments and evaluated at the pooled p25 (31) and p75 (66% of GDP); DK bands (lag 2).

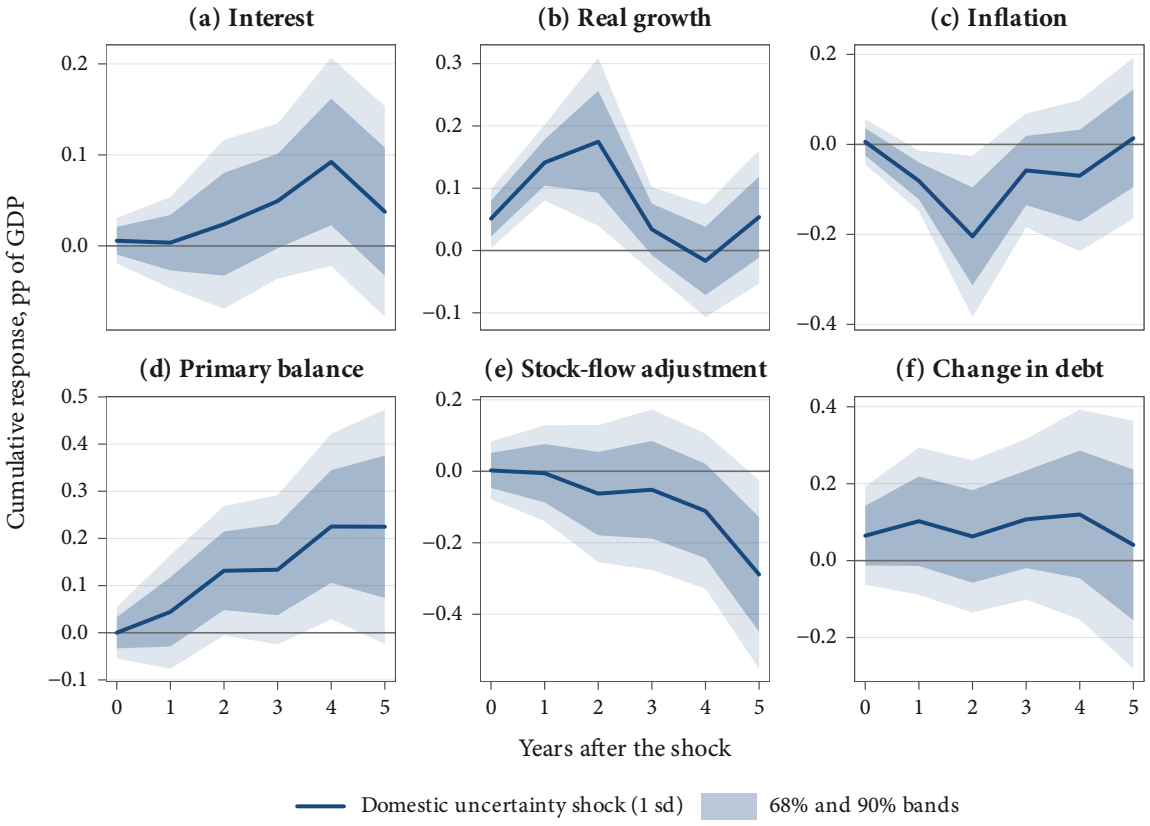


Figure C.3. Responses of the debt-dynamics components to a domestic uncertainty shock

Notes: Cumulative responses, in percentage points of GDP, to a one-standard-deviation domestic uncertainty shock (the residual of country WUI on its own lags and global WUI) from local projections with country and year fixed effects and lagged controls; DK bands (lag 2); 132 economies, 3,367 country-years at h = 0.

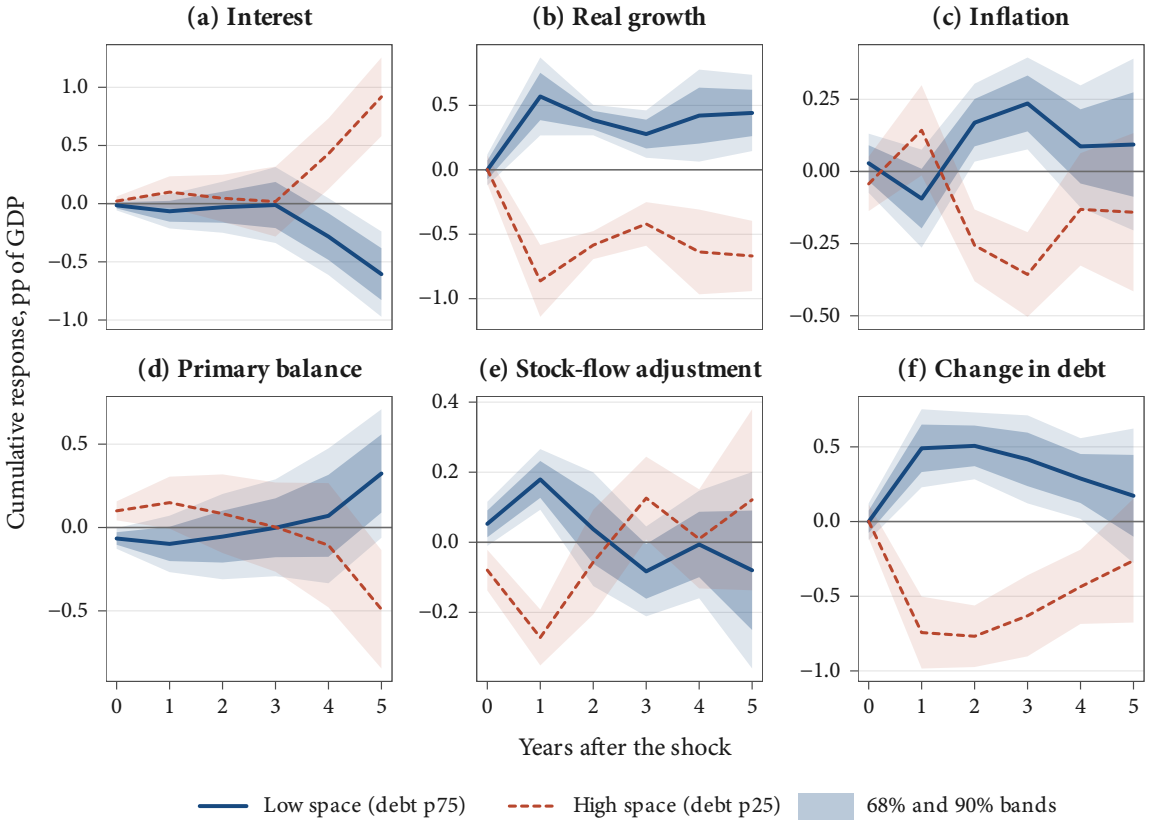


Figure C.4. Responses of the debt-dynamics components to a pandemic-purged global uncertainty shock

Notes: As Figure 3, with the global shock replaced by the standardised AR(2) residual of the pandemic-purged global WUI (eps_glob_np); 132 economies, 3,367 country-years at h = 0; DK bands (lag 2).

Table C.2. Fiscal space at risk: the six economies of Figure 2 and the group medians

			Debt added, pp of GDP				Regime responses	
	Debt_{t-1}, % of GDP	$\tilde{d}_i$ (sd)	$h = 2$	$h = 5$	$h = 2$, year effects	F	Interest rate, $h = 2$ (pp)	Primary spending, $h = 5$ (% of GDP)
Brazil	84.0	1.02	7.12 (1.46)	5.59 (1.66)	1.67 (0.36)	0.73	0.09 (0.04)	0.79 (0.28)
United Kingdom	99.9	1.52	8.06 (1.48)	6.03 (1.87)	2.49 (0.54)	0.51	−0.05 (0.05)	0.16 (0.22)
Italy	134.7	2.62	10.10 (1.58)	6.99 (2.36)	4.29 (0.92)	0.77	0.11 (0.04)	0.91 (0.31)
Türkiye	23.6	−0.89	3.58 (1.58)	3.92 (1.20)	−1.46 (0.31)	0.58	0.00 (0.04)	0.37 (0.21)
United States	122.3	2.22	9.37 (1.53)	6.64 (2.18)	3.65 (0.78)	0.85	0.16 (0.05)	1.14 (0.39)
South Africa	73.2	0.68	6.49 (1.46)	5.29 (1.54)	1.11 (0.24)	0.66	0.04 (0.04)	0.59 (0.23)
Advanced economies, median (28 economies)	64.9	0.42	6.01 (1.50)	5.06 (1.45)	0.68 (0.21)	0.42	−0.10 (0.06)	−0.10 (0.39)
Emerging markets, median (58 economies)	53.0	0.04	5.30 (1.49)	4.73 (1.34)	0.06 (0.18)	0.61	0.01 (0.04)	0.44 (0.26)
Low-income developing countries, median (46 economies)	49.2	−0.08	5.08 (1.50)	4.63 (1.32)	−0.13 (0.12)	0.73	0.09 (0.05)	0.80 (0.30)

Notes: Entries are the responses to a one-sd global uncertainty shock implied by the stage-2 estimates for each economy at its latest sample year; delta-method Driscoll–Kraay standard errors (lag 2) in parentheses. Debt added: cumulative change in the debt ratio (pp of GDP) from the direct-effect projection of Table C.1, $\beta^{(h,\ell)} + \theta^{(h,\ell)}\tilde{d}_i$, with $\tilde{d}_i$ the economy's lagged debt at its latest sample year standardised as in Figure 3 (mean 51.8, sd 31.7% of GDP) and $\beta^{(h,\ell)}$, $\theta^{(h,\ell)}$ the level and interaction coefficients of that projection for the change in debt; the year-effects column uses the slope of Figure 3 alone ($\theta^{(h,\ell)}\tilde{d}_i$, the response relative to the average economy). Regime responses: $F\,\mathbf{b}_{\mathrm{V}}^{(h)} + (1-F)\,\mathbf{b}_{\mathrm{S}}^{(h)}$ from the smooth-transition model of Figure 5, the exposure gradient (an economy one sd more open than the mean against the mean-exposure economy in the same regime) at the economy's regime weight F, the logistic transition $F(z_{i,t-1})$ at its latest standardised vulnerability (vulnerable near 1; $\kappa = 1.5$), with $\mathbf{b}_{\mathrm{V}}^{(h)}$ and $\mathbf{b}_{\mathrm{S}}^{(h)}$ the vulnerable- and safe-regime responses to the global shock times exposure. Group medians are taken element by element over the economies of the group.

C.2.3 The reaction function (M2). Table C.3 reports the reaction function of Table 2 by ordinary least squares (OLS), with the first-stage statistics of the instrumented columns. By OLS the debt–uncertainty interaction is −0.0025 (SE 0.0017) against −0.0034 by instrumental variables, so the instrumenting changes the interaction little. The gap coefficient is where the two differ, 0.157 by OLS against 0.595 instrumented, which is the simultaneity between the balance and the cycle that the instruments remove.

Table C.4 asks whether the opposite signs of the debt interaction across the columns of Table 2, negative with the country WUI and positive with the foreign-outlet index, are a property of the source of uncertainty. It enters a domestic and a global or foreign measure together, each interacted with debt. With the country WUI and the global index the two interactions are −0.0043 (SE 0.0023) and 0.0040 (SE 0.0021), a difference that is rejected at the 5% level, and with the GDELT domestic and foreign indices on 954 country-years the difference is −0.0170 (SE 0.0076), rejected at the 5% level as well. The split carries two qualifications. The GDELT difference does not survive the exclusion of 2020–2021, where it is −0.0005 (SE 0.0063), and the calendar-year decomposition of the WUI into a domestic residual and the global index gives a difference that is not rejected at the 10% level. The pattern holds in the point estimates of 4 of the five two-source columns, and Section 3.3 reports it with these qualifications.

C.2.4 Model-based debt limits: a non-identification note. The proposal of this project asked for debt limits in the sense of Ghosh et al. (2013), the debt ratio at which the long-run primary-balance reaction $\mu_i(d;U)$ crosses the interest–growth line $k_i d$ from above, with $k_i = (r_i - g_i)/(1 + g_i)$, and for the fiscal space those limits imply at low and at high uncertainty. The exercise is reported in Table C.5 and Figure C.5, and it does not identify the limits in this panel, for three reasons. First, the interest–growth differential, taken as each economy's own mean over 2010–2025, is non-positive for 122 of the 127 economies, so a reaction function with a positive slope at the relevant debt ratios never crosses the line from above. Second, the cubic of the emerging market and developing economies has no fiscal-fatigue bend, since its leading coefficient is positive, so no limit exists for them whatever the differential: finite limits at both uncertainty levels exist for 3 of 58 emerging markets and for none of the low-income economies. Third, the advanced-economy cubic does bend, and 23 of 28 advanced economies have finite limits, but the median limit is 308 percent of GDP at low uncertainty, and the 90% block-bootstrap band on the median space removed by uncertainty runs from −75 to 37 points of GDP. Türkiye, with debt at 23.5 percent of GDP and an own differential of −17.9 percent, has no finite limit at either level. A common differential of one percentage point, the long-run value of Ghosh et al. (2013), gives finite limits for fewer economies still. The main text therefore reports fiscal space at risk, which reads the robust slopes of the decomposition and the regime models at each economy's own position and makes no claim about a limit.

Table C.3. The fiscal reaction function by OLS, and first-stage statistics of Table 2

	Primary balance, % of GDP					
	(1)	(2)	(3)	(4)	(5)	(6)
Uncertainty U_t (sd)	−0.0630	−0.0407	−0.0595	−0.0408	−0.320	−0.701
	(0.102)	(0.117)	(0.115)	(0.107)	(0.117)	(0.166)
$\text{Debt}_{t-1} \times U_t$	−0.00254	−0.00202	−0.00249	−0.00157	−0.00133	0.00457
	(0.00167)	(0.00177)	(0.00242)	(0.00125)	(0.00269)	(0.00225)
Output gap (% of trend GDP)	0.157	0.128	0.156	0.102	0.277	0.252
	(0.0548)	(0.0525)	(0.0544)	(0.0391)	(0.0432)	(0.0504)
Output gap $\times U_t$	0.0265	0.0220	0.0149	0.00847	0.0445	0.0384
	(0.0204)	(0.0175)	(0.0168)	(0.0170)	(0.0143)	(0.0185)
Debt_{t-1} (% of GDP)	0.0354	0.0360	0.0352	0.0417	0.0572	0.101
	(0.00936)	(0.0108)	(0.00968)	(0.00851)	(0.103)	(0.0808)
Debt^2_{t-1} ($\times 10^{-3}$)	−0.147	−0.139	−0.149	−0.176	−0.0465	−0.497
	(0.0631)	(0.0833)	(0.0658)	(0.0615)	(0.913)	(0.707)
Debt^3_{t-1} ($\times 10^{-6}$)	0.236	0.219	0.245	0.273	−0.560	0.716
	(0.108)	(0.141)	(0.114)	(0.113)	(2.36)	(1.79)
$\text{Primary balance}_{t-1}$	0.512	0.510	0.512	0.506	0.522	0.492
	(0.0616)	(0.0610)	(0.0622)	(0.0621)	(0.0503)	(0.0664)
ΔAE: U_t		−0.334				
		(0.208)				
ΔAE: $\text{Debt}_{t-1} \times U_t$		0.00239				
		(0.00292)				
ΔAE: Output gap		0.194				
		(0.0342)				
ΔAE: Output gap $\times U_t$		0.00404				
		(0.0488)				
ΔAE: Debt_{t-1}		−0.00679				
		(0.0166)				
ΔAE: Debt^2_{t-1}		−0.0354				
		(0.181)				
ΔAE: Debt^3_{t-1}		0.167				
		(0.546)				
Fixed effects	Country	Country	Country	Country	Country	Country
Observations	3,834	3,834	3,833	3,594	954	1,110
Economies	127	127	127	127	114	124
Years	1990–2025	1990–2025	1990–2025	1990–2025	2016–2025	2016–2025
First stages of Table 2 (IV)						
Robust first-stage F: output gap	174.3	509.3	73.5	35.7	6,020.1	1,219.1
Robust first-stage F: output gap $\times U_t$	43.8	411.4	28.8	24.6	42.9	127.3
Robust first-stage F: ΔAE output gap		144.2				
Robust first-stage F: ΔAE output gap $\times U_t$		156.7				
First-stage F: output gap	93.6	49.6	91.5	37.7	86.0	89.4
First-stage F: output gap $\times U_t$	184.7	94.9	196.5	142.6	54.2	87.8
First-stage F: ΔAE output gap		173.3				
First-stage F: ΔAE output gap $\times U_t$		254.0				
Cragg–Donald statistic	82.9	37.1	93.5	37.2	38.2	41.7

Notes: Entries are OLS coefficients of the specification of Table 2 with the output gap and its interaction not instrumented, primary balance in percent of GDP; Driscoll–Kraay standard errors (lag 2) in parentheses. The bottom block reports, for the IV columns of Table 2, the Driscoll–Kraay-robust first-stage Wald F and the conventional first-stage F of each instrumented variable and the Cragg–Donald statistic. Columns, uncertainty measures and controls as in Table 2.

Table C.4. The source of uncertainty in the fiscal reaction function: domestic against global or foreign

	Primary balance, % of GDP					
	(1)	(2)	(3)	(4)	(5)	(6)
Uncertainty U_t (sd)	0.107					
	(0.104)					
$\text{Debt}_{t-1} \times U_t$	−0.00343					
	(0.00172)					
Country / domestic uncertainty U_t^d (sd)		0.231	0.0789	0.128	0.215	−0.00672
		(0.125)	(0.0851)	(0.125)	(0.189)	(0.141)
$\text{Debt}_{t-1} \times U_t^d$		−0.00427	−0.000439	−0.00393	−0.00693	0.00125
		(0.00228)	(0.00170)	(0.00218)	(0.00408)	(0.00446)
Global / foreign uncertainty U_t^g (sd)		−0.627	−0.548	−0.157	−0.668	−0.261
		(0.211)	(0.215)	(0.184)	(0.174)	(0.136)
$\text{Debt}_{t-1} \times U_t^g$		0.00399	0.00233	0.000465	0.0101	0.00179
		(0.00206)	(0.00163)	(0.00249)	(0.00433)	(0.00243)
Output gap (% of trend GDP)	0.595	0.533	0.524	0.575	0.407	0.213
	(0.0634)	(0.0648)	(0.0672)	(0.0716)	(0.0791)	(0.0746)
Output gap $\times U_t$	0.00646					
	(0.0325)					
Output gap $\times U_t^d$		0.0138	−0.0355	0.0340	0.0847	−0.0474
		(0.0265)	(0.0365)	(0.0337)	(0.0363)	(0.0375)
Output gap $\times U_t^g$		−0.0159	−0.00700	−0.0681	0.0235	0.0185
		(0.0324)	(0.0368)	(0.0371)	(0.0370)	(0.0196)
Debt_{t-1} (% of GDP)	0.0263	0.0285	0.0289	0.0293	0.0741	0.114
	(0.0123)	(0.0120)	(0.0118)	(0.0130)	(0.110)	(0.0967)
Debt_{t-1}^2 ($\times 10^{-3}$)	−0.118	−0.129	−0.138	−0.141	−0.281	−0.690
	(0.0935)	(0.0917)	(0.0901)	(0.101)	(1.00)	(0.936)
Debt_{t-1}^3 ($\times 10^{-6}$)	0.235	0.251	0.280	0.267	0.351	1.58
	(0.149)	(0.151)	(0.151)	(0.158)	(2.71)	(2.56)
$\text{Primary balance}_{t-1}$	0.508	0.506	0.507	0.504	0.547	0.570
	(0.0579)	(0.0568)	(0.0565)	(0.0603)	(0.0544)	(0.0234)
Difference and economic size						
Debt $\times (U_t^d - U_t^g)$		−0.00826	−0.00277	−0.00440	−0.0170	−0.000540
		(0.00394)	(0.00234)	(0.00415)	(0.00765)	(0.00634)
		[0.036]	[0.236]	[0.290]	[0.026]	[0.932]
p10→p90 of U^d: Δ response to debt (pp per 10 pp)	−0.0739	−0.0921	−0.00962	−0.0855	−0.177	0.0296
	(0.0372)	(0.0493)	(0.0373)	(0.0473)	(0.104)	(0.106)
p10→p90 of U^g: Δ response to debt (pp per 10 pp)		0.0841	0.0491	0.00565	0.273	0.0431
		(0.0435)	(0.0344)	(0.0302)	(0.117)	(0.0587)
Fixed effects	Country	Country	Country	Country	Country	Country
Instrumented output gap (first-stage F)	43.8	154.6	53.5	107.0	204.8	218.4
Cragg–Donald statistic	82.9	37.0	37.4	51.9	22.2	10.9
Observations	3,834	3,834	3,834	3,723	954	748
Economies	127	127	127	127	114	111
Years	1990–2025	1990–2025	1990–2025	1992–2025	2016–2025	2016–2025

Columns: (1) WUI (Table 2, col. 1); (2) WUI (fiscal year) and global WUI; (3) WUI domestic component and global WUI; (4) WUI (fiscal year) and the global shock; (5) GDELT: domestic and foreign FPU; (6) GDELT, excl. 2020–21. Notes: Entries are coefficients of the fiscal reaction function of Table 2 with two uncertainty measures entering together, each interacted with lagged debt and with the output gap; primary balance in percent of GDP; Driscoll–Kraay standard errors (lag 2) in parentheses, p-values in brackets. Column (1) repeats Table 2, column (1). Column (2) adds the global WUI index (wui_glob (mean 0.193, sd 0.094)) to the country index, so that the country interaction is identified from the country-specific part of uncertainty; column (3) replaces the country index by its domestic component (the calendar-year country index net of its loading on the global index, wui_dom (mean 0.004, sd 0.133)); column (4) pairs the country index with the identified global uncertainty shock ε_t^g (sd units); columns (5)–(6) use the GDELT domestic-outlet and foreign-outlet fiscal-policy-uncertainty indices (fpu_dom_z (mean -0.087, sd 0.895); fpu_for_z (mean -0.116, sd 0.871); 2016–2025, non-thin country-years), column (6) without 2020–2021. The middle block reports the difference between the two debt interactions with its standard error and p-value, and the change in the marginal response of the primary balance to debt (pp of GDP per 10 pp of debt) when each measure moves from its 10th to its 90th percentile. Instruments: world growth excluding the own economy, its product with 1995–2004 openness and the lagged gap, each alone and multiplied by each uncertainty measure (9 instruments for 3 instrumented variables); the first-stage row reports the smallest DK-robust first-stage Wald F across the instrumented variables. Controls and the cubic in debt as in Table 2; country fixed effects.

Table C.5. Implied debt limits and fiscal space by economy

Economy	Group	Debt	$r-g$	d^*(p10)	Band	d^*(p90)	Band	Space (p10)	Space (p90)	Δ space	Band
Australia	AE	50.6 (2024)	−3.0	337	[82, 624]	359	[124, 654]	286	308	22	[−18, 93]
Austria	AE	79.2 (2024)	−1.7	268	[64, 389]	296	[35, 467]	189	217	28	[−52, 107]
Belgium	AE	103.9 (2024)	−1.6	299	[136, 390]	322	[132, 416]	196	218	23	[−19, 50]
Canada	AE	110.0 (2024)	−4.2	403	[159, 721]	421	[159, 733]	293	311	18	[−11, 49]
Switzerland	AE	39.4 (2025)	−1.7	309	[123, 486]	331	[107, 456]	270	292	22	[−33, 61]
Czech Republic	AE	43.3 (2024)	−2.7	307	[13, 489]	333	[129, 485]	264	289	26	[−18, 183]
Germany	AE	62.9 (2025)	−2.2	315	[132, 421]	337	[120, 447]	252	275	23	[−16, 101]
Denmark	AE	30.5 (2024)	−3.9	405	[164, 702]	423	[165, 725]	375	392	17	[−8, 56]
Finland	AE	82.4 (2024)	−2.6	330	[129, 530]	352	[98, 619]	248	270	22	[−17, 90]
United Kingdom	AE	102.3 (2025)	−1.8	257	[13, 416]	287	[8, 461]	154	185	31	[−52, 279]
Greece	AE	155.4 (2024)	2.0	8	[3, 54]		[1, 48]	−147			[−33, 2]
Hong Kong Special Administrative Region, People's Republic of China	AE	9.2 (2024)	−2.3	268	[121, 499]	299	[148, 577]	258	290	32	[−17, 75]
Ireland	AE	38.3 (2024)	−5.8	446	[165, 625]	462	[169, 680]	407	424	17	[−9, 50]
Israel	AE	68.5 (2025)	−1.7	290	[128, 388]	315	[116, 424]	222	246	24	[−16, 101]
Italy	AE	137.1 (2025)	0.7	151	[90, 190]	74	[86, 197]	14	−63	−77	[−81, 129]
Korea, Republic of	AE	49.7 (2024)	−5.7	450	[164, 624]	466	[165, 656]	400	417	16	[−6, 57]
Latvia, Republic of	AE	46.2 (2024)	−2.5	297	[10, 585]	324	[135, 618]	251	277	26	[−22, 208]
Netherlands, The	AE	43.7 (2024)	−2.4	335	[144, 510]	356	[139, 526]	291	312	21	[−16, 61]
Norway	AE	52.8 (2024)	−11.1	593	[206, 1,243]	605	[211, 1,281]	540	552	12	[−4, 56]
New Zealand	AE	54.7 (2025)	−3.1	374	[155, 555]	393	[161, 656]	319	338	19	[−10, 98]
Portugal	AE	93.5 (2024)	−0.8	0	[27, 260]		[13, 316]	−93			[−134, 234]
Slovak Republic	AE	59.7 (2024)	−2.3	274	[16, 464]	304	[108, 473]	214	244	30	[−45, 133]
Slovenia, Republic of	AE	66.6 (2024)	−1.7	268	[31, 452]	295	[17, 543]	201	229	28	[−53, 200]
Sweden	AE	33.9 (2024)	−3.9	392	[158, 523]	411	[159, 593]	358	377	19	[−12, 82]
United States	AE	123.9 (2025)	−2.5	308	[118, 544]	333	[73, 618]	184	209	25	[−20, 94]
Bulgaria	EM	27.1 (2025)	−5.8	437	[162, 564]	454	[167, 568]	410	427	17	[−12, 54]
Croatia, Republic of	EM	57.4 (2024)	−1.5	249	[28, 349]	279	[20, 391]	191	222	30	[−48, 190]
Lithuania, Republic of	EM	38.0 (2024)	−4.1	377	[137, 576]	397	[145, 681]	339	359	20	[−12, 84]
Mexico	EM	59.1 (2024)	1.2		[80, 1,354]	174	[82, 1,532]		115		[−128, 32]

Notes: Entries are the implied debt limit $d^*(U)$ and fiscal space $d^*(U) - d_{\text{last}}$, in percent of GDP, at the 10th and 90th percentiles of fiscal-year uncertainty, for every economy with a finite limit at at least one level; the limit is the largest root of $\mu_i(d;U) = \frac{r_i-g_i}{1+g_i} d$ at which the long-run reaction μ_i (column 2 of Table 2, country effect, controls at country means, gap at zero) crosses the interest–growth line from above. Debt is the last WEO actual (year in parentheses); $r-g$ in percent is the 2010–2025 country mean of the effective interest rate minus nominal growth, divided by one plus growth (group median for 2 economies without five years of data); the limits use this differential. Bands are 5th–95th percentiles of a block bootstrap over economies (500 draws). Δ space is space at p90 minus space at p10. Economies without a finite limit at either level (98) are omitted.

C.2.5 The cyclicality model (M3). Figure C.6 is the dynamic counterpart of Table 3: state-dependent local projections of the primary balance, revenue and primary spending on the two shocks in downturns and in upturns, with the state $D_{i,t-1}$ an indicator of a downturn in the year before the shock (the lagged Hamilton gap below zero) and the lower-order terms $D_{i,t-1}$, $\tilde{d}_{i,t-1}D_{i,t-1}$ and $\varepsilon_t^{\mathrm{g}}D_{i,t-1}$ as controls. The low-space economy relative to the high-space economy shows a primary-balance response of 0.71 (SE 0.39) points of GDP two years after a global shock in downturns, beyond 1.645 standard errors of zero, and −0.39 (SE 0.29) in upturns, within 1.645 standard errors of zero. The domestic shock moves the balance by less than its standard error in either state. In the point estimates the dynamic version therefore shows the pattern the proposal expected and Table 3 does not find, the indebted economy tightening relative to the economy with space after a global shock in a downturn, and it rests on one coefficient at the 10% level.

Table C.6 and Figure C.7 carry the weak-instrument reading of the cyclicality null. The triple interaction is 0.010 (SE 0.029) by two-stage least squares with twelve instruments, 0.001 by limited-information maximum likelihood (LIML), −0.039 (SE 0.031) by the fitted-gap just-identified estimator and 0.024 with the lagged gap as the single base instrument, so the point estimates stay within one standard error of one another across estimators. The 90% S set with the fitted-gap instruments is bounded, from −0.0875 to 0.0225, whereas the sets with twelve instruments and with the single weak instrument are the whole grid. A sign switch of the high-uncertainty cyclical response between the high-space corner, 0.342 (SE 0.105), and the low-space corner would need a value of ϕ_{UF} outside the grid, far beyond the bounded set. The null of Section 3.3 is therefore not an artefact of weak instruments.

C.2.6 Regimes of vulnerability (M4). Table C.7 is the sharp alternative to the smooth transition of Figure 5: a panel threshold on lagged debt for three outcomes at two horizons, with the regime indicator and its products with the global shock and with exposure as regime-level terms. The hypothesis of no threshold is rejected at the 5% level in 4 of the 6 equations and at the 10% level in 4. The estimated thresholds lie between 20 and 25 percent of GDP, and the 95% intervals reach the lower edge of the search range, the 10th percentile of lagged debt at 18.6 percent, in 2 of the equations. The rejection comes from the regime-level terms. The level response to the global shock below the threshold differs from the response above it by −1.13 (SE 0.41) percentage points of GDP for the primary balance and 0.81 (SE 0.36) for primary spending in the year of the shock, and by 1.08 (SE 0.45) percentage points for the effective interest rate two years on. The below-minus-above difference in the exposure gradient is within 1.645 standard errors of zero in 6 of the 6 equations; for the primary balance at two years the gradient is 0.17 (SE 0.20) below the threshold of 25 percent of GDP against 0.01 (SE 0.12) above it. The sharp model therefore finds a regime at a low debt ratio in the level response to the global shock, the sharp counterpart of Table C.8 and of the slope of Figure 3, and none in the exposure gradient. The regime evidence of the main text rests on the smooth-transition differences.

Table C.8 reports the smooth model's level difference, the coefficient on $F(z_{i,t-1})\varepsilon_t^{\mathrm{g}}$, by horizon and outcome. At mean exposure a one-standard-deviation global shock adds 7.16 (SE 2.83) points of GDP more debt in the vulnerable regime than in the safe one after two years, a difference beyond 1.645 standard errors of zero at 5 of the six horizons. The level difference in the effective interest rate is −0.80 (SE 0.32) percentage points at two years and within 1.645 standard errors of zero at the other horizons, so the price that the vulnerable regime pays in Figure 5 is paid through exposure, not at mean exposure.

Figure C.8 completes Figure 5 with the debt and revenue outcomes, the two alternative transition

Table C.6. The cyclicality model under weak-instrument-robust inference

	Change in the primary balance, pp of GDP				
	(1)	(2)	(3)	(4)	(5)
Coefficients					
Output gap ϕ	0.236	0.315	0.356	−0.123	0.0743
	(0.107)	(0.158)	(0.104)	(0.297)	(0.0314)
Output gap × U (ϕ_U)	0.00114	−0.00517	0.00806	0.0483	0.0120
	(0.0251)	(0.0281)	(0.0164)	(0.0533)	(0.0144)
Output gap × $\tilde{d}$ (ϕ_F)	0.0174	0.0186	0.0445	0.0374	0.0244
	(0.0246)	(0.0273)	(0.0248)	(0.0332)	(0.00943)
Output gap × U × $\tilde{d}$ (ϕ_{UF})	0.00972	0.000713	−0.0391	0.0243	−0.00827
	(0.0294)	(0.0362)	(0.0315)	(0.0445)	(0.0116)
Implied cyclical response at the corners (pp of GDP per pp of gap)					
Low U, high space	0.228	0.307	0.305	−0.169	0.0470
	(0.111)	(0.157)	(0.111)	(0.300)	(0.0365)
Low U, low space	0.239	0.326	0.377	−0.148	0.0776
	(0.118)	(0.178)	(0.111)	(0.319)	(0.0379)
High U, high space	0.222	0.301	0.342	−0.132	0.0665
	(0.104)	(0.152)	(0.105)	(0.301)	(0.0286)
High U, low space	0.245	0.320	0.370	−0.0826	0.0876
	(0.104)	(0.152)	(0.0988)	(0.283)	(0.0300)
S-statistic confidence sets					
S set for ϕ_{UF}, 90%	[−0.600, 0.600] (open)		[−0.0875, 0.0225]	[−0.600, 0.600] (open)	
S set for ϕ_{UF}, 95%	[−0.600, 0.600] (open)		[−0.100, 0.0375]	[−0.600, 0.600] (open)	
Instruments	12	12	4	4	
Robust first-stage F (min)	24.5		35.5	4.7	
Cragg–Donald statistic	9.5		24.2	2.7	
LIML eigenvalue		1.010			
Fixed effects	Country, year	Country, year	Country, year	Country, year	Country, year
Observations	3,832	3,832	3,832	3,832	3,832
Economies	127	127	127	127	127
Years	1990–2025	1990–2025	1990–2025	1990–2025	1990–2025

Columns: (1) 2SLS, 12 instr.; (2) LIML, 12 instr.; (3) just-identified: fitted gap ×$(1, U, \tilde{d}, U\tilde{d})$; (4) just-identified: lagged gap ×$(1, U, \tilde{d}, U\tilde{d})$; (5) OLS. Notes: Entries are the gap coefficients of the pooled cyclicality model of Table 3 (change in the primary balance, pp of GDP; country and year effects; controls as in Table 3) under alternative estimators; Driscoll–Kraay standard errors (lag 2) in parentheses. Column (1) is Table 3, column (1): the four gap terms instrumented by world growth excluding the own economy, its product with 1995–2004 openness and the lagged gap, each multiplied by $(1, U, \tilde{d}, U\tilde{d})$, $\tilde{d}$ lagged debt standardised within the sample. Column (2) is LIML on the same instruments. Column (3) is just-identified with generated instruments: the fitted gap from the first stage on the three base instruments (with the exogenous regressors and the fixed effects), multiplied by $(1, U, \tilde{d}, U\tilde{d})$ (Wooldridge, 2010); no standard-error adjustment is needed for instruments that are fitted values. Column (4) is just-identified with the single base instrument that has the largest first-stage F on its own. Column (5) is OLS. The middle block reports the implied cyclical response at the 25th and 75th percentiles of U and of $\tilde{d}$ (high space = 25th percentile of debt). The S-set rows give the set of values of ϕ_{UF} not rejected by the S statistic of Stock and Wright (2000), the two-step GMM objective with the Driscoll–Kraay weighting matrix, the other gap coefficients concentrated out at each value in the subset form of Kleibergen and Mavroeidis (2009), with χ^2 critical values on one degree of freedom in the just-identified columns and nine in column 1; a set marked open reaches the edge of the grid $[-0.6, 0.6]$. The robust first-stage F is the smallest Driscoll–Kraay-robust first-stage Wald F across the instrumented variables.

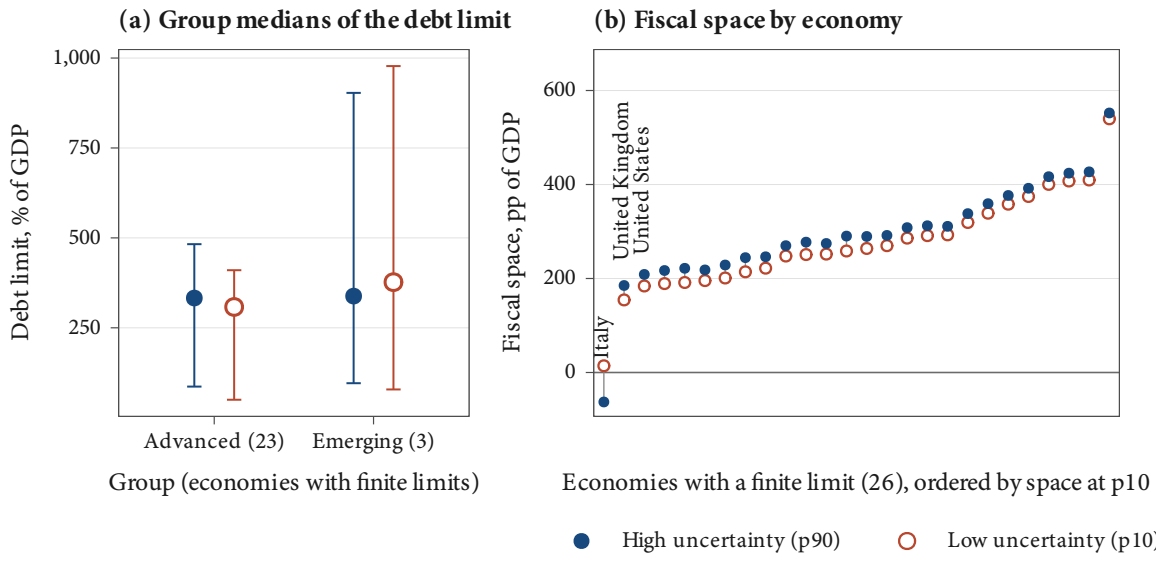


Figure C.5. Implied debt limits and fiscal space under low and high uncertainty

Notes: Panel (a): group medians of the implied debt limit $d^*(U)$, percent of GDP, at the 10th and 90th percentiles of fiscal-year WUI, with 90% block-bootstrap bands (500 draws); groups without finite limits omitted. Panel (b): fiscal space, $d^*(U)$ minus the last actual debt ratio, 26 economies with finite limits at both levels. Column (2) of Table 2, $r - g$ at 2010–2025 country means.

speeds and the three income groups. After a global shock the exposure gradient of the debt response does not separate the regimes beyond the first year: the difference is 0.51 (SE 1.07) points of GDP at two years, and its largest value, −1.81 (SE 0.80) at $h = 1$, is the one horizon at which it is beyond 1.645 standard errors of zero. Vulnerable economies add debt after the shock whatever their exposure, through the level difference of Table C.8. The two-year primary-balance difference is 0.17 (SE 0.60) with $\kappa = 1$ and −0.19 (SE 0.26) with $\kappa = 3$ against −0.01 at the headline speed, so the transition speed does not decide the result. By income group the same difference is within 1.645 standard errors of zero in each of the three groups, and widest in the low-income economies.

C.2.7 Fiscal space as insurance (M5). Figure C.9 shows the three insurance outcomes of the bottom row of Figure 5 with their bands. It adds the version with exposure measured by the foreign-currency share of debt and the output response by income group. Output after a global shock is −0.59 (SE 0.61) log points in the high-space regime and 0.48 (SE 0.64) in the low-space regime after two years, and the largest difference between them, −1.43 (SE 1.59) at $h = 4$, is within one standard error of zero. The insurance inequality holds at 0 of the six horizons for output, 1 for the effective rate and 5 for the rating, which is the count Section 3.4 reports. With foreign-currency exposure on 50 economies, and within each income group, the high-minus-low output difference after two years stays within one standard error of zero, so no exposure measure and no group shows the output insurance.

C.2.8 The pandemic. Three figures and one table support Table 4 and Figure 6. Figure C.10 plots above-the-line support against 2019 debt for the 64 economies with IMF measures; the slope is 0.060 (SE 0.019) percent of GDP per point of debt, the raw version of the debt coefficient of Table 4. Figure C.11 reports the coefficients on 2020 pandemic uncertainty from the event study, kept off Figure 6

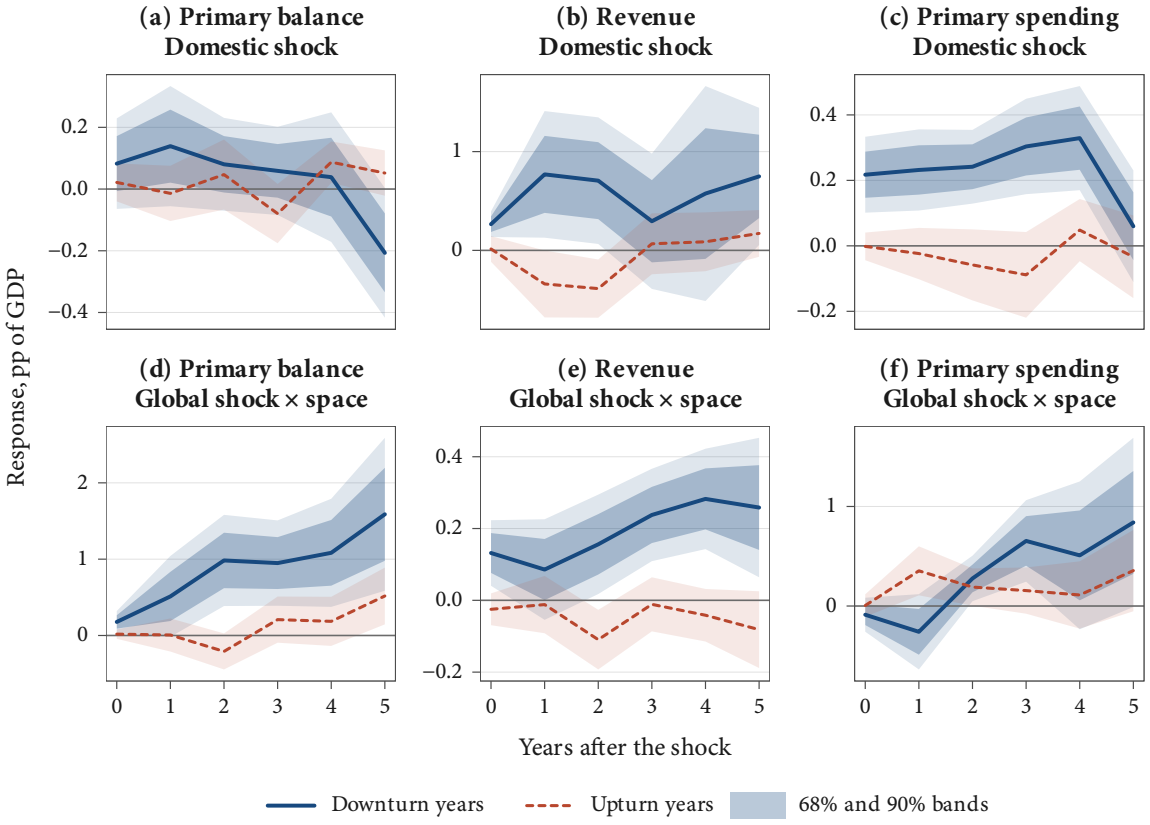


Figure C.6. Fiscal responses to uncertainty shocks in downturns and upturns

Notes: Local projections of $y_{i,t+h} - y_{i,t-1}$ on the shock interacted with the lagged state $D_{i,t-1}$ (downturn: Hamilton gap below zero), with country and year effects, lagged controls and the lower-order terms $D_{i,t-1}$, $\tilde{d}_{i,t-1}D_{i,t-1}$, $\varepsilon_t^g D_{i,t-1}$. Panels (a)–(c): one-sd domestic shock; (d)–(f): global shock ε_t^g times lagged standardised debt $\tilde{d}_{i,t-1}$, low-space (p75) minus high-space (p25) response, 0.9 sd of debt. DK bands (lag 2).

because their scale differs: one standard deviation of the 2020 pandemic index is associated with a 2021 debt change of −0.73 (SE 0.93) points of GDP, within one standard error of zero. Figure C.12 estimates the event study by income group. The 2021 debt coefficient on 2019 space is −3.17 (SE 0.97) in the advanced economies and imprecise in the other two groups, so the pooled coefficient of Figure 6 is precise only for the advanced economies. Table C.9 gives the 2020 composition of uncertainty. The pandemic component is a median 55% of the country WUI across the 139 economies, from 46% in the advanced economies to 83% in the low-income economies, which is why the purged series exist.

C.2.9 Status or space. Table C.10 races the debt–uncertainty interaction of the reaction function against interactions of uncertainty with advanced-economy status, with the sovereign rating and with the foreign-currency share, and does the same for the triple interaction of the cyclicality model. The debt interaction is −0.0034 (SE 0.0017) alone and −0.0026 (SE 0.0018) with the status interaction added, so status does not absorb it. With the rating interaction it is −0.0012 (SE 0.0020), and with the foreign-currency share on 752 country-years it loses its sign, where the smaller sample rather than the added regressor is what changes. In the cyclicality model the triple interaction with status is within one standard error of zero like the debt version. Table C.11 extends the race, and Appendix C.3.10 reads it.

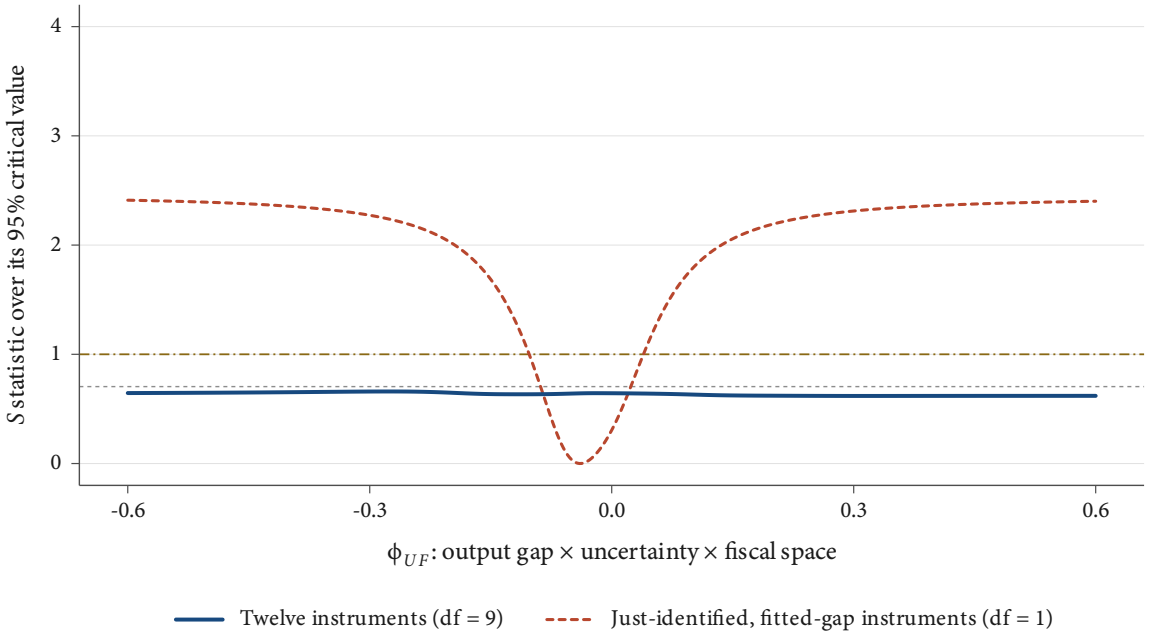


Figure C.7. *S*-statistic confidence sets for the triple interaction of the cyclicality model

Notes: The *S* statistic of Stock and Wright (2000) (two-step GMM, Driscoll–Kraay weighting matrix, the other gap coefficients concentrated out) as a function of ϕ_{UF}, divided by its 95% critical value (dash-dotted line; the dotted line is the 90% value for one degree of freedom), for the twelve-instrument specification of Table 3 and the just-identified fitted-gap specification. Values below the line are not rejected.

Table C.7. Threshold estimates: responses below and above an estimated debt threshold

	Primary balance		Primary spending		Effective interest rate	
	$h = 0$	$h = 2$	$h = 0$	$h = 2$	$h = 0$	$h = 2$
Threshold $\hat{\tau}$ (debt, % of GDP)	21.4	25.1	19.7	20.1	24.2	21.5
95% interval	[18.6, 24.5]	[20.1, 25.1]	[18.6, 25.1]	[19.1, 20.1]	[20.4, 29.6]	[20.8, 21.5]
Share of country-years below $\hat{\tau}$	0.13	0.19	0.11	0.12	0.15	0.12
F_1 (no threshold)	48.2	58.7	54.1	38.8	25.5	27.9
Bootstrap p	0.003	0.020	0.000	0.167	0.113	0.023
Domestic shock, debt $\leq \hat{\tau}$	0.413	0.535	−0.177	−0.397	0.231	0.206
	(0.232)	(0.273)	(0.130)	(0.177)	(0.104)	(0.152)
Domestic shock, debt $> \hat{\tau}$	0.0103	0.00450	0.0313	0.0620	0.00230	0.0257
	(0.0311)	(0.0748)	(0.0323)	(0.0690)	(0.0163)	(0.0459)
Global shock × exposure, debt $\leq \hat{\tau}$	−0.0639	0.174	0.273	0.206	−0.318	0.0257
	(0.179)	(0.198)	(0.123)	(0.247)	(0.220)	(0.483)
Global shock × exposure, debt $> \hat{\tau}$	−0.00901	0.0137	0.0420	0.0291	−0.0131	0.0169
	(0.0386)	(0.116)	(0.0446)	(0.159)	(0.0182)	(0.0559)
Global shock × $\mathbb{1}(d_{i,t-1} \leq \hat{\tau})$ (regime level)	−1.13	−0.144	0.806	−0.281	0.371	1.08
	(0.405)	(0.760)	(0.357)	(0.795)	(0.135)	(0.446)
Observations	2,845	2,609	2,845	2,609	2,774	2,537
Economies	122	121	122	121	121	120

Notes: Entries are Hansen (1999) panel-threshold estimates of $y_{i,t+h} - y_{i,t-1} = \alpha_i^{(h)} + \delta_t^{(h)} + \mathbf{b}_1^{(h)\prime}\mathbf{s}_{i,t}\,\mathbb{1}(d_{i,t-1} \leq \tau) + \mathbf{b}_2^{(h)\prime}\mathbf{s}_{i,t}\,\mathbb{1}(d_{i,t-1} > \tau) + \boldsymbol{\lambda}^{(h)\prime}\mathbf{m}_{i,t}^{\tau} + \Gamma^{(h)\prime}X_{i,t-1} + u_{i,t+h}^{(h)}$, where y is the outcome (percent of GDP; the effective interest rate in pp), $\alpha_i^{(h)}$ and $\delta_t^{(h)}$ are country and year effects, $\mathbb{1}(\cdot)$ is the indicator function, τ the threshold on lagged debt $d_{i,t-1}$ (percent of GDP), $\mathbf{b}_1^{(h)}$ and $\mathbf{b}_2^{(h)}$ the responses below and above it to the shocks $\mathbf{s}_{i,t} = (\varepsilon_{i,t}^{\mathrm{d}}, \varepsilon_t^{\mathrm{g}} E_i)'$ (the domestic uncertainty shock and the global shock times exposure E_i), $\boldsymbol{\lambda}^{(h)}$ the coefficients on $\mathbf{m}_{i,t}^{\tau} = (z_{i,t-1},\ \mathbb{1}(d_{i,t-1} \leq \tau),\ \mathbb{1}(d_{i,t-1} \leq \tau)\varepsilon_t^{\mathrm{g}},\ \mathbb{1}(d_{i,t-1} \leq \tau)E_i)'$, the lagged standardised vulnerability $z_{i,t-1}$ and the regime-level terms (the sharp counterparts of the lower-order terms of the smooth model), $\Gamma^{(h)}$ the coefficients on the lagged controls $X_{i,t-1}$ of Figure 5, and $u_{i,t+h}^{(h)}$ the projection error at horizon h, serially correlated by construction and correlated across economies within a year. The threshold minimises the sum of squared residuals over a grid of the 10th–90th percentiles of lagged debt (1-point steps); the 95% interval is the set of grid points with $LR_1(\tau) \leq 7.35$, LR_1 the likelihood-ratio statistic for the threshold; F_1 tests the absence of a threshold (the two regime differences and the three regime-level terms jointly zero), with a fixed-regressor cluster-wild bootstrap by economy (300 draws). Coefficients at $\hat{\tau}$ with Driscoll–Kraay standard errors (lag 2) in parentheses; the regime-level row is the coefficient on $\mathbb{1}(d_{i,t-1} \leq \hat{\tau})\varepsilon_t^{\mathrm{g}}$, the difference in the level response to the global shock below the threshold.

Table C.8. The regime difference in the level response to the global shock: coefficient on $F(z_{i,t-1})\,\varepsilon_t^{g}$ by horizon

	Primary balance	Primary spending	Effective interest rate	Gross debt	Revenue
$h = 0$	1.10	−0.615	−0.149	0.564	0.475
	(0.382)	(0.312)	(0.138)	(0.888)	(0.193)
$h = 1$	2.27	−1.76	−0.214	7.33	0.477
	(1.10)	(1.03)	(0.336)	(1.35)	(0.467)
$h = 2$	1.17	−0.637	−0.797	7.16	0.497
	(1.48)	(1.16)	(0.315)	(2.83)	(0.503)
$h = 3$	0.244	0.298	−0.541	6.57	0.515
	(1.51)	(1.29)	(0.354)	(2.66)	(0.480)
$h = 4$	0.701	−0.122	−0.284	8.80	0.536
	(1.15)	(1.02)	(0.428)	(3.77)	(0.683)
$h = 5$	0.965	0.410	−0.397	8.30	1.32
	(1.20)	(1.55)	(0.399)	(4.81)	(0.784)
Observations ($h = 0$)	2,845	2,845	2,774	2,845	2,845

Notes: Entries are the coefficients $\lambda_{Fe}^{(h)}$ on $F(z_{i,t-1})\,\varepsilon_t^{g}$ in the smooth-transition local projections of Figure 5, by horizon h (rows) and outcome (columns), in the units of the outcome (percent of GDP; the effective interest rate in pp) per standard deviation of the global shock. The coefficient is the difference between the level response to the global shock of an economy in the vulnerable regime ($F = 1$) and of one in the safe regime ($F = 0$), at mean exposure; the regime responses of Figure 5 are the exposure gradients, the difference between an economy one sd more open than the mean and the mean-exposure economy in the same regime. Country and year effects, the lagged controls of Figure 5, the state $z_{i,t-1}$ (lagged standardised vulnerability) and the lower-order terms $F(z_{i,t-1})$ and $F(z_{i,t-1})E_i$, E_i exposure and F the logistic transition with speed $\kappa = 1.5$; Driscoll–Kraay standard errors (lag 2) in parentheses; 122 economies.

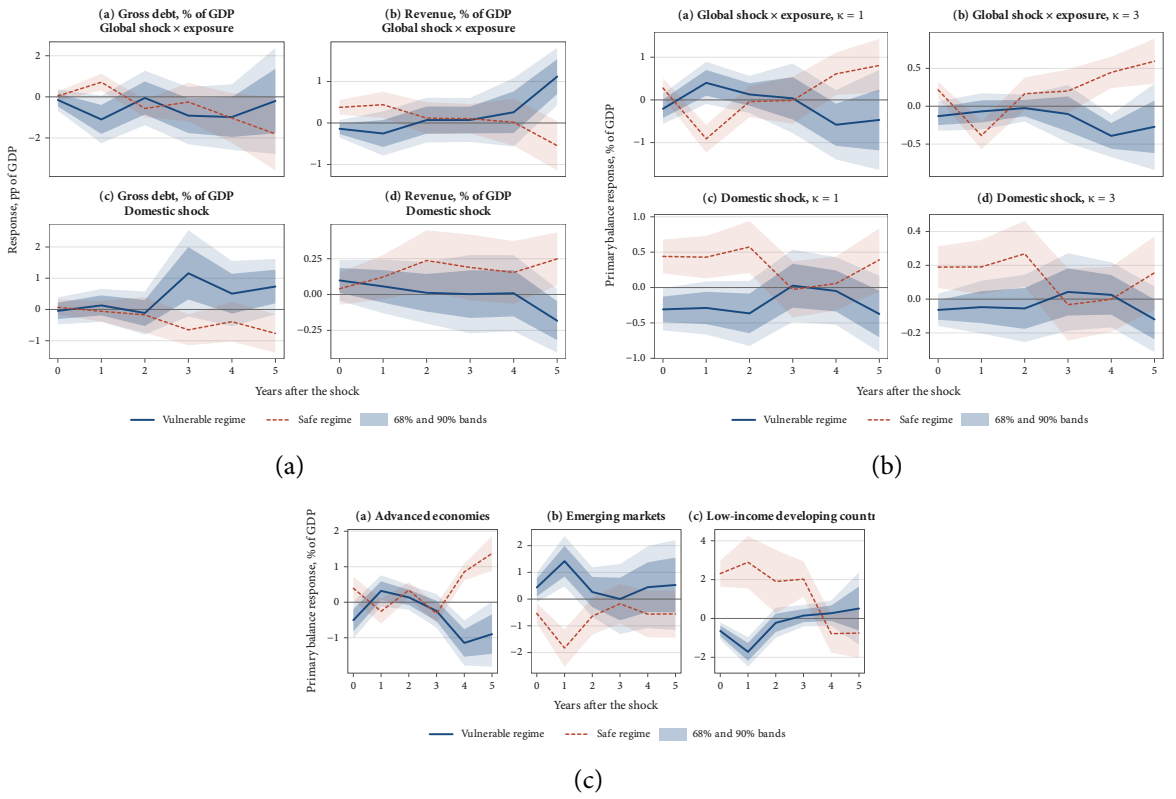


Figure C.8. Regime responses: other outcomes, transition speed and income groups

Notes: As Figure 5. Panel set (a): gross debt and revenue. Panel set (b): the primary-balance responses with the transition speed $\kappa = 1$ and $\kappa = 3$. Panel set (c): the primary-balance response to the global shock × exposure by income group (AE 28, EM 58, LIDC 36 economies). DK bands (lag 2).

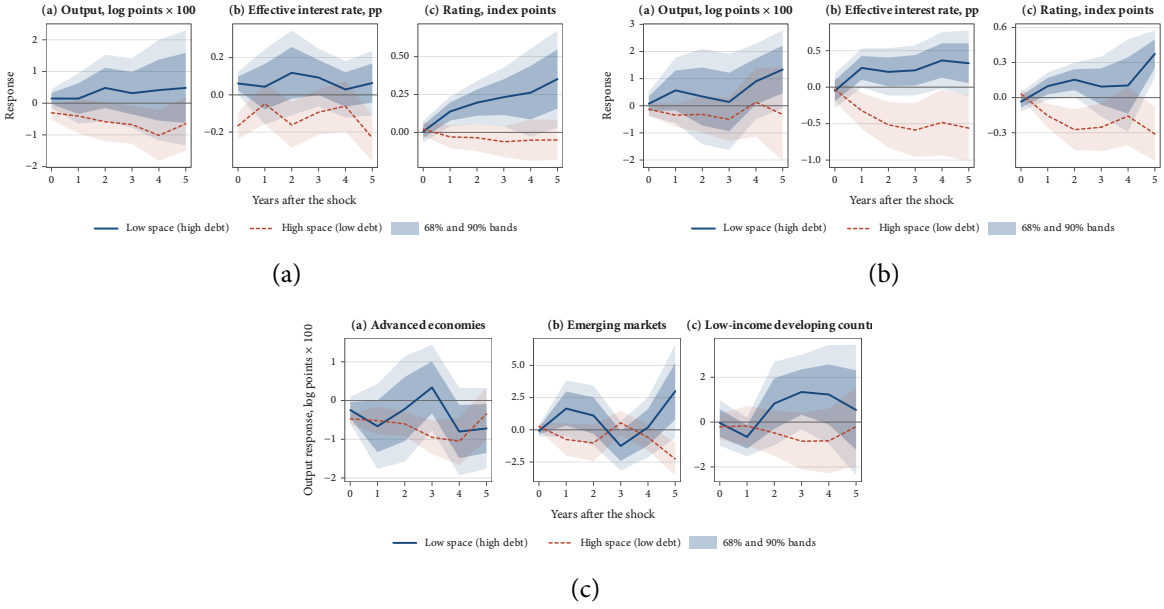


Figure C.9. Fiscal space as insurance: responses to the global shock by space regime

Notes: Smooth-transition local projections with the regime indexed by lagged debt (high space: F near 1 at low debt, $\kappa = 1.5$), country and year effects, lagged controls and the lower-order regime terms. Panel set (a): global shock × trade exposure (129 economies); (b): × FX-debt exposure (50 economies); (c): output by income group. DK bands (lag 2).

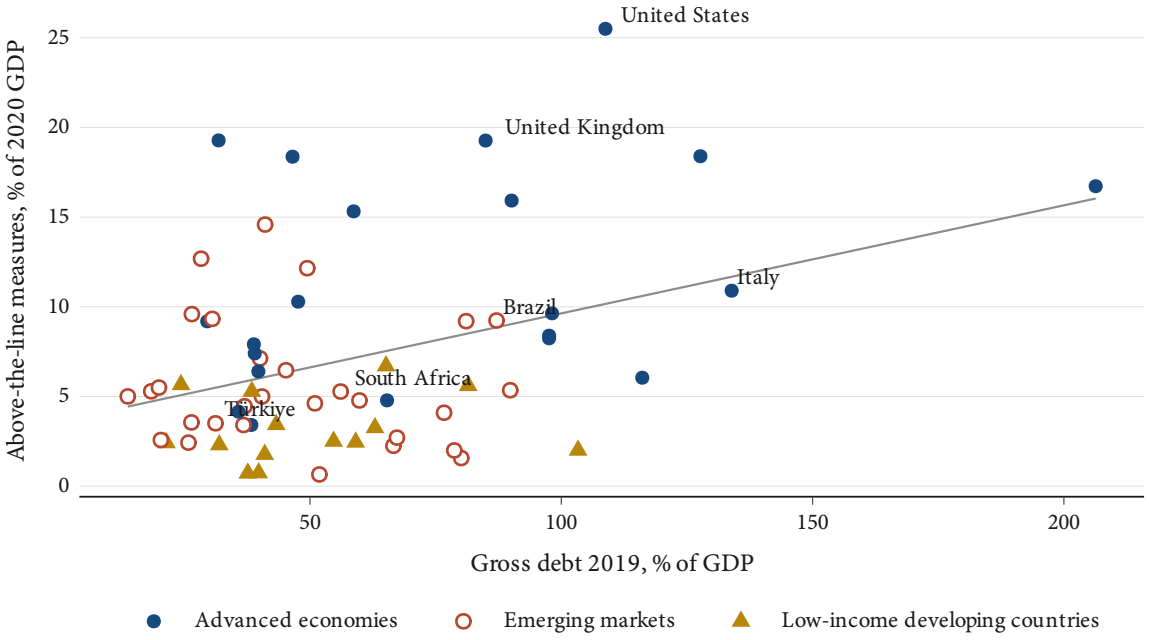


Figure C.10. Pandemic fiscal support and pre-pandemic debt

Notes: IMF above-the-line COVID-19 measures (January 2020–September 2021, percent of 2020 GDP) against the 2019 gross debt ratio, 64 economies; the grey line is the OLS fit (slope 0.060, SE 0.019, per pp of debt). The six economies of Figure 2 are labelled.

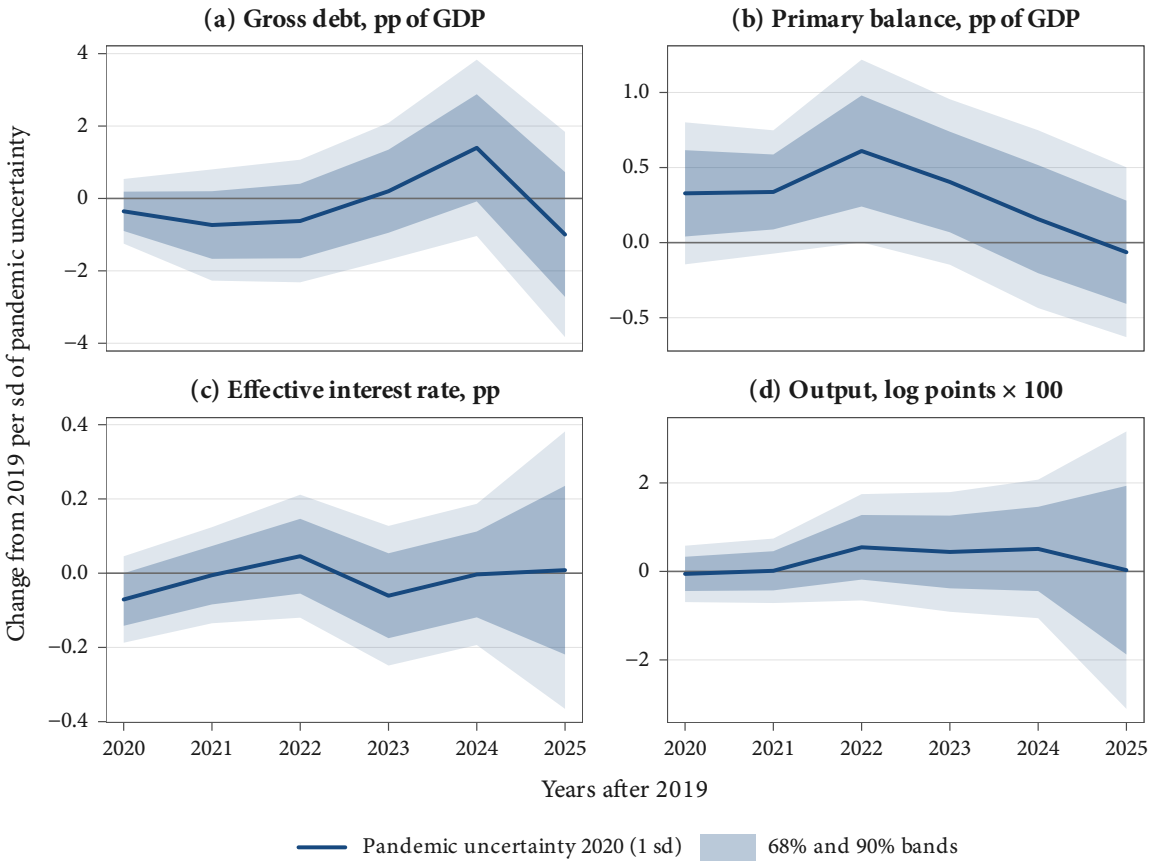


Figure C.11. Pandemic uncertainty in 2020 and outcomes after 2020

Notes: As Figure 6, the coefficient on the standardised 2020 World Pandemic Uncertainty Index (sd 0.135) by horizon, HC1 bands.

Table C.9. The pandemic composition of uncertainty in 2020, by country group

Group	Economies	WUI	WPUI	Pandemic share	Share > 0.5	Pandemic discussion
Advanced economies	30	0.332	0.165	0.46	0.43	2.69
Emerging markets	59	0.200	0.101	0.52	0.53	2.77
Low-income developing countries	50	0.323	0.287	0.83	0.66	3.29
All	139	0.282	0.163	0.55	0.55	2.99

Notes: Entries are group medians of the 2020 calendar-year World Uncertainty Index (WUI, mentions of uncertainty per thousand words), the World Pandemic Uncertainty Index (WPUI, same scale), the pandemic share min(WPUI/WUI, 1), the share of economies whose pandemic share exceeds one half, and the pandemic-discussion index (WPUI T2, same scale), over the economies with both indices; groups by the IMF classification of the WPUI.

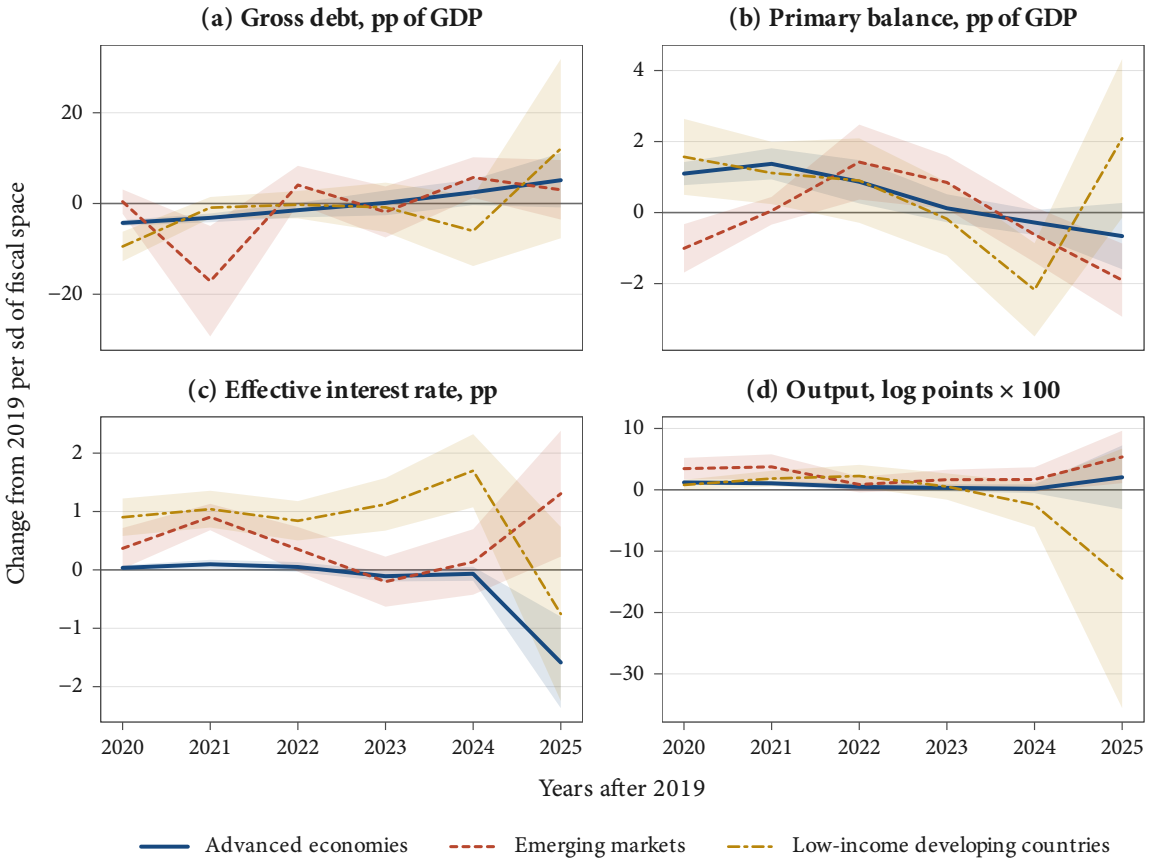


Figure C.12. Pre-pandemic fiscal space and outcomes after 2020, by income group

Notes: As Figure 6, with 2019 fiscal space interacted with the three income groups (group indicators included); 68% HC1 bands.

Table C.10. Status or space: uncertainty interactions with fiscal space against the AE indicator, the FX share and the rating

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: fiscal reaction function (Table 2, pooled), primary balance						
$\text{Debt}_{t-1} \times U$	−0.00343	−0.00255	0.000707	−0.00122	0.000428	
	(0.00172)	(0.00183)	(0.00292)	(0.00195)	(0.00315)	
AE × U		−0.255			0.0371	
		(0.155)			(0.228)	
$\text{FX share}_{t-1} \times U$			0.00434		0.00354	
			(0.00190)		(0.00302)	
$\text{Rating}_{t-1} \times U$				−0.0119	−0.000743	
				(0.0128)	(0.0172)	
U	0.107	0.117	−0.173	0.161	−0.158	
	(0.104)	(0.106)	(0.189)	(0.204)	(0.306)	
Output gap × U	0.00646	0.00630	−0.0242	−0.00290	−0.0231	
	(0.0325)	(0.0322)	(0.0573)	(0.0179)	(0.0587)	
Fixed effects	Country	Country	Country	Country	Country	
Instrumented output gap (first-stage F)	43.8	39.3	76.0	91.4	140.6	
Observations	3,834	3,834	752	2,823	744	
Economies	127	127	48	114	46	
Years	1990–2025	1990–2025	1991–2025	1991–2025	1991–2025	
Panel B: cyclical response (Table 3, pooled), change in the primary balance						
Output gap × U × $\tilde{d}$	0.00972	0.0137	0.00241	0.0197	0.00597	−0.00261
	(0.0294)	(0.0251)	(0.0204)	(0.0216)	(0.0253)	(0.0253)
Output gap × U × AE		0.00238			0.116	
		(0.0810)			(0.0969)	
Output gap × U × FX share			0.000225		0.00136	
			(0.00154)		(0.00138)	
Output gap × U × rating				−0.00368	−0.00702	
				(0.00778)	(0.0110)	
Output gap	0.236	0.0904	0.175	−0.0945	−0.304	0.237
	(0.107)	(0.102)	(0.103)	(0.0865)	(0.221)	(0.121)
Output gap × U	0.00114	−0.00735	−0.0211	0.0193	−0.0308	0.00653
	(0.0251)	(0.0399)	(0.0822)	(0.0927)	(0.135)	(0.0281)
Output gap × $\tilde{d}$	0.0174	0.0207	0.0255	0.0534	0.0771	0.0204
	(0.0246)	(0.0266)	(0.0296)	(0.0231)	(0.0349)	(0.0253)
Fixed effects	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year
Instrumented output gap (first-stage F)	24.5	63.3	326.0	134.3	13,812.9	19.6
Observations	3,832	3,832	747	2,822	739	3,592
Economies	127	127	48	114	46	127
Years	1990–2025	1990–2025	1996–2025	1991–2025	1996–2025	1990–2025

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses. Panel A adds to the pooled column of Table 2 the interactions of fiscal-year uncertainty U (sd) with the advanced-economy indicator, the lagged foreign-currency share of government debt (percent) and the lagged sovereign rating (1–21), one at a time and jointly (the level terms are included; AE is absorbed by the country effects); the samples shrink with the coverage of the FX share. Panel B adds to the pooled column of Table 3 the triple interactions of the output gap with U and the same variables (with the lower-order terms, instrumented as in Table 3; $\tilde{d}$ is lagged debt standardised as there); column (6) is the pooled specification of Table 3 with pandemic-purged uncertainty and 2020–2021 excluded. First-stage row: the smallest DK-robust first-stage Wald F across the instrumented variables.

Table C.11. Status or space, extended: uncertainty interactions with fiscal space against status, market access, institutions and rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	AE	EM	LIDC
Panel A: fiscal reaction function (Table 2, pooled), primary balance											
$\text{Debt}_{t-1} \times U$	−0.00343	−0.00255	0.000707	−0.00122	−0.000845	−0.00316	−0.00319	0.00191	0.00225	−0.000134	−0.00494
	(0.00172)	(0.00183)	(0.00292)	(0.00195)	(0.00218)	(0.00179)	(0.00177)	(0.00555)	(0.00325)	(0.00329)	(0.00275)
AE × U		−0.255						−0.776			
		(0.155)						(0.467)			
$\text{FX share}_{t-1} \times U$			0.00434					0.000508			
			(0.00190)					(0.00465)			
$\text{Rating}_{t-1} \times U$				−0.0119				0.0166			
				(0.0128)				(0.0247)			
$\text{Gov. effectiveness}_{t-1} \times U$					−0.183			0.232			
					(0.0779)			(0.278)			
Floating regime × U						−0.286		−0.432			
						(0.0972)		(0.186)			
Fiscal rule × U							−0.205	0.237			
							(0.206)	(0.301)			
U	0.107	0.117	−0.173	0.161	−0.0175	0.200	0.245	−0.121	−0.170	−0.146	0.306
	(0.104)	(0.106)	(0.189)	(0.204)	(0.141)	(0.120)	(0.160)	(0.673)	(0.221)	(0.178)	(0.180)
Output gap × U	0.00646	0.00630	−0.0242	−0.00290	0.00941	0.00178	0.00570	−0.0232	−0.00361	0.0404	−0.0455
	(0.0325)	(0.0322)	(0.0573)	(0.0179)	(0.0212)	(0.0323)	(0.0326)	(0.0641)	(0.0635)	(0.0594)	(0.0592)
Fixed effects	Country	Country	Country	Country	Country	Country	Country	Country	Country	Country	Country
Instrumented output gap (first-stage F)	43.8	39.3	76.0	91.4	53.1	43.6	46.7	158.8	178.9	36.2	16.3
Observations	3,834	3,834	752	2,823	2,889	3,822	3,834	643	931	1,762	1,141
Economies	127	127	48	114	127	127	127	44	28	58	41
Years	1990–2025	1990–2025	1991–2025	1991–2025	1997–2023	1990–2025	1990–2025	1997–2023	1990–2025	1990–2025	1990–2025
Panel B: cyclical response (Table 3, pooled), change in the primary balance											
Output gap × $U \times \tilde{d}$	0.00972	0.0137	0.00241	0.0197	0.0147	0.00820	0.00830	0.0401	0.0456	−0.0189	0.0137
	(0.0294)	(0.0251)	(0.0204)	(0.0216)	(0.0286)	(0.0315)	(0.0287)	(0.0476)	(0.0305)	(0.0345)	(0.0476)
Output gap × U × AE		0.00238						0.546			
		(0.0810)						(0.211)			
Output gap × U × FX share			0.000225					0.00121			
			(0.00154)					(0.00250)			
Output gap × U × rating				−0.00368				−0.0236			
				(0.00778)				(0.0146)			
Output gap × U × gov. effectiveness					0.0170			−0.152			
					(0.0286)			(0.0945)			
Output gap × U × floating						0.0926		0.295			
						(0.0510)		(0.147)			
Output gap × U × fiscal rule							0.0111	−0.0311			
							(0.0696)	(0.0632)			
Output gap	0.236	0.0904	0.175	−0.0945	0.128	0.244	0.229	−0.308	0.226	0.144	0.0997
	(0.107)	(0.102)	(0.103)	(0.0865)	(0.0972)	(0.112)	(0.0871)	(0.290)	(0.123)	(0.0995)	(0.0384)
Output gap × U	0.00114	−0.00735	−0.0211	0.0193	−0.0215	−0.0250	−0.00337	−0.115	−0.0654	0.0230	−0.0187
	(0.0251)	(0.0399)	(0.0822)	(0.0927)	(0.0152)	(0.0345)	(0.0421)	(0.168)	(0.0703)	(0.0514)	(0.0302)
Output gap × $\tilde{d}$	0.0174	0.0207	0.0255	0.0534	0.00660	0.0145	0.0213	0.0392	0.0106	0.0362	0.0294
	(0.0246)	(0.0266)	(0.0296)	(0.0231)	(0.0242)	(0.0237)	(0.0263)	(0.0517)	(0.0363)	(0.0328)	(0.0492)
Output gap × $U \times \tilde{d}$, fitted-gap just-identified	−0.0391	−0.0417	−0.000333	0.0210	−0.0296	−0.0346	−0.0378	0.0168	−0.0125	−0.0256	0.0725
	(0.0315)	(0.0291)	(0.0180)	(0.0174)	(0.0225)	(0.0294)	(0.0311)	(0.0308)	(0.0313)	(0.0283)	(0.0404)
Fixed effects	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year	Country, year
Instrumented output gap (first-stage F)	24.5	63.3	326.0	134.3	75.5	31.7	32.0	82,115,197,658.4	26.0	19.0	9.1
Robust first-stage F (min), fitted-gap	35.5	21.2	4.7	11.5	26.0	30.0	21.3	97.2	6.0	17.5	13.7
Observations	3,832	3,832	747	2,822	2,888	3,820	3,832	643	930	1,761	1,141
Economies	127	127	48	114	127	127	127	44	28	58	41
Years	1990–2025	1990–2025	1996–2025	1991–2025	1997–2023	1990–2025	1990–2025	1997–2023	1990–2025	1990–2025	1990–2025

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses. Panel A adds to the pooled column of Table 2 the interactions of fiscal-year uncertainty U (sd) with the advanced-economy indicator, the lagged foreign-currency share of government debt (percent), the lagged sovereign rating (1–21), lagged WGI government effectiveness, the floating-exchange-rate indicator and the fiscal-rule indicator (zero-filled, with its missing indicator also interacted), one at a time (columns 2–7) and jointly (column 8); the level terms are included where not absorbed;

C.3 Robustness

Table C.12 collects, in one row per check, the four objects under test across the 78 checks of this section, with the baseline in the first row and the checks grouped by family. K1 is the debt–uncertainty interaction of the pooled reaction function, in points of primary balance per point of debt per standard deviation of uncertainty. K2 is the triple interaction of the cyclicality model by the fitted-gap just-identified estimator, with the two-stage least squares estimate beside it. K3 is the debt response to a one-standard-deviation global shock at low minus high fiscal space, in points of GDP, at two and at five years, together with the effective-rate difference between the vulnerable and the safe regime at two years, in percentage points. K4 is the group medians of the debt a shock adds at two years relative to the average economy. A check re-estimates one baseline with one change and reports each object in the baseline's units, so that each row is read against the first. Across the families the two-year debt differential keeps its sign in 26 of 32 checks and the regime difference in 27 of 33, K2 is within 1.645 standard errors of zero in 49 of 63, and K1 keeps its sign in 40 of 59. The family subsections give the checks one by one.

Table C.12. The key estimates across robustness checks

Check	K1: debt × U (M2)	K2: ϕ_{UF}, fitted gap (M3)	K3: debt, $h = 2$ (M1)	K3: debt, $h = 5$ (M1)	K3: interest rate, $h = 2$ (M4)	K4: median debt added, AE / EM / LIDC	N (M2 / M3 / M1)
Baseline	−0.00343 (0.00172)	−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	0.68 / 0.06 / −0.13	3,834 / 3,832 / 2,766
Uncertainty measures (Table C.7, Figure C.8)							
Simple-average global WUI	0.00406 (0.00185)	−0.0336 (0.0227)	1.02 (0.941)	0.463 (0.696)	0.466 (0.215)	0.38 / 0.04 / −0.07	3,834 / 3,832 / 2,766
First-PC global WUI (1996–2025)	0.00583 (0.00181)	−0.0485 (0.0251)	0.986 (1.11)	0.348 (0.653)	0.645 (0.335)	0.37 / 0.03 / −0.07	3,443 / 3,441 / 2,434
Published GDP-weighted WUI	0.00227 (0.00162)	−0.0698 (0.0309)	1.84 (0.378)	0.760 (0.714)	0.602 (0.139)	0.69 / 0.06 / −0.13	3,834 / 3,832 / 2,766
GEPU (Baker–Bloom–Davis)	0.00515 (0.00142)	−0.0507 (0.0220)	1.47 (0.314)	1.05 (0.535)	0.449 (0.118)	0.55 / 0.05 / −0.11	3,355 / 3,353 / 2,367
Global geopolitical risk (GPR)	0.00138 (0.00175)	−0.00430 (0.0280)	−0.346 (0.307)	−0.00168 (0.219)	−0.0559 (0.102)	−0.13 / −0.01 / 0.03	3,834 / 3,832 / 2,766
VIX	−0.00520 (0.00187)	−0.0637 (0.0167)	0.492 (0.231)	0.416 (0.304)	0.274 (0.0794)	0.18 / 0.02 / −0.04	3,834 / 3,832 / 2,766
GDELT world FPU (2016–2025)	−0.000570 (0.00140)	−0.00796 (0.0148)	0.366 (0.695)	−0.364 (0.362)	0.389 (0.132)	0.14 / 0.01 / −0.03	1,110 / 1,110 / 751
Country geopolitical risk (GPR, 43 economies)	0.00163 (0.00361)	0.0405 (0.0462)					1,401 / 1,400 / —
GDELT domestic FPU (2016–2025)	−0.00119 (0.00344)	−0.0230 (0.0209)					954 / 954 / —
GDELT foreign FPU (2016–2025)	0.00583 (0.00268)	−0.0398 (0.0108)					1,110 / 1,110 / —
GDELT non-fiscal EPU, domestic (2016–2025)	0.000375 (0.00167)	−0.00299 (0.00924)					954 / 954 / —
Two-source: WUI: country (fiscal year) and global	−0.00427 (0.00228)						3,834 / — / —
Two-source: GDELT: domestic and foreign FPU	−0.00693 (0.00408)						954 / — / —
Two-source: GPR: country and global	0.000414 (0.00342)						1,401 / — / —
Timing (Table C.8)							
Calendar-year WUI	−0.00382 (0.00172)	−0.0393 (0.0322)					3,834 / 3,832 / —
Pre-budget WUI (two quarters before the fiscal year)	−0.00334 (0.00237)	−0.0262 (0.0284)					3,833 / 3,831 / —
Lagged calendar-year WUI	−0.00354 (0.00226)	−0.0374 (0.0286)					3,833 / 3,831 / —
Fourth-quarter WUI	−0.00223 (0.00143)	−0.0628 (0.0319)					3,834 / 3,832 / —
GDELT domestic FPU, calendar basis (2016–2025)	−0.00119 (0.00344)	−0.0230 (0.0209)					954 / 954 / —
GDELT domestic FPU, fiscal-year basis	−0.00131 (0.00350)	−0.0224 (0.0210)					954 / 954 / —
GDELT domestic FPU, pre-budget basis	−0.000797 (0.00286)	0.0121 (0.0167)					841 / 841 / —
Pandemic (Table C.9, Figure C.7)							
Pandemic-purged uncertainty everywhere (wui_np, eps_glob_np)	−0.00284 (0.00161)	−0.0457 (0.0335)	1.27 (0.342)	0.434 (0.689)	0.413 (0.211)	0.48 / 0.04 / −0.09	3,834 / 3,832 / 2,766
2020–2021 excluded	−0.00293 (0.00173)	−0.0431 (0.0318)	0.919 (0.399)	0.654 (0.516)	0.693 (0.220)	0.34 / 0.03 / −0.07	3,594 / 3,592 / 2,354
2020–2022 excluded	−0.00269 (0.00164)	−0.0392 (0.0329)	1.01 (0.402)	0.654 (0.516)	0.677 (0.208)	0.38 / 0.03 / −0.09	3,475 / 3,473 / 2,259
2009 excluded	−0.00391 (0.00173)	−0.0379 (0.0305)	1.95 (0.375)	0.343 (0.667)	0.597 (0.168)	0.73 / 0.07 / −0.14	3,709 / 3,707 / 2,471
2009 and 2020–2021 excluded	−0.00327 (0.00171)	−0.0422 (0.0300)	1.17 (0.399)	0.00276 (0.467)	0.739 (0.328)	0.44 / 0.04 / −0.09	3,469 / 3,467 / 2,059
World pandemic uncertainty added (direct-effect specifications)	−0.00182 (0.00192)						3,443 / — / —
Pandemic-discussion index as a control (M2, M3)	−0.00207 (0.00187)	−0.0293 (0.0318)					3,442 / 3,440 / —
Samples (Table C.10, Figure C.9)							
Advanced economies only	0.00225 (0.00325)	−0.0125 (0.0313)	2.02 (0.581)	−0.977 (0.761)	−0.0658 (0.297)	0.76 / — / —	931 / 930 / 773
Emerging markets only	−0.000134 (0.00329)	−0.0256 (0.0283)	−0.299 (1.02)	1.36 (1.54)	1.05 (0.275)	— / −0.01 / —	1,762 / 1,761 / 1,275
Low-income developing countries only	−0.00494 (0.00275)	0.0725 (0.0404)	3.51 (1.46)	5.74 (4.17)	0.384 (0.413)	— / — / −0.26	1,141 / 1,141 / 718
Commodity exporters excluded (59 economies dropped)	−0.00303 (0.00280)	−0.00149 (0.0229)	1.84 (0.597)	0.406 (0.630)	−0.0740 (0.140)	0.86 / 0.46 / −0.39	2,228 / 2,226 / 1,781
Small states excluded (median population below 1 million; 0 economies dropped)	−0.00343 (0.00172)	−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	0.68 / 0.06 / −0.13	3,834 / 3,832 / 2,766
1990–2007	−0.000869 (0.00469)	−0.0657 (0.0646)	0.158 (0.419)	1.15 (1.10)	0.419 (0.246)	−0.04 / −0.08 / −0.07	1,731 / 1,729 / 998
2008–2019	−0.000587 (0.00224)	0.00765 (0.00987)	0.939 (0.576)	0.00166 (0.376)	0.268 (0.406)	0.25 / −0.07 / −0.20	1,484 / 1,484 / 1,038
2020–2025	−0.00152 (0.00299)	0.0206 (0.0423)	0.880 (0.632)		0.245 (0.424)	0.33 / 0.03 / −0.07	618 / 618 / 328
Economies with at least 25 years (123 economies)	−0.00154 (0.00187)	0.0268 (0.0237)	1.84 (0.433)	0.615 (0.694)	0.603 (0.142)	0.83 / 0.19 / −0.18	3,597 / 3,595 / 2,632
Outcomes (Table C.11)							
Cyclically adjusted primary balance (approximation)	−0.00291 (0.00145)	−0.0844 (0.0440)					3,782 / 3,780 / —
WEO structural primary balance	−0.00139 (0.00169)	−0.0288 (0.0150)					1,874 / 1,872 / —
Revenue (M3 outcome)		−0.0189 (0.0143)					— / 3,780 / —
Primary spending (M3 outcome)		0.0185 (0.0245)					— / 3,780 / —
Fiscal space (Table C.12, Figure C.10)							
Lagged debt (M1, M3, M5 baseline); debt also as the M4 and Figure 4 regime variable		−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.244 (0.243)	0.68 / 0.06 / −0.13	— / 3,832 / 2,766
Composite vulnerability index (M4 baseline)		0.0156 (0.0141)	1.06 (0.441)	1.38 (1.19)	0.599 (0.130)	−0.23 / 0.25 / 0.56	— / 2,839 / 2,239
Sovereign rating (sign reversed)		0.00308 (0.0210)	−0.753 (0.798)	0.917 (2.09)	0.400 (0.172)	0.48 / −0.22 / −0.52	— / 2,822 / 2,216
Foreign-currency share of government debt		0.0175 (0.0247)	−1.22 (0.410)	−1.34 (0.620)	−0.258 (0.245)	0.69 / −0.30 / −1.01	— / 747 / 635
Debt maturing within 12 months, % of GDP		0.0346 (0.0176)	0.498 (0.181)	−0.158 (0.533)	0.0692 (0.288)	0.10 / −0.16 / −0.35	— / 1,366 / 997
Average maturity of sovereign debt (sign reversed)		−0.0125 (0.0135)	−0.188 (0.224)	−0.266 (0.413)	−0.167 (0.112)	0.04 / 0.02 / 0.01	— / 927 / 687

Table C.12 (continued)

Check	K1: debt × U (M2)	K2: ϕ_{UF}, fitted gap (M3)	K3: debt, $h = 2$ (M1)	K3: debt, $h = 5$ (M1)	K3: interest rate, $h = 2$ (M4)	K4: median debt added, AE / EM / LIDC	N (M2 / M3 / M1)
First-PC composite (mixed loadings)		−0.00962 (0.0260)	−1.49 (0.467)	0.551 (1.50)	0.111 (0.181)	0.99 / −0.28 / −0.89	— / 2,839 / 2,239
Estimation (Table C.13, Figure C.11)							
M2 by system GMM (two-step, Windmeijer, collapsed lags 2–4)	−0.0457 (0.0887)						8,763 / — / —
HP-filter output gap in place of the Hamilton gap (M2, M3)	−0.000973 (0.00244)	−0.123 (0.113)					3,794 / 3,792 / —
WEO output gap in place of the Hamilton gap (M2, M3)	0.00434 (0.00269)	0.0242 (0.0269)					760 / 759 / —
Country-specific linear trends (M2, M3)	−0.00102 (0.00190)	−0.0492 (0.0262)					3,834 / 3,832 / —
Three lags of the controls (M1, M4)			1.93 (0.433)	0.615 (0.673)	0.681 (0.164)	0.72 / 0.07 / −0.14	— / — / 2,682
The outcome's own lag added (M1, M4)			1.92 (0.444)	0.698 (0.698)	0.299 (0.148)	0.72 / 0.07 / −0.14	— / — / 2,743
Transition speed $\kappa = 1$ (M4)					0.726 (0.165)	0.68 / 0.06 / −0.13	— / — / —
Transition speed $\kappa = 3$ (M4)					0.474 (0.104)	0.68 / 0.06 / −0.13	— / — / —
M1 without the outlier rule			2.88 (0.928)	1.72 (1.51)		1.08 / 0.29 / −0.11	— / — / 3,518
Output gap not instrumented (OLS, M2 and M3)	−0.00254 (0.00167)	−0.00827 (0.0116)					3,834 / 3,832 / —
M1 with the components winsorised at the 1st/99th percentiles, no outlier rule			1.49 (0.390)	0.480 (1.01)		0.56 / 0.15 / −0.06	— / — / 3,518
Hansen threshold on lagged debt in place of the smooth transition (M4)					−0.00878 (0.467)		— / — / —
Inference (Table C.14)							
Two-way clustering by economy and year (K1, K2 unchanged; SEs in Table C.14)	−0.00343 (0.00196)	−0.0391 (0.0348)					3,834 / 3,832 / —
Wild cluster bootstrap by year (SEs by year; p in Table C.14)	−0.00343 (0.00203)	−0.0391 (0.0296)					3,834 / 3,832 / —
Wild cluster bootstrap by economy	−0.00343 (0.00191)	−0.0391 (0.0368)					3,834 / 3,832 / —
GDELT (Table C.15)							
Stable-outlet-panel indices (domestic; foreign in the split)	0.000370 (0.00425)	−0.0643 (0.0208)					954 / 954 / —
Native-language domestic index	−0.00271 (0.00340)	−0.0178 (0.0230)					919 / 919 / —
Own-country-mention domestic index	0.0000800 (0.00382)	−0.0276 (0.0259)					954 / 954 / —
Economies with an outlet-composition break dropped (39 economies)	−0.00181 (0.00453)	−0.0273 (0.0222)					740 / 740 / —
Thin country-years kept (190 added)	0.000355 (0.00245)	−0.00875 (0.0254)					1,102 / 1,102 / —
All-outlet index (domestic and foreign together)	0.00522 (0.00367)	−0.0452 (0.0187)					954 / 954 / —
Domestic minus foreign FPU gap as the regressor	0.0101 (0.00452)	−0.0198 (0.0208)					954 / 954 / —
Fiscal-attention index (coverage, no uncertainty)	0.00673 (0.00373)	−0.0241 (0.0123)					954 / 954 / —
Status or space (Table C.5b)							
U × AE added	−0.00255 (0.00183)	−0.0417 (0.0291)					3,834 / 3,832 / —
U × FX share added	0.000707 (0.00292)	−0.000333 (0.0180)					752 / 747 / —
U × rating added	−0.00122 (0.00195)	0.0210 (0.0174)					2,823 / 2,822 / —
U × government effectiveness added	−0.000845 (0.00218)	−0.0296 (0.0225)					2,889 / 2,888 / —
U × floating regime added	−0.00316 (0.00179)	−0.0346 (0.0294)					3,822 / 3,820 / —
U × fiscal rule added	−0.00319 (0.00177)	−0.0378 (0.0311)					3,834 / 3,832 / —
All six interactions added	0.00191 (0.00555)	0.0168 (0.0308)					643 / 643 / —

Notes: Entries are the four objects under test re-estimated with one change per row (estimate, with the Driscoll–Kraay standard error in parentheses, lag 2). K1 is the interaction of lagged debt with uncertainty in the pooled reaction function of Table 2 (pp of primary balance per pp of debt per sd of U); K2 the triple interaction of the output gap with uncertainty and fiscal space in the pooled cyclicality model of Table 3, by the fitted-gap just-identified estimator of Table C.6; K3 the cumulative debt response to a one-sd global uncertainty shock at low minus high fiscal space (lagged debt at the stage-2 p75 minus p25) from the local projections of Figure 3, and the effective-interest-rate difference between the vulnerable and the safe regime at $h = 2$ from the smooth-transition model of Figure 5; K4 the group medians of the debt added at $h = 2$ relative to the mean-debt economy from Figure 4. Driscoll–Kraay standard errors (lag 2) in parentheses; an empty cell means the check does not alter that model. The inference rows repeat the baseline estimates with the standard error of the alternative covariance or bootstrap. Rows are grouped by family; each family's own table or figure carries the full specification. The GDELT rows are 2016–2025.

C.3.1 Uncertainty measures. The baseline measures global uncertainty by the GDP-weighted WUI, and the question is whether another reading of global or of country uncertainty gives the same objects. Table C.13 re-estimates the reaction function and the cyclicality model with each alternative in place of the fiscal-year WUI, and Figure C.13 re-estimates the debt differential with the shock of each global series, the standardised innovation of the same autoregression fitted to that series. The two-year debt differential keeps its sign for 6 of the 7 global series. The exception is geopolitical risk, with −0.35 (SE 0.31) points of GDP, the one global series that neither erodes space nor separates the regimes, and the VIX gives the smallest positive value, 0.49 (SE 0.23). The debt interaction of the reaction function is the fragile object in this family. It is negative for 3 of the 11 alternative measures, and a global series that varies by year only carries the positive global interaction of Table C.4 rather than the country interaction, so the sign switches are the two-source reading. The cyclicality null survives with every measure. The index of non-fiscal economic policy uncertainty built from the same GDELT pipeline gives a debt interaction of 0.0004 (SE 0.0017), smaller in absolute value than the fiscal index gives. The two-source split holds in the point estimates and is rejected at the 5% level for the WUI and the GDELT pairs but not for the geopolitical pair, the 3 pairs with a domestic and a global reading.

Table C.13. The reaction function and the cyclicality model across uncertainty measures

	Reaction function (M2)						Cyclicality model (M3)				
Measure	U (sd)	Debt$_{t-1} \times U$	Gap $\times U$	First-stage F	N	Econ.	ϕ_{UF}, fitted gap	ϕ_{UF}, 2SLS	High U, high space	High U, low space	N
Panel A: one measure in place of fiscal-year WUI											
Fiscal-year WUI; GDP-weighted global WUI shock (baseline)	0.107 (0.104)	−0.00343 (0.00172)	0.00646 (0.0325)	43.8	3,834	127	−0.0391 (0.0315)	0.00972 (0.0294)	0.342 (0.105)	0.370 (0.0988)	3,832
Simple-average global WUI	−0.777 (0.160)	0.00406 (0.00185)	−0.0172 (0.0281)	60.4	3,834	127	−0.0336 (0.0227)	−0.00237 (0.0285)	0.340 (0.112)	0.371 (0.104)	3,832
First-PC global WUI (1996–2025)	−0.767 (0.171)	0.00583 (0.00181)	−0.0103 (0.0267)	84.8	3,443	127	−0.0485 (0.0251)	0.0152 (0.0253)	0.324 (0.125)	0.354 (0.116)	3,441
Published GDP-weighted WUI	−0.548 (0.214)	0.00227 (0.00162)	−0.0113 (0.0364)	340.2	3,834	127	−0.0698 (0.0309)	−0.0215 (0.0265)	0.344 (0.113)	0.368 (0.105)	3,832
GEPU (Baker–Bloom–Davis)	−0.823 (0.186)	0.00515 (0.00142)	0.0222 (0.0133)	108.7	3,355	127	−0.0507 (0.0220)	0.0141 (0.0212)	0.233 (0.126)	0.253 (0.111)	3,353
Global geopolitical risk (GPR)	−0.0970 (0.171)	0.00138 (0.00175)	−0.0228 (0.0694)	12.3	3,834	127	−0.00430 (0.0280)	−0.0232 (0.0201)	0.299 (0.0964)	0.361 (0.0976)	3,832
VIX	0.438 (0.227)	−0.00520 (0.00187)	−0.0138 (0.0370)	30.7	3,834	127	−0.0637 (0.0167)	−0.0507 (0.0231)	0.332 (0.0998)	0.345 (0.108)	3,832
GDELT world FPU (2016–2025)	−0.411 (0.0936)	−0.000570 (0.00140)	0.0709 (0.0217)	2,683.4	1,110	124	−0.00796 (0.0148)	−0.00903 (0.0248)	0.0264 (0.0635)	0.0636 (0.0546)	1,110
Country geopolitical risk (GPR, 43 economies)	−0.275 (0.217)	0.00163 (0.00361)	−0.0145 (0.0424)	17.8	1,401	42	0.0405 (0.0462)	0.0830 (0.0693)	0.336 (0.114)	0.370 (0.101)	1,400
GDELT domestic FPU (2016–2025)	−0.0878 (0.126)	−0.00119 (0.00344)	0.105 (0.0385)	42.9	954	114	−0.0230 (0.0209)	−0.0196 (0.0390)	−0.00502 (0.0593)	0.0230 (0.0463)	954
GDELT foreign FPU (2016–2025)	−0.514 (0.0799)	0.00583 (0.00268)	0.0474 (0.0389)	127.3	1,110	124	−0.0398 (0.0108)	−0.0274 (0.0387)	0.0345 (0.0642)	0.0585 (0.0449)	1,110
GDELT non-fiscal EPU, domestic (2016–2025)	−0.0993 (0.116)	0.000375 (0.00167)	0.105 (0.0379)	50.8	954	114	−0.00299 (0.00924)	−0.0426 (0.0334)	−0.0101 (0.0499)	0.0248 (0.0433)	954
Panel B: two sources together (debt × domestic, debt × global or foreign, difference)											
WUI: country (fiscal year) and global	−0.00427 (0.00228)	0.00399 (0.00206)	−0.00826 (0.00394) [0.036]	154.6	3,834	127					
GDELT: domestic and foreign FPU	−0.00693 (0.00408)	0.0101 (0.00433)	−0.0170 (0.00765) [0.026]	204.8	954	114					
GPR: country and global	0.000414 (0.00342)	0.00265 (0.00201)	−0.00223 (0.00445) [0.616]	20.4	1,401	42					

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses, from the pooled fiscal reaction function of Table 2 (columns 2–7: primary balance in percent of GDP; the debt interaction in pp per pp of debt per sd of U) and the pooled cyclicality model of Table 3 (columns 8–12: change in the primary balance; ϕ_{UF} by the fitted-gap just-identified estimator of Table C.6 and by 2SLS with twelve instruments; the implied cyclical response under high uncertainty, pp per pp of gap, at the 25th and 75th percentiles of lagged debt), with the uncertainty measure of each row in place of fiscal-year WUI. Every measure is standardised within the main sample. A series that varies by year only enters the cyclicality model without a main effect (absorbed by the year effects). GEPU is the Baker–Bloom–Davis global index (annual means of non-imputed months); GDELT indices are 2016–2025 (domestic: non-thin country-years). Panel B re-estimates the two-source reaction function of Table C.4 for each pair: column 2 reports debt × the country or domestic measure, column 3 debt × the global or foreign measure, column 4 their difference with its standard error and p-value in brackets. First-stage F: the smallest DK-robust first-stage Wald F across the instrumented variables.

C.3.2 Timing of the uncertainty measure. The fiscal-year index aligns uncertainty with the budget year, and the question is whether the alignment matters. Table C.14 re-estimates the two models with the calendar-year index, the pre-budget index, the lagged calendar index and the fourth-quarter index, and with the GDELT domestic index on its calendar, fiscal-year and pre-budget bases. The debt interaction lies between −0.0038 and −0.0022 across the 4 WUI timings, each within one baseline standard error of the baseline, and the cyclicality null survives in each. The GDELT pre-budget basis correlates with the calendar index at 0.32 against 0.98 for the fiscal-year basis, and its debt interaction falls to less than half the baseline, as the calendar-basis GDELT interaction already does.

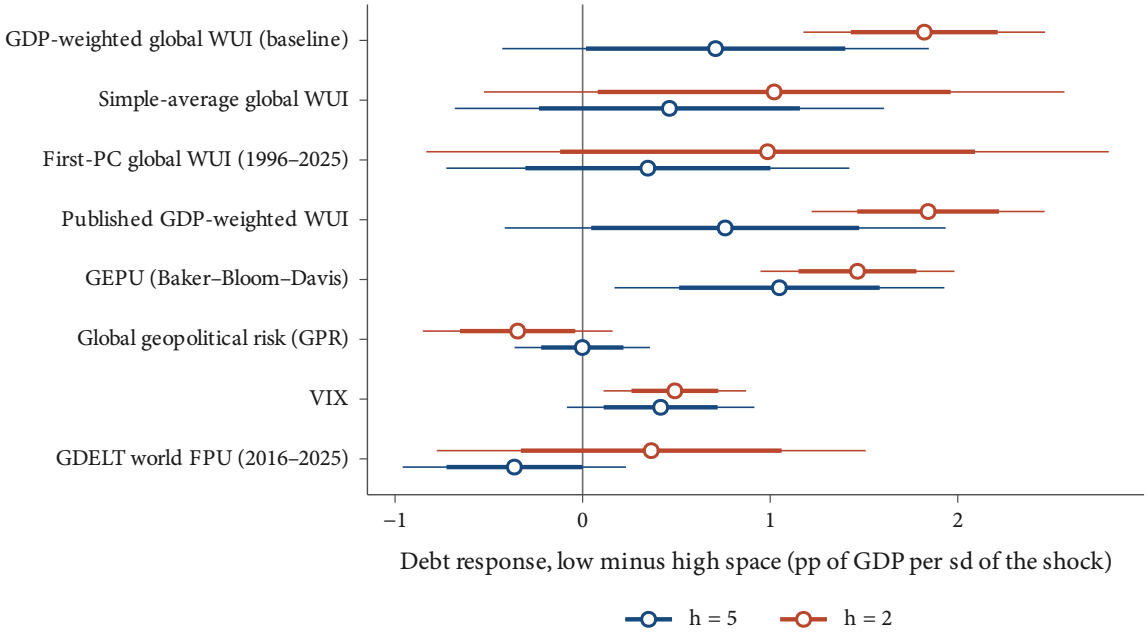


Figure C.13. The debt response to a global uncertainty shock across measures of global uncertainty

Notes: The Figure 3 debt response at low space (debt p75 = 66% of GDP) minus high space (p25 = 31%), pp of GDP per sd of the global shock, with the shock recomputed from each series (standardised AR(2) residual; GDELT: standardised level). Thick bars 68%, thin bars 90% DK bands (lag 2). Cumulative local projections with country and year effects.

Table C.14. The reaction function and the cyclicality model across timings of the uncertainty measure

	Reaction function (M2)						Cyclicality model (M3)				
Measure	U (sd)	$\text{Debt}_{t-1} \times U$	Gap $\times U$	First-stage F	N	Econ.	ϕ_{UF}, fitted gap	ϕ_{UF}, 2SLS	High U, high space	High U, low space	N
Fiscal-year WUI (baseline)	0.107 (0.104)	−0.00343 (0.00172)	0.00646 (0.0325)	43.8	3,834	127	−0.0391 (0.0315)	0.00972 (0.0294)	0.342 (0.105)	0.370 (0.0988)	3,832
Calendar-year WUI	0.153 (0.0999)	−0.00382 (0.00172)	0.00455 (0.0313)	63.6	3,834	127	−0.0393 (0.0322)	0.00854 (0.0301)	0.341 (0.105)	0.369 (0.0990)	3,832
Pre-budget WUI (two quarters before the fiscal year)	0.0900 (0.145)	−0.00334 (0.00237)	−0.0123 (0.0260)	28.8	3,833	127	−0.0262 (0.0284)	0.0427 (0.0340)	0.321 (0.104)	0.364 (0.0973)	3,831
Lagged calendar-year WUI	0.0512 (0.157)	−0.00354 (0.00226)	0.00695 (0.0222)	29.8	3,833	127	−0.0374 (0.0286)	0.0187 (0.0330)	0.336 (0.105)	0.367 (0.0973)	3,831
Fourth-quarter WUI	0.105 (0.104)	−0.00223 (0.00143)	0.00495 (0.0371)	36.8	3,834	127	−0.0628 (0.0319)	−0.00240 (0.0207)	0.348 (0.103)	0.363 (0.0986)	3,832
GDELT domestic FPU, calendar basis (2016–2025)	−0.0878 (0.126)	−0.00119 (0.00344)	0.105 (0.0385)	42.9	954	114	−0.0230 (0.0209)	−0.0196 (0.0390)	−0.00502 (0.0593)	0.0230 (0.0463)	954
GDELT domestic FPU, fiscal-year basis	−0.0826 (0.138)	−0.00131 (0.00350)	0.113 (0.0395)	42.6	954	114	−0.0224 (0.0210)	−0.0124 (0.0393)	−0.00320 (0.0577)	0.0246 (0.0453)	954
GDELT domestic FPU, pre-budget basis	0.139 (0.282)	−0.000797 (0.00286)	−0.0398 (0.0249)	1,122.5	841	109	0.0121 (0.0167)	0.0333 (0.0290)	−0.0769 (0.0603)	−0.0451 (0.0525)	841

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses, from the pooled fiscal reaction function of Table 2 (columns 2–7; primary balance in percent of GDP) and the pooled cyclicality model of Table 3 (columns 8–12; change in the primary balance; ϕ_{UF} by the fitted-gap just-identified estimator of Table C.6 and by 2SLS; the implied cyclical response under high uncertainty at the 25th and 75th percentiles of lagged debt), with the uncertainty measure timed as stated in each row, standardised within the main sample. Fiscal year: WUI averaged over the economy's fiscal year (WEO labelling); pre-budget: the two quarters before the fiscal year starts; lagged: the previous calendar year; fourth quarter: the last quarter of the calendar year. GDELT rows: the domestic fiscal-policy-uncertainty index on the calendar, fiscal-year and pre-budget (six months before the fiscal year) bases, 2016–2025, non-thin country-years; the pre-budget basis is correlated 0.32 with the calendar index (fiscal-year basis 0.98). First-stage F: the smallest DK-robust first-stage Wald F across the instrumented variables.

C.3.3 The pandemic years. The pandemic years carry the largest uncertainty readings and the largest common debt surge of the sample, and the question is whether the estimates are those two years. Table C.15 purges the pandemic component from every uncertainty regressor at once, excludes 2020–2021, 2020–2022, 2009, and 2009 with 2020–2021, adds the world pandemic index to the specifications without year effects, and adds the pandemic-discussion index as a control. Figure C.14 repeats the regime model with the purged shock. With the purged index and shock the two-year debt differential is 1.27 (SE 0.34), within one baseline standard error of the baseline, and the regime difference 0.41 (SE 0.21), within 1.645, and the debt interaction is within one as well. Without 2020–2021 the debt differential is 0.92 (SE 0.40), half the baseline, and without 2009 it is 1.95 (SE 0.37); it keeps its sign in all 5 sample exclusions. The regime difference strengthens when the pandemic years are excluded, and the cyclicality null survives in every row.

Table C.15. The key estimates and the pandemic years

		WUI two-source split							
Check	K1: debt × U	Debt × U^d	Debt × U^g	Difference [p]	K2: ϕ_{UF}	K3: debt, $h = 2$	K3: debt, $h = 5$	K3: interest rate, $h = 2$	N (M2 / M3 / M1)
Baseline (fiscal-year WUI, global WUI shock, 1990–2025)	−0.00343 (0.00172)	−0.00427 (0.00228)	0.00399 (0.00206)	−0.00826 (0.00394) [0.036]	−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	3,834 / 3,832 / 2,766
Pandemic-purged uncertainty everywhere (wui_np, eps_glob_np)	−0.00284 (0.00161)	−0.00347 (0.00214)	0.00314 (0.00183)	−0.00662 (0.00362) [0.068]	−0.0457 (0.0335)	1.27 (0.342)	0.434 (0.689)	0.413 (0.211)	3,834 / 3,832 / 2,766
2020–2021 excluded	−0.00293 (0.00173)	−0.00342 (0.00205)	0.00355 (0.00195)	−0.00697 (0.00359) [0.052]	−0.0431 (0.0318)	0.919 (0.399)	0.654 (0.516)	0.693 (0.220)	3,594 / 3,592 / 2,354
2020–2022 excluded	−0.00269 (0.00164)	−0.00340 (0.00204)	0.00357 (0.00201)	−0.00698 (0.00361) [0.054]	−0.0392 (0.0329)	1.01 (0.402)	0.654 (0.516)	0.677 (0.208)	3,475 / 3,473 / 2,259
2009 excluded	−0.00391 (0.00173)	−0.00471 (0.00233)	0.00404 (0.00204)	−0.00875 (0.00393) [0.026]	−0.0379 (0.0305)	1.95 (0.375)	0.343 (0.667)	0.597 (0.168)	3,709 / 3,707 / 2,471
2009 and 2020–2021 excluded	−0.00327 (0.00171)	−0.00327 (0.00176)	0.00414 (0.00197)	−0.00741 (0.00320) [0.021]	−0.0422 (0.0300)	1.17 (0.399)	0.00276 (0.467)	0.739 (0.328)	3,469 / 3,467 / 2,059
World pandemic uncertainty added (direct-effect specifications)	−0.00182 (0.00192)	−0.00324 (0.00233)	0.00483 (0.00203)	−0.00807 (0.00384) [0.036]					3,443 / — / —
Pandemic-discussion index as a control (M2, M3)	−0.00207 (0.00187)	−0.00351 (0.00225)	0.00496 (0.00203)	−0.00847 (0.00378) [0.025]	−0.0293 (0.0318)				3,442 / 3,440 / —

Notes: Entries are the objects under test with one change to the treatment of the pandemic years. K1 is the interaction of lagged debt with uncertainty in the pooled reaction function of Table 2 (pp of primary balance per pp of debt per sd of U); K2 the triple interaction of the output gap with uncertainty and fiscal space in the pooled cyclicality model of Table 3, by the fitted-gap just-identified estimator of Table C.6; K3 the cumulative debt response to a one-sd global uncertainty shock at low minus high fiscal space (lagged debt at the stage-2 p75 minus p25) from the local projections of Figure 3, and the effective-interest-rate difference between the vulnerable and the safe regime at $h = 2$ from the smooth-transition model of Figure 5; K4 the group medians of the debt added at $h = 2$ relative to the mean-debt economy from Figure 4. Driscoll–Kraay standard errors (lag 2) in parentheses; an empty cell means the check does not alter that model. Columns 3–5 re-estimate the two-source reaction function of Table C.4 with country WUI (U^d) and global WUI (U^g) (pandemic-purged versions in the first row); the difference carries its standard error and p-value. Excluded years are dropped from every model and from every cumulative window of the local projections. World pandemic uncertainty (WPUI, published global index) enters the reaction function as a regressor in the sixth row (the year effects of the other models absorb it); the pandemic-discussion index (share of pandemic words, country level) enters M2 and M3 as a control in the last row (both available from 1996).

C.3.4 Samples. Table 2 rejects the equality of the advanced and the other reaction functions, and the question is where each object lives. Table C.16 re-estimates every model on the three income groups alone, without commodity exporters, without small states, on three sub-periods and on the economies with at least 25 years, and Figure C.15 plots the debt interaction by decade. The two-year debt differential is 2.02 (SE 0.58) in the advanced economies, −0.30 (SE 1.02) in the emerging markets and 3.51 (SE 1.46) in the low-income economies, so the pooled estimate is carried by the first and the third group, while the regime difference is sharpest in the emerging markets, 1.05 (SE 0.27). The debt interaction by group and by decade is within one standard error of zero in every cell. Without the 59 commodity exporters the differential is 1.84 (SE 0.60) but the regime difference falls to −0.07 (SE 0.14), so the price of debt after a global shock is paid where commodity revenues move with the shock. No economy of the main sample has a median population below one million, so the small-state row repeats the baseline and is kept on record. Over 1990–2007 the differential is 0.16 (SE 0.42), so the pre-crisis years alone do not carry the erosion result, whereas the exclusion of both common surges in Table C.15 leaves it at 1.17 (SE 0.40). The cyclicality triple interaction differs from zero at the 10% level in 1 of the 9 subsamples.

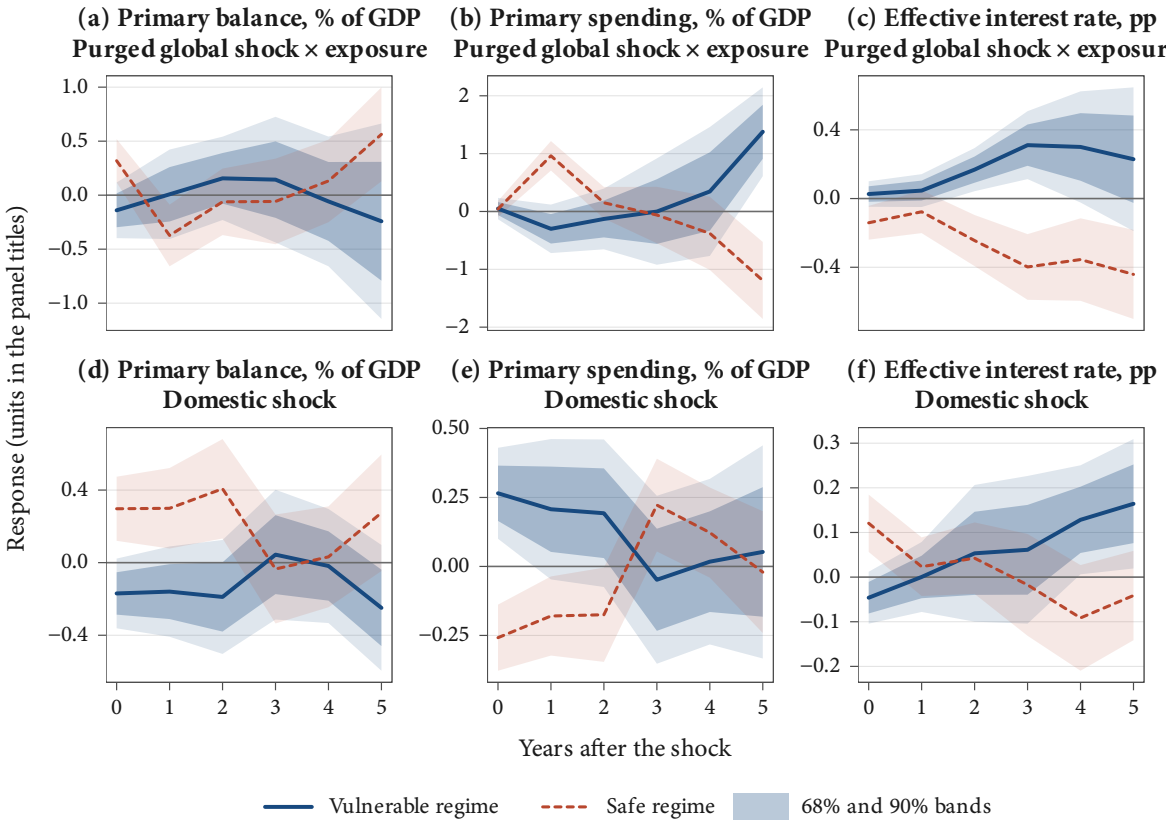


Figure C.14. Regime responses with the pandemic-purged global uncertainty shock (M4)

Notes: As Figure 5, rows 1–2, with the global shock replaced by the standardised AR(2) residual of the pandemic-purged global WUI (eps_glob_np); regime weight $F(z)$ on lagged standardised vulnerability, $\kappa = 1.5$. DK bands (lag 2); 122 economies, 2,845 country-years at h = 0.

Table C.16. The key estimates across subsamples

Check	K1: debt × U (M2)	K2: ϕ_{UF}, fitted gap (M3)	K3: debt, $h = 2$ (M1)	K3: debt, $h = 5$ (M1)	K3: interest rate, $h = 2$ (M4)	K4: median debt added, AE / EM / LIDC	N (M2 / M3 / M1)
Baseline (pooled, 1990–2025)	−0.00343 (0.00172)	−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	0.68 / 0.06 / −0.13	3,834 / 3,832 / 2,766
Advanced economies only	0.00225 (0.00325)	−0.0125 (0.0313)	2.02 (0.581)	−0.977 (0.761)	−0.0658 (0.297)	0.76 / — / —	931 / 930 / 773
Emerging markets only	−0.000134 (0.00329)	−0.0256 (0.0283)	−0.299 (1.02)	1.36 (1.54)	1.05 (0.275)	— / −0.01 / —	1,762 / 1,761 / 1,275
Low-income developing countries only	−0.00494 (0.00275)	0.0725 (0.0404)	3.51 (1.46)	5.74 (4.17)	0.384 (0.413)	— / — / −0.26	1,141 / 1,141 / 718
Commodity exporters excluded (59 economies dropped)	−0.00303 (0.00280)	−0.00149 (0.0229)	1.84 (0.597)	0.406 (0.630)	−0.0740 (0.140)	0.86 / 0.46 / −0.39	2,228 / 2,226 / 1,781
Small states excluded (median population below 1 million; 0 economies dropped)	−0.00343 (0.00172)	−0.0391 (0.0315)	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	0.68 / 0.06 / −0.13	3,834 / 3,832 / 2,766
1990–2007	−0.000869 (0.00469)	−0.0657 (0.0646)	0.158 (0.419)	1.15 (1.10)	0.419 (0.246)	−0.04 / −0.08 / −0.07	1,731 / 1,729 / 998
2008–2019	−0.000587 (0.00224)	0.00765 (0.00987)	0.939 (0.576)	0.00166 (0.376)	0.268 (0.406)	0.25 / −0.07 / −0.20	1,484 / 1,484 / 1,038
2020–2025	−0.00152 (0.00299)	0.0206 (0.0423)	0.880 (0.632)		0.245 (0.424)	0.33 / 0.03 / −0.07	618 / 618 / 328
Economies with at least 25 years (123 economies)	−0.00154 (0.00187)	0.0268 (0.0237)	1.84 (0.433)	0.615 (0.694)	0.603 (0.142)	0.83 / 0.19 / −0.18	3,597 / 3,595 / 2,632

Notes: Entries are the objects under test re-estimated on each subsample. K1 is the interaction of lagged debt with uncertainty in the pooled reaction function of Table 2 (pp of primary balance per pp of debt per sd of U); K2 the triple interaction of the output gap with uncertainty and fiscal space in the pooled cyclicality model of Table 3, by the fitted-gap just-identified estimator of Table C.6; K3 the cumulative debt response to a one-sd global uncertainty shock at low minus high fiscal space (lagged debt at the stage-2 p75 minus p25) from the local projections of Figure 3, and the effective-interest-rate difference between the vulnerable and the safe regime at $h = 2$ from the smooth-transition model of Figure 5; K4 the group medians of the debt added at $h = 2$ relative to the mean-debt economy from Figure 4. Driscoll–Kraay standard errors (lag 2) in parentheses; an empty cell means the check does not alter that model. Groups follow the IMF WPUI classification with the stage-1 fallback; fiscal space in M1 is standardised and evaluated at the pooled p25 and p75 of lagged debt in every subsample, so that the K3 columns are comparable, while M3 standardises fiscal space within the subsample. Commodity exporters: economies with a 1995–2004 commodity export share above 50% (18 economies in the sample have no share and are kept). Period subsamples drop the other years from every model and from every cumulative window. The K4 medians of a group subsample are those of the group itself.

C.3.5 Outcomes. The reaction function and the cyclicality model are about the discretionary balance, and the question is whether the headline primary balance misreads it. Table C.17 replaces the balance by the cyclically adjusted primary balance, by the WEO structural primary balance, available for 70 economies, and, in the cyclicality model, by revenue and by primary spending. The debt interaction is −0.0029 (SE 0.0015) with the cyclically adjusted balance and −0.0014 (SE 0.0017) with the structural balance, so the adjustment leaves its sign and halves it on the smaller sample. The triple interaction is −0.084 (SE 0.044) with the adjusted balance, which differs from zero at the 10% level unlike the baseline, and within one standard error of zero for revenue and for primary spending. Table C.18 projects the overall-balance contribution to debt, which absorbs net interest and is reported outside the identity. At five years it is −0.12 (SE 0.51), within one standard error of the sum of the primary-balance and interest contributions of Figure 3, −0.13, as the identity implies.

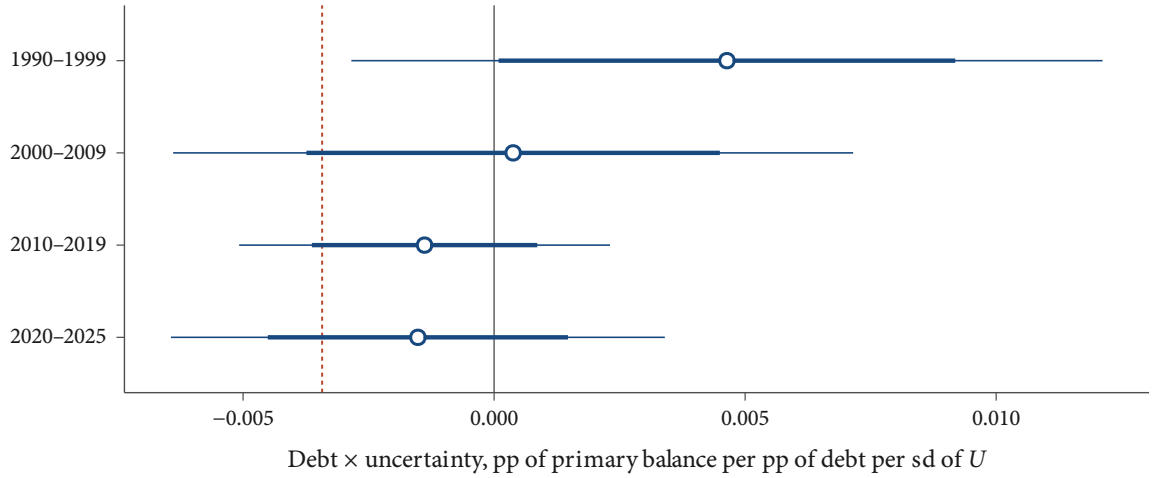


Figure C.15. The debt–uncertainty interaction of the reaction function by decade

Notes: The interaction of lagged debt with fiscal-year uncertainty in the pooled reaction function of Table 2, estimated on each decade separately (country effects; the output gap and its interaction instrumented as in Table 2). Thick bars 68%, thin bars 90% DK bands (lag 2); the dashed line is the full-sample estimate (-0.0034). Country-years: 1990–1999 766, 2000–2009 1,211, 2010–2019 1,234, 2020–2025 618.

Table C.17. The reaction function and the cyclicality model with other fiscal outcomes

	Reaction function (M2)						Cyclicality model (M3)				
Measure	U (sd)	$\text{Debt}_{t-1} \times U$	Gap $\times U$	First-stage F	N	Econ.	ϕ_{UF}, fitted gap	ϕ_{UF}, 2SLS	High U, high space	High U, low space	N
Primary balance (baseline)	0.107 (0.104)	−0.00343 (0.00172)	0.00646 (0.0325)	43.8	3,834	127	−0.0391 (0.0315)	0.00972 (0.0294)	0.342 (0.105)	0.370 (0.0988)	3,832
Cyclically adjusted primary balance (approximation)	0.0930 (0.0867)	−0.00291 (0.00145)	0.00618 (0.0404)	42.3	3,782	127	−0.0844 (0.0440)	0.0434 (0.0345)	0.843 (0.217)	0.849 (0.215)	3,780
WEO structural primary balance	−0.0144 (0.0991)	−0.00139 (0.00169)	0.0220 (0.0204)	61.3	1,874	70	−0.0288 (0.0150)	0.00274 (0.0142)	−0.0259 (0.0459)	−0.0450 (0.0433)	1,872
Revenue (M3 outcome)							−0.0189 (0.0143)	0.0111 (0.0304)	0.0368 (0.0709)	0.0459 (0.0657)	3,780
Primary spending (M3 outcome)							0.0185 (0.0245)	0.000973 (0.0434)	−0.301 (0.0813)	−0.323 (0.0812)	3,780

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses, from the pooled reaction function of Table 2 (columns 2–7) and the pooled cyclicality model of Table 3 (columns 8–12; ϕ_{UF} by the fitted-gap just-identified estimator and by 2SLS; the implied cyclical response under high uncertainty at the 25th and 75th percentiles of lagged debt), with the dependent variable of each row in place of the primary balance (and its change in M3). The cyclically adjusted primary balance is the stage-1 approximation $pb - (rev/100)\,gap$ (4,277 country-years, 137 economies in the main sample, against 4,327 and 139 for the primary balance); the WEO structural primary balance is the WEO structural balance plus net interest (1,973 country-years, 70 economies). In the component rows the change in revenue or in primary spending (percent of GDP) replaces the change in the primary balance in M3, with the same regressors and instruments; a positive gap coefficient means a procyclical component. Total investment is not in the panel. First-stage F: the smallest DK-robust first-stage Wald F across the instrumented variables.

C.3.6 The measure of fiscal space. Debt is the measure of space in the decomposition, cyclicality and insurance models and the composite is the regime variable of the regime model, and the question is whether another measure changes the objects. Table C.19 re-estimates the cyclicality model with the composite, the rating, the foreign-currency share, debt maturing within a year, the average maturity and the first principal component in place of debt, each oriented so that a higher value is less space, and Figure C.16 re-estimates the insurance model by measure. The triple interaction lies between −0.012 and 0.035 across the 6 alternative measures and differs from zero at the 10% level in 1. The two-year debt differential keeps its sign with 2 of the 6 alternative measures, the composite, 1.06 (SE 0.44), and debt maturing within a year, and switches it with the rating and with the foreign-currency share, −1.22 (SE 0.41) on 52 economies, so the erosion result is a result about the debt stock, not about market access. With debt in place of the composite as the regime variable the regime difference in the effective rate is 0.24 (SE 0.24), less than half the baseline and within one standard error of zero. The composite's rating and currency components therefore carry part of the price. The rollover measures start in 2012 and their rows rest on the shorter sample.

Table C.18. The overall-balance contribution to debt after a global uncertainty shock

	$\theta^{(h,\ell)}$, pp of GDP per sd of debt					
Component	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
Overall balance ($-ob$)	−0.146	−0.0952	0.185	0.323	−0.179	−0.124
	(0.0731)	(0.231)	(0.334)	(0.392)	(0.466)	(0.507)
Primary balance ($-pb$, Figure 3)	−0.110	0.0167	0.221	0.395	0.646	0.975
	(0.0627)	(0.233)	(0.376)	(0.414)	(0.530)	(0.593)
Interest (Figure 3)	−0.0364	−0.114	−0.0393	−0.0749	−0.829	−1.10
	(0.0505)	(0.209)	(0.302)	(0.426)	(0.439)	(0.569)
Observations	3,367	3,033	2,766	2,547	2,352	2,178

Notes: Entries are the interaction coefficients $\theta^{(h,\ell)}$ on $\varepsilon_t^g \tilde{d}_{i,t-1}$ (the global shock times lagged standardised debt) of the cumulative local projection of Figure 3, in pp of GDP per sd of lagged debt, for the overall-balance contribution to the change in debt ($-ob_t$, with $ob = pb-$ net interest), estimated separately because the five-component identity of Figure 3 would then count interest twice; the primary-balance and interest contributions of Figure 3 are repeated for comparison. Country and year effects, lagged controls, Driscoll–Kraay standard errors (lag 2) in parentheses; the M1 sample (132 economies).

Table C.19. The cyclicality model across measures of fiscal space

				Implied cyclical response at the corners					
Measure of fiscal space	Output gap ϕ	ϕ_{UF}, fitted gap	ϕ_{UF}, 2SLS	Low U, high space	Low U, low space	High U, high space	High U, low space	N	Econ.
Lagged debt (M1, M3, M5 baseline); debt also as the M4 and Figure 4 regime variable	0.356 (0.104)	−0.0391 (0.0315)	0.00972 (0.0294)	0.305 (0.111)	0.377 (0.111)	0.342 (0.105)	0.370 (0.0988)	3,832	127
Composite vulnerability index (M4 baseline)	0.315 (0.0654)	0.0156 (0.0141)	0.0232 (0.0295)	0.348 (0.0678)	0.295 (0.0655)	0.327 (0.0683)	0.296 (0.0679)	2,839	121
Sovereign rating (sign reversed)	0.241 (0.0579)	0.00308 (0.0210)	0.0245 (0.0358)	0.335 (0.0622)	0.155 (0.0551)	0.326 (0.0644)	0.153 (0.0608)	2,822	114
Foreign-currency share of government debt	0.0842 (0.164)	0.0175 (0.0247)	0.00806 (0.0437)	0.183 (0.145)	0.0157 (0.181)	0.142 (0.168)	0.0135 (0.170)	747	48
Debt maturing within 12 months, % of GDP	0.0552 (0.0541)	0.0346 (0.0176)	0.0364 (0.0183)	0.0501 (0.0581)	0.0483 (0.0526)	0.0447 (0.0643)	0.0798 (0.0564)	1,366	110
Average maturity of sovereign debt (sign reversed)	−0.0405 (0.0466)	−0.0125 (0.0135)	−0.0337 (0.0152)	−0.0560 (0.0442)	−0.0314 (0.0595)	−0.0309 (0.0354)	−0.0253 (0.0433)	927	84
First-PC composite (mixed loadings)	0.293 (0.0724)	−0.00962 (0.0260)	0.00982 (0.0379)	0.358 (0.0704)	0.238 (0.0782)	0.353 (0.0751)	0.216 (0.0772)	2,839	121

Notes: Entries are coefficients of the pooled cyclicality model of Table 3 by the fitted-gap just-identified estimator of Table C.6 (2SLS for ϕ_{UF} in column 4), with the fiscal-space measure of each row standardised within the sample and oriented so that higher means less space; Driscoll–Kraay standard errors (lag 2) in parentheses. Columns 5–8 report the implied cyclical response $\phi + \phi_U U + \phi_F \tilde{d} + \phi_{UF} U \tilde{d}$ of Table 3, with the standardised measure of each row in place of $\tilde{d}$ (pp of GDP per pp of gap), at the 25th and 75th percentiles of U and of the measure (high space = 25th percentile of the oriented measure). Lagged debt stays in the controls when the measure is not debt. Coverage in the main sample: debt 4,263 country-years, 139 economies, 1990–2025; composite 3,067 country-years, 131 economies, 1991–2025; rating 3,011 country-years, 123 economies, 1991–2025; FX share 769 country-years, 52 economies, 1991–2025; debt due 1,487 country-years, 120 economies, 2012–2025; maturity 955 country-years, 90 economies, 2012–2025; first PC 3,067 country-years, 131 economies, 1991–2025. Correlation of each oriented measure with lagged debt: debt 1.00, composite 0.61, rating -0.00, FX share -0.35, debt due 0.50, maturity -0.08, first PC -0.55. The first-PC composite has mixed-sign loadings (debt, deficit and rollover needs against rating and FX share), so its orientation is not that of vulnerability (stage-1 build log).

C.3.7 Estimation. Table C.20 changes one estimation choice at a time, and Figure C.17 splits the regime model at the estimated debt threshold of each horizon. The reaction function has the lagged balance on the right with country effects, the setting of Nickell (1981), so it was re-estimated by the two-step system GMM of Blundell and Bond (1998) with the correction of Windmeijer (2005) and collapsed lags, which gives a debt interaction of −0.046 (SE 0.089) with 24 instruments and a Hansen J with $p = 0.051$. The standard error is 51 times the baseline's, so the estimator is too imprecise to discriminate and the baseline lies inside its 90% band. With the Hodrick–Prescott gap the interaction is −0.0010 (SE 0.0024) and with the WEO gap, on 23 advanced economies, it switches sign, the one switch in the family; by OLS and with country-specific trends it keeps its sign and shrinks. For the local projections, three lags of the controls and the outcome's own lag leave the two-year debt differential within one baseline standard error of the baseline. Without the outlier rule the differential is 2.88 (SE 0.93), and with the components winsorised at the 1st and 99th percentiles in place of the rule it is 1.49 (SE 0.39), so the rule removes variance rather than the result. In the regime model the effective-rate difference is 0.73 (SE 0.17) with $\kappa = 1$ and 0.47 (SE 0.10) with $\kappa = 3$, so it keeps its sign under both speeds and is largest when the regimes blend most, whereas the above-minus-below difference at the sharp debt threshold, with the regime-level terms of Table C.7, is −0.01 (SE 0.47), within one standard error of zero: the sharp split finds the regime in the level of the response, not in the exposure gradient.

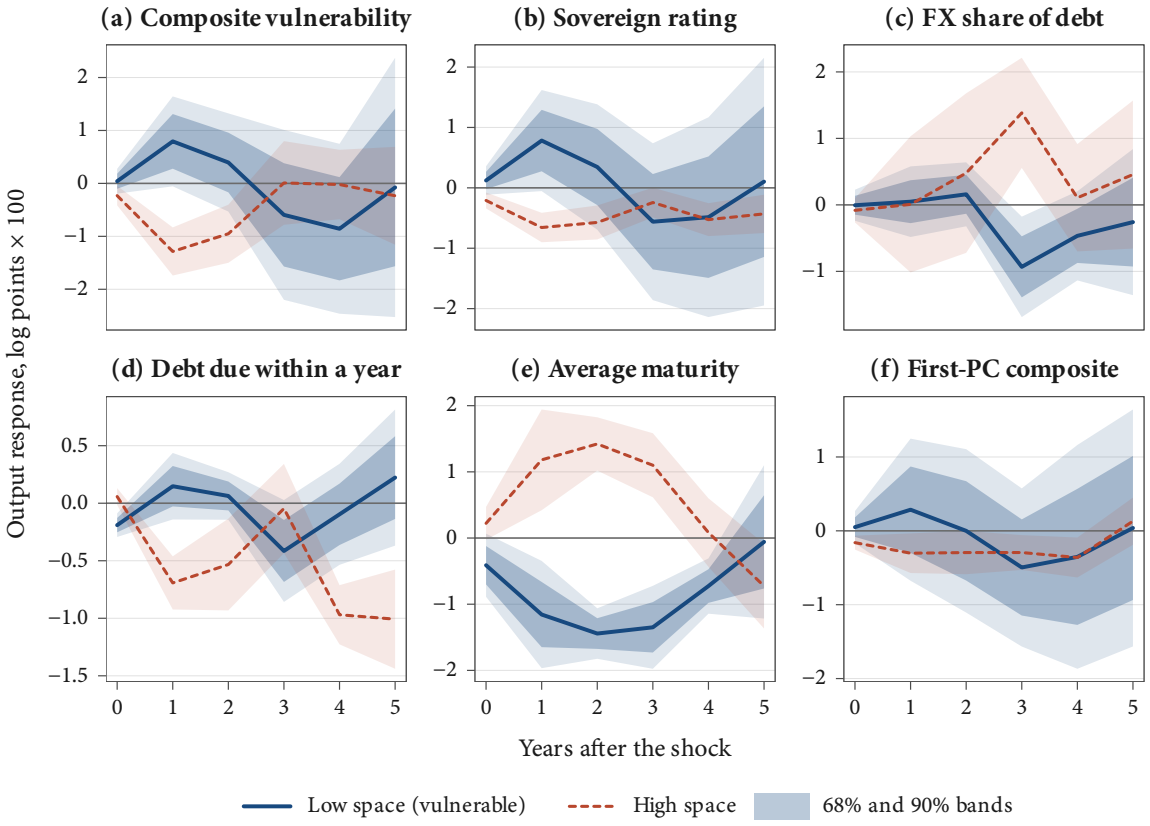


Figure C.16. Fiscal space as insurance across measures of fiscal space (M5)

Notes: As Figure 5, row 3 (output after the global shock × exposure), with the regime indexed by each measure of fiscal space in place of lagged debt (F near 1: high space; $\kappa = 1.5$); country and year effects, lagged controls, DK bands (lag 2). Economies: composite 122, rating 115, FX share 48, debt due 112, maturity 85, first PC 122.

Table C.20. The key estimates across estimation choices

Check	K1: debt × U (M2)	K2: ϕ_{UF}, fitted gap (M3)	S set (90%) for ϕ_{UF}	K3: debt, $h = 2$ (M1)	K3: debt, $h = 5$ (M1)	K3: interest rate, $h = 2$ (M4)	K4: median debt added, AE / EM / LIDC	N (M2 / M3 / M1)
Baseline	−0.00343 (0.00172)	−0.0391 (0.0315)	[−0.0850, 0.0200]	1.82 (0.392)	0.709 (0.691)	0.599 (0.130)	0.68 / 0.06 / −0.13	3,834 / 3,832 / 2,766
M2 by system GMM (two-step, Windmeijer, collapsed lags 2–4)	−0.0457 (0.0887)							8,763 / — / —
HP-filter output gap in place of the Hamilton gap (M2, M3)	−0.000973 (0.00244)	−0.123 (0.113)	[−0.320, 0.105]					3,794 / 3,792 / —
WEO output gap in place of the Hamilton gap (M2, M3)	0.00434 (0.00269)	0.0242 (0.0269)	[−0.0200, 0.0750]					760 / 759 / —
Output gap not instrumented (OLS, M2 and M3)	−0.00254 (0.00167)	−0.00827 (0.0116)						3,834 / 3,832 / —
Country-specific linear trends (M2, M3)	−0.00102 (0.00190)	−0.0492 (0.0262)						3,834 / 3,832 / —
Three lags of the controls (M1, M4)				1.93 (0.433)	0.615 (0.673)	0.681 (0.164)	0.72 / 0.07 / −0.14	— / — / 2,682
The outcome's own lag added (M1, M4)				1.92 (0.444)	0.698 (0.698)	0.299 (0.148)	0.72 / 0.07 / −0.14	— / — / 2,743
Transition speed $\kappa = 1$ (M4)						0.726 (0.165)	0.68 / 0.06 / −0.13	— / — / —
Transition speed $\kappa = 3$ (M4)						0.474 (0.104)	0.68 / 0.06 / −0.13	— / — / —
Hansen threshold on lagged debt in place of the smooth transition (M4)						−0.00878 (0.467)		— / — / —
M1 without the outlier rule				2.88 (0.928)	1.72 (1.51)		1.08 / 0.29 / −0.11	— / — / 3,518
M1 with the components winsorised at the 1st/99th percentiles, no outlier rule				1.49 (0.390)	0.480 (1.01)		0.56 / 0.15 / −0.06	— / — / 3,518

Notes: Entries are the objects under test with one change in the estimation. K1 is the interaction of lagged debt with uncertainty in the pooled reaction function of Table 2 (pp of primary balance per pp of debt per sd of U); K2 the triple interaction of the output gap with uncertainty and fiscal space in the pooled cyclicality model of Table 3, by the fitted-gap just-identified estimator of Table C.6; K3 the cumulative debt response to a one-sd global uncertainty shock at low minus high fiscal space (lagged debt at the stage-2 p75 minus p25) from the local projections of Figure 3, and the effective-interest-rate difference between the vulnerable and the safe regime at $h = 2$ from the smooth-transition model of Figure 5; K4 the group medians of the debt added at $h = 2$ relative to the mean-debt economy from Figure 4. Driscoll–Kraay standard errors (lag 2) in parentheses; an empty cell means the check does not alter that model. The S-set column is the 90% set for ϕ_{UF} not rejected by the S statistic of Stock and Wright (2000) with the fitted-gap instruments (two-step GMM with the Driscoll–Kraay weighting matrix, the other gap coefficients concentrated out, grid $[-0.6, 0.6]$ by 0.005; not computed with country trends or by OLS). System GMM: plm::pgmm, two-step with the Windmeijer correction, lagged primary balance, output gap and gap × U instrumented by their lags 2–4 (collapsed) in the differenced and level equations, the external instruments of Table 2 as standard instruments, 24 instruments; Hansen $J = 15.5$ (df 8, $p = 0.051$), Arellano–Bond AR(1) $p = 0.021$, AR(2) $p = 0.972$. The HP gap uses a smoothing parameter of 6.25 on annual real GDP; the WEO gap is the IMF's output gap (23 economies). Three lags of the controls: $X_{i,t-1}, X_{i,t-2}, X_{i,t-3}$. The threshold row reports the above-minus-below response at the Hansen threshold of Figure C.17 in the M4 column. Winsorisation: each debt-dynamics component capped at its 1st and 99th percentiles of the complete M1 sample (456 component-years changed) and cumulated; no outlier rule.

C.3.8 Inference. The baseline covariance is Driscoll–Kraay with two lags, and the question is whether the verdicts depend on it. Table C.21 reports the baseline estimates with the Bartlett lag at 1 and 4, with two-way clustering by economy and year, and with the wild cluster bootstrap by year and by economy (the wild restricted efficient bootstrap of Davidson and MacKinnon (2010) for the instrumented coefficients), together with the block-bootstrap bands of the Figure 4 medians. The debt interaction's p-value lies between 0.037 and 0.062 across the Driscoll–Kraay lags, is 0.089 under two-way clustering, and is 0.094 by the bootstrap by year and 0.063 by economy, so the interaction is rejected at the 10% level under every choice and at the 5% level under the four-lag covariance alone, the verdict of Section 3.3. The triple interaction's bootstrap p-values are 0.518 by year and 0.479 by economy, so the null survives under both clusterings. For the projections the Bartlett lag is also set to $h + 1$, the order of the moving average that the projection error follows. The lag leaves the standard error of the two-year interest-rate regime difference of Figure 5 at 0.13, as under the baseline lag. It lowers that of the five-year difference, 0.75, from 0.48 to 0.40, so that the five-year difference is rejected at the 10% level under the longer lag and not under the baseline. For the Figure 4 medians the block bootstrap over economies gives a 90% band for the debt added in the advanced economies from −0.10 to 2.10 points of GDP, which includes zero, while the ordering of the advanced above the low-income median holds in 94% of the draws.

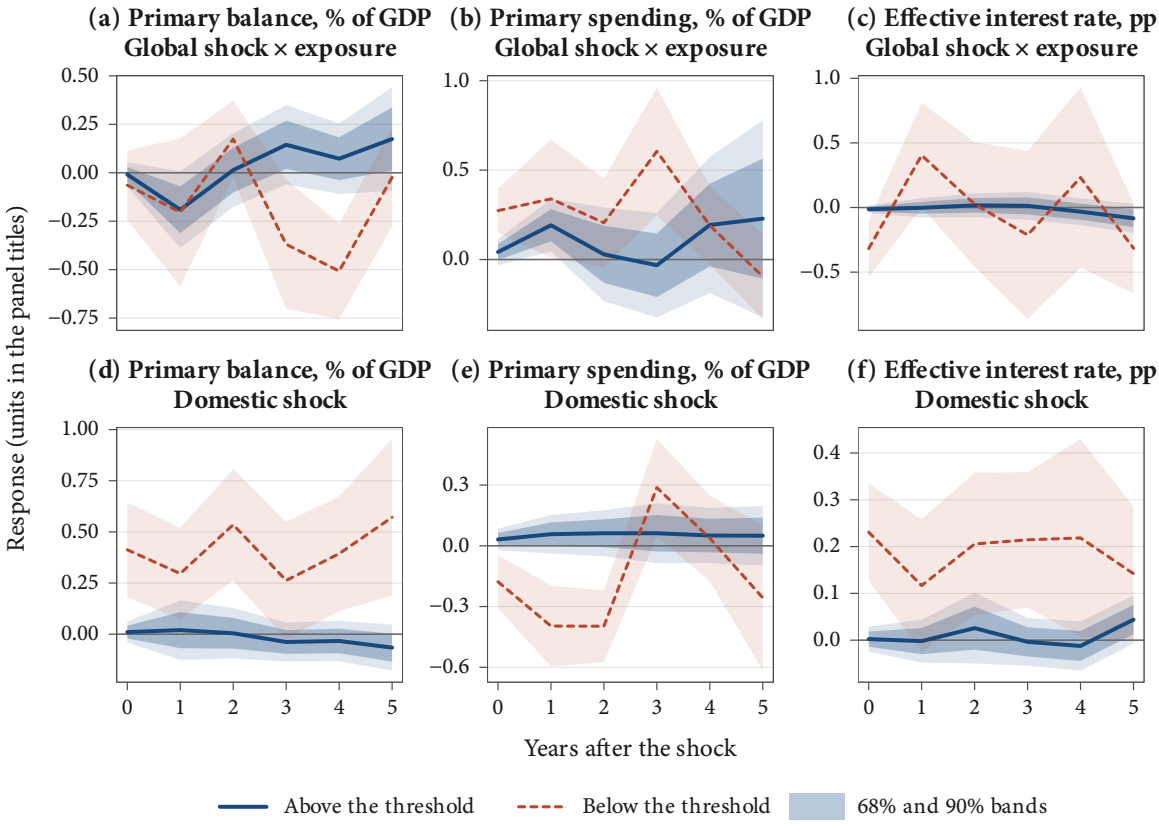


Figure C.17. Responses below and above an estimated debt threshold (M4, sharp alternative)

Notes: Local projections of $y_{i,t+h} - y_{i,t-1}$ with the shocks interacted with an indicator of lagged debt above the Hansen (1999) threshold, re-estimated at each horizon (grid: 10th–90th percentiles; thresholds 18–25% of GDP); country and year effects, lagged controls, the indicator and its products with the global shock and exposure; DK bands (lag 2). Global shock: one sd, exposure gradient (one sd of openness).

Table C.21. The key estimates under alternative inference

Object	Estimate	Standard error [p-value]							N
		DK, lag 1	DK, lag 2	DK, lag 4	DK, lag $h+1$	Two-way cluster	Wild bootstrap, year	Wild bootstrap, economy	
K1: debt × U (M2, IV)	−0.00343	0.00177 [0.062]	0.00172 [0.055]	0.00158 [0.037]		0.00196 [0.089]	0.00203 [0.094]	0.00191 [0.063]	3,834
K2: ϕ_{UF} (M3, fitted-gap just-identified)	−0.0391	0.0314 [0.221]	0.0315 [0.223]	0.0300 [0.202]		0.0348 [0.270]	0.0296 [0.518]	0.0368 [0.479]	3,832
M1 direct effect on debt, $\beta^{(h,\ell)}$ (global shock, $h = 2$)	5.23	1.34 [0.000]	1.49 [0.001]	1.63 [0.003]	1.56 [0.002]	1.16 [0.000]	1.16 [0.010]	0.311 [0.000]	2,766
M1 direct effect on debt, $\theta^{(h,\ell)}$ (shock × $\tilde{d}_{i,t-1}$, $h = 2$)	1.86	0.331 [0.000]	0.337 [0.000]	0.341 [0.000]	0.339 [0.000]	0.319 [0.000]	0.308 [0.027]	0.361 [0.000]	2,766
M1 direct effect on debt, $\beta^{(h,\ell)}$ ($h = 5$)	4.70	1.32 [0.001]	1.33 [0.001]	1.09 [0.000]	0.941 [0.000]	1.29 [0.001]	1.26 [0.001]	0.583 [0.000]	2,178
M1 direct effect on debt, $\theta^{(h,\ell)}$ ($h = 5$)	0.875	0.509 [0.097]	0.550 [0.123]	0.558 [0.128]	0.544 [0.119]	0.513 [0.099]	0.438 [0.026]	0.620 [0.160]	2,178
M4 interest-rate regime difference, $\mathbf{b}_{\mathrm{V}}^{(h)} - \mathbf{b}_{\mathrm{S}}^{(h)}$ on $\varepsilon_t^{\mathrm{g}} E_i$ ($h = 2$)	0.599	0.131 [0.000]	0.130 [0.000]	0.122 [0.000]	0.127 [0.000]	0.252 [0.018]	0.158 [0.003]	0.344 [0.092]	2,536
M4 interest-rate regime difference ($h = 5$)	0.752	0.472 [0.111]	0.479 [0.116]	0.437 [0.085]	0.395 [0.057]	0.513 [0.142]	0.470 [0.143]	0.496 [0.152]	2,196

Notes: Entries are standard errors with p-values in brackets for the baseline estimates (column 2) under alternative inference: Driscoll–Kraay with Bartlett lags 1, 2 (the baseline) and 4, and, for the local projections, lag $h + 1$, the order of the moving average that the projection error at horizon h follows by construction; two-way clustering by economy and year; the wild cluster bootstrap by year and by economy (999 Rademacher draws, null imposed; the standard error shown is the cluster-robust one by that dimension and the p-value the bootstrap one). For the IV coefficients the bootstrap is the wild restricted efficient (WRE) bootstrap of Davidson and MacKinnon (2010) on the within-transformed system, with the reduced form of the instrumented gap terms resampled with the same weights. K1 is the debt × uncertainty interaction of Table 2, column 1; K2 the triple interaction of Table 3 by the fitted-gap just-identified estimator of Table C.6; the direct effects are those of Table C.1 for the change in debt (pp of GDP per sd of the global shock $\varepsilon_t^{\mathrm{g}}$, and per sd of lagged standardised debt $\tilde{d}_{i,t-1}$ for the interaction), without year effects; the M4 rows are the effective-interest-rate regime difference of Figure 5 (vulnerable minus safe, pp per sd of the global shock for an economy one sd more open than the mean), with the wild bootstrap of the difference by the reparameterisation that enters the sum and the difference of the two regime regressors.

C.3.9 The GDELT index. The headline GDELT column uses the all-outlet domestic index on non-thin country-years, and the question is whether its construction drives the two-source split. Table C.22 re-estimates the two models and the split with the stable-outlet-panel indices, the native-language index, the own-country-mention index, the sample without the 39 economies with a composition break, the sample with the 190 thin country-years kept, the all-outlet index, the domestic-minus-foreign gap as one regressor, and a fiscal-attention index built without the uncertainty filter. The domestic debt interaction lies between −0.0027 and 0.0004 across the 5 domestic variants, each within one standard error of zero. The split keeps its signs in all 6 pairs and the difference is rejected at the 5% level in 4, so the split does not depend on the outlet panel, the language or the mention rule. The fiscal-attention index gives a debt interaction of 0.0067 (SE 0.0037), not smaller than the headline domestic interaction in absolute value, which is one reason the domestic GDELT results are read as patterns.

Table C.22. The reaction function and the cyclicality model across GDELT index variants, 2016–2025

		Two-source split							
Variant	K1: debt × U	Debt × U^d	Debt × U^f	Difference [p]	K2: ϕ_{UF}, fitted gap	ϕ_{UF}, 2SLS	N (M2)	Econ.	N (M3)
Domestic FPU, non-thin country-years (Table 2, column 5)	−0.00119 (0.00344)	−0.00693 (0.00408)	0.0101 (0.00433)	−0.0170 (0.00765) [0.026]	−0.0230 (0.0209)	−0.0196 (0.0390)	954	114	954
Stable-outlet-panel indices (domestic; foreign in the split)	0.000370 (0.00425)	−0.00599 (0.00392)	0.0105 (0.00335)	−0.0165 (0.00597) [0.006]	−0.0643 (0.0208)	−0.0329 (0.0471)	954	114	954
Native-language domestic index	−0.00271 (0.00340)	−0.00596 (0.00188)	0.00859 (0.00367)	−0.0145 (0.00441) [0.001]	−0.0178 (0.0230)	−0.0142 (0.0344)	919	109	919
Own-country-mention domestic index	0.0000800 (0.00382)	−0.00448 (0.00283)	0.00855 (0.00335)	−0.0130 (0.00462) [0.005]	−0.0276 (0.0259)	−0.0228 (0.0385)	954	114	954
Economies with an outlet-composition break dropped (39 economies)	−0.00181 (0.00453)	−0.00659 (0.00447)	0.00707 (0.00441)	−0.0137 (0.00723) [0.059]	−0.0273 (0.0222)	−0.0126 (0.0455)	740	84	740
Thin country-years kept (190 added)	0.000355 (0.00245)	−0.00373 (0.00314)	0.00773 (0.00358)	−0.0115 (0.00618) [0.064]	−0.00875 (0.0254)	−0.00254 (0.0357)	1,102	123	1,102
All-outlet index (domestic and foreign together)	0.00522 (0.00367)				−0.0452 (0.0187)	−0.0356 (0.0520)	954	114	954
Domestic minus foreign FPU gap as the regressor	0.0101 (0.00452)				−0.0198 (0.0208)	−0.0403 (0.00905)	954	114	954
Fiscal-attention index (coverage, no uncertainty)	0.00673 (0.00373)				−0.0241 (0.0123)	−0.107 (0.0655)	954	114	954

Notes: Entries are coefficients with Driscoll–Kraay standard errors (lag 2) in parentheses, from the pooled reaction function of Table 2 (K1: debt × U, pp of primary balance per pp of debt per sd) and the pooled cyclicality model of Table 3 (K2: ϕ_{UF} by the fitted-gap just-identified estimator and by 2SLS), 2016–2025, with the GDELT index of each row as U (standardised within the main sample). Columns 3–5 re-estimate the two-source reaction function of Table C.4 with the domestic variant of the row and its foreign counterpart (the stable-panel foreign index in the panel row, the headline foreign index otherwise); the difference carries its standard error and p-value. Rows: the headline domestic index on non-thin country-years; indices built on the stable panel of outlets; articles in the country's own languages; articles mentioning the country; economies whose all-outlet and stable-panel indices differ by more than a factor of two in some year dropped; thin cells (fewer than 2,000 articles or 20 fiscal-uncertainty articles) kept; the all-outlet view; the domestic-minus-foreign gap; and the share of fiscal-economic articles without the uncertainty filter (fiscal attention).

C.3.10 Status or space, extended. Table C.11 extends the race of Table C.10 with interactions of uncertainty with government effectiveness, a floating exchange rate and a fiscal rule, one at a time and all six together. The debt interaction keeps its sign in 5 of the 7 extended columns and is 0.0019 (SE 0.0055) with all six interactions on the 643 country-years that report every variable, too imprecise to discriminate. Two of the added interactions carry their own weight: uncertainty interacted with government effectiveness is −0.18 (SE 0.08) and with a floating regime −0.29 (SE 0.10), both rejected at the 5% level, while the fiscal-rule interaction is within one standard error of zero. The triple interaction of the cyclicality model differs from zero at the 10% level in 0 of the 7 columns.

C.3.11 Infeasible checks. Two checks of the robustness plan could not be run as planned. The decomposition model at five years is not estimable on the 2020–2025 subsample, because the window leaves too few country-years for the cumulative projection, so the cell of Table C.16 is left empty and the two-year entry stands. The cyclicality of total investment was not estimated, because the investment series of the proposal is not in the panel; revenue and primary spending take its place in Table C.17 and in Figure C.6, and the insurance model's outcomes are output, the effective rate and the rating. Two further checks reduced to the baseline. No economy of the main sample has a median population below one million, so the small-state exclusion repeats the baseline row of Table C.16. The thin-cell flags of the foreign and all-outlet GDELT views are never set, so the planned variant that drops thin cells is the baseline, and the variant that keeps the thin domestic cells is reported instead.

C.4 A Bayesian re-estimation of the local projections

The local projections of Figures 3, 5 and 6 and their appendix counterparts were re-estimated under the prior of Ferreira et al. (2025), as the project implemented it for a panel: a Normal prior with mean zero on the shock coefficients, scaled as in that paper, a flat prior on the fixed effects and the controls, the Driscoll–Kraay sandwich as the likelihood information, and the tightness of the prior at each horizon sampled from its marginal posterior under a Gamma hyperprior. Table C.23 compares the frequentist and the Bayesian versions exhibit by exhibit over the 160 response paths that have a counterpart. The posterior medians differ from the point estimates by a mean of 0.10 standard errors, the 90% credible bands are 0.94 times the width of the frequentist bands at the median, and no point estimate whose 90% band excludes zero under either method changes sign. The headline estimates are unchanged: the two-year debt interaction of Figure 3 is 1.64 with a posterior standard deviation of 0.35, the effective-rate responses of Figure 5 are 0.31 in the vulnerable and −0.35 in the safe regime, and the peak output difference of the insurance model still includes zero. The event study is where the prior binds, since it rests on a cross-section of about a hundred economies. The 2020 debt coefficient of Figure 6 shrinks to −2.55 (1.35) and the 2021 coefficient to −3.76 (4.18), so the pandemic conclusions keep their signs and their 2020 precision and lose some of their 2021 magnitude. Of the 4 designs run, only the one that tempers the likelihood to the number of years alone would change conclusions, and the Driscoll–Kraay variances say that it overstates the dependence for every family but the growth channel. No ledger conclusion of the main text changes under the main design.

Table C.23. Bayesian against frequentist local projections, by exhibit

Exhibit	Paths	Horizons	Mean |Δ|/SE	Max |Δ|/SE	Shrinkage	90% width ratio	68% width ratio	LP: 90% excl. 0	BLP: 90% excl. 0	Sign changes	λ median [min, max]
Figure 3	18	108	0.04	0.18	2%	0.94	0.93	45	45	0	0.38 [0.31, 0.42]
Figure C.1	30	180	0.05	0.24	4%	0.93	0.92	60	62	0	0.38 [0.31, 0.45]
Figure C.2	6	36	0.02	0.06	0%	0.95	0.94	7	9	0	0.37 [0.29, 0.39]
Figure C.7	18	108	0.04	0.13	2%	0.93	0.92	42	42	0	0.38 [0.30, 0.42]
Figure C.3	12	72	0.05	0.22	3%	0.93	0.93	7	11	1	0.26 [0.08, 0.37]
Figure 5 (rows 1–2)	18	108	0.22	0.87	15%	0.87	0.85	34	26	1	0.13 [0.04, 0.31]
Figure C.4a	12	72	0.13	0.50	15%	0.85	0.85	8	13	3	0.12 [0.03, 0.32]
Figure C.4b	8	48	0.21	0.91	17%	0.84	0.82	8	8	1	0.12 [0.04, 0.32]
Figure C.4c	6	36	0.10	0.30	6%	0.89	0.87	10	10	0	0.29 [0.13, 0.38]
Figure 5 (row 3) / C.5a	9	54	0.04	0.15	3%	0.91	0.90	7	8	1	0.24 [0.10, 0.36]
Figure C.5b	9	54	0.17	0.46	13%	0.98	0.98	18	15	0	0.28 [0.18, 0.36]
Figure C.5c	6	36	0.07	0.19	7%	0.90	0.89	2	4	0	0.27 [0.12, 0.37]
Figure 6 / C.6b	8	48	0.22	0.68	19%	0.90	0.89	10	7	1	0.31 [0.22, 0.40]

Notes: Entries compare, over the response paths and horizons of each exhibit, the Bayesian local projections (posterior medians and posterior bands; shock coefficients shrunk towards zero under the Minnesota-scaled prior with $\lambda(h)$ sampled from its marginal posterior, Gaussian likelihood of the OLS estimates with the Driscoll–Kraay covariance, $h+1$ lags) with the stage-2 local projections (OLS, DK lag 2). Δ is the posterior median minus the OLS estimate in units of the OLS standard error; shrinkage is 1 − median/OLS averaged over the horizons where the OLS estimate exceeds its standard error; width ratios are posterior-band widths over DK confidence-band widths; the exclusion counts are horizons whose 90% band excludes zero; λ summarises the posterior median tightness across paths and horizons.

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